

Solar Apsidal U-Turn Resonance in Real GTO/HEO Objects

A TLE-based reproduction of Wang & Gurfil (2017), Advances in Space Research

59, 2101-2116, applied to 23 real, independently tracked objects.

Resonance condition (paper Eq. 12):

$$a (1-e^2)^{4/7} = \lambda_{crit}(i)$$

$$\lambda_{crit}(i) = [3 J_2 \sqrt{\mu} R_E^2 (5\cos^2 i - 2\cos i - 1) / (4 dM_{solar}/dt)]^{2/7}$$

Resonance angle plotted (proxy for the papers solar-azimuth U-turn):

$$\psi(t) = \text{mod}(RAAN(t) + \arg p(t) - L_{sun}(t), 360 \text{ deg})$$

$a(t)$, $e(t)$, $i(t)$, $RAAN(t)$, $\arg p(t)$ computed directly from each objects real

Space-Track TLE mean elements -- no propagation.

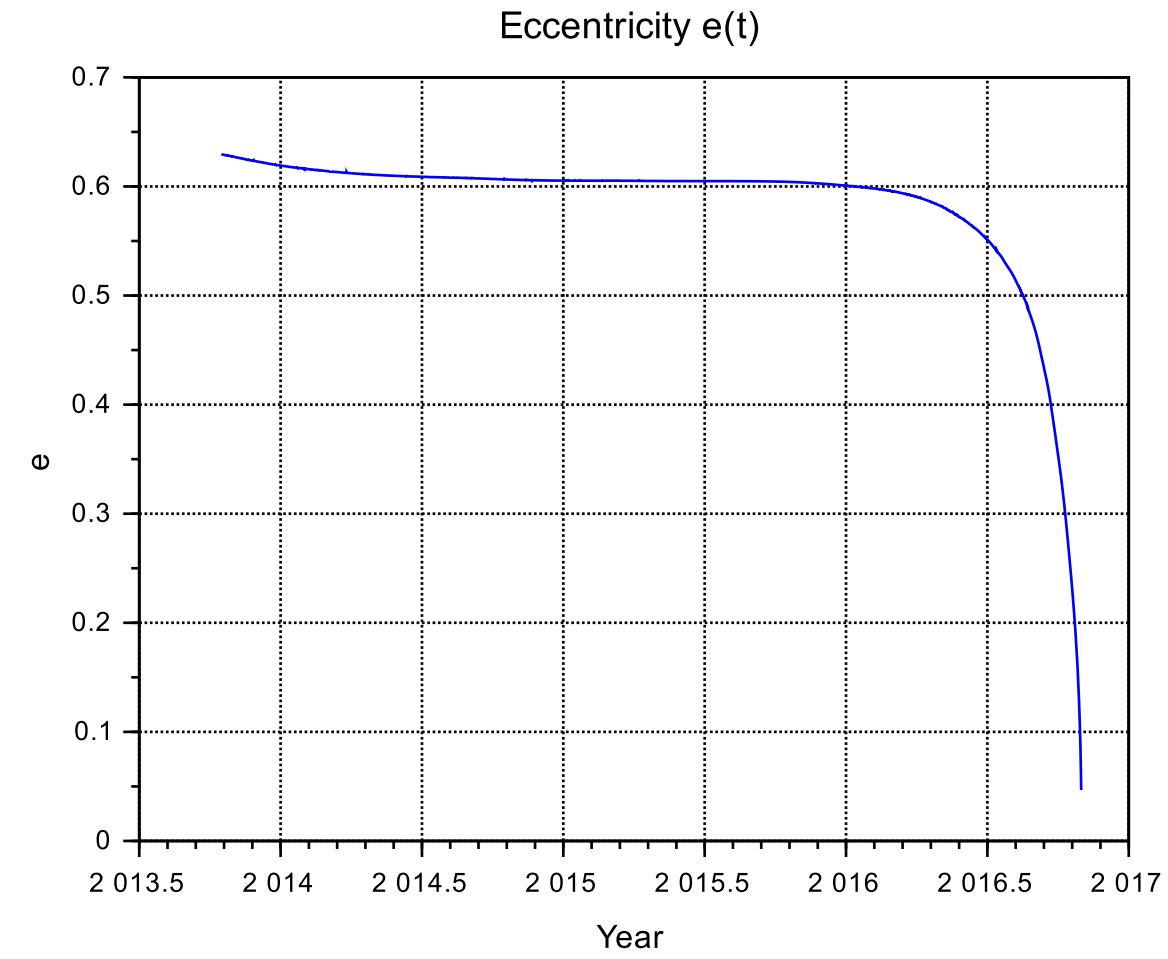
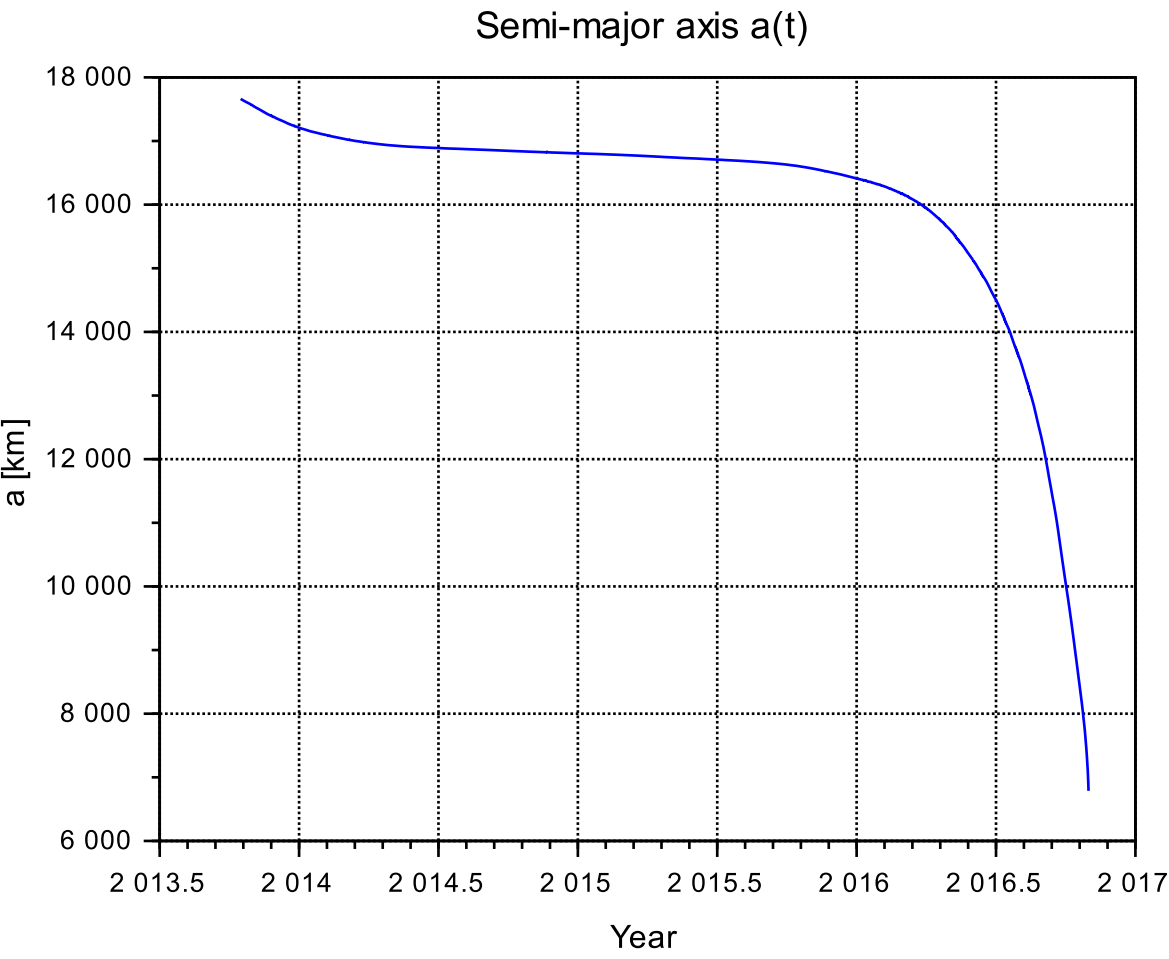
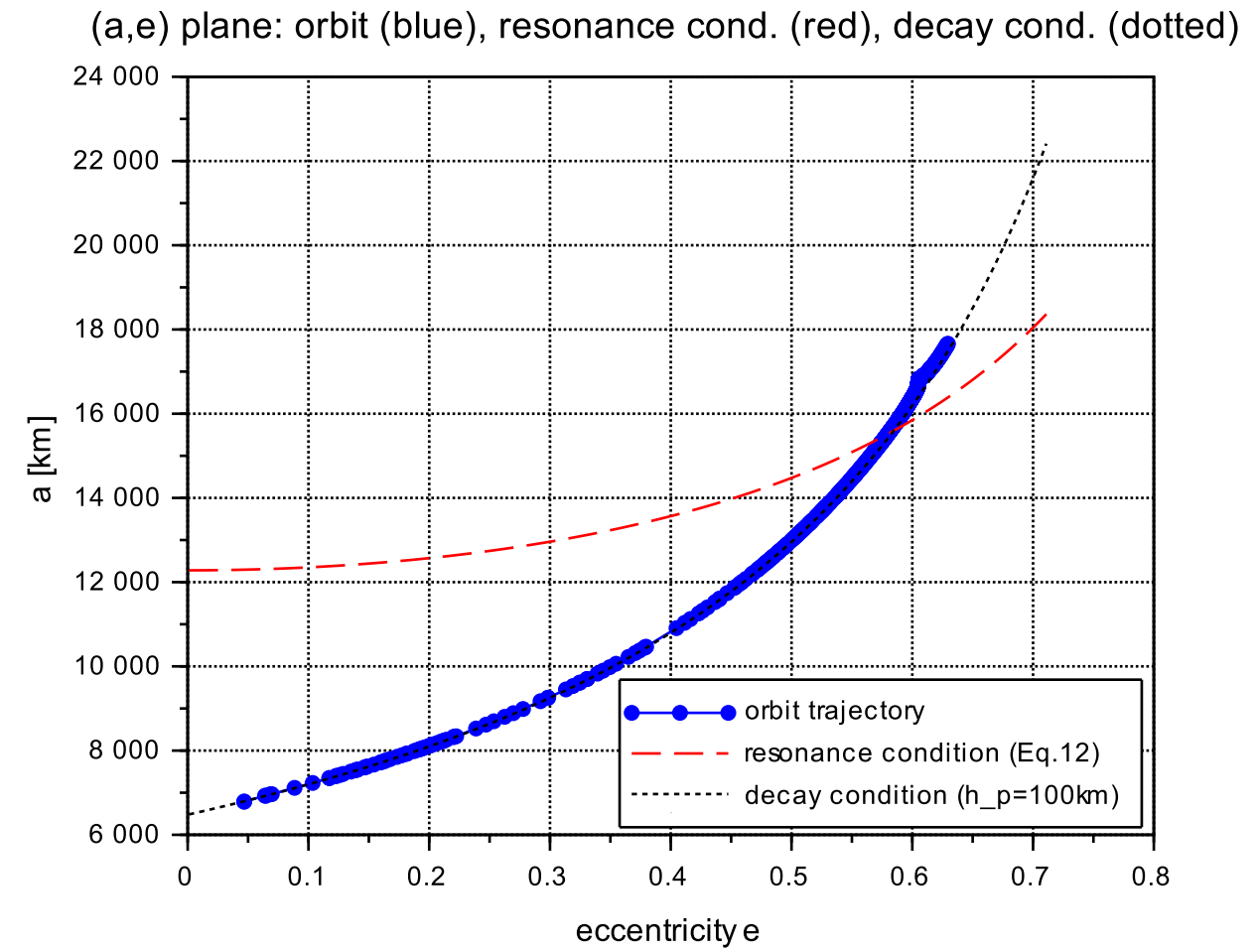
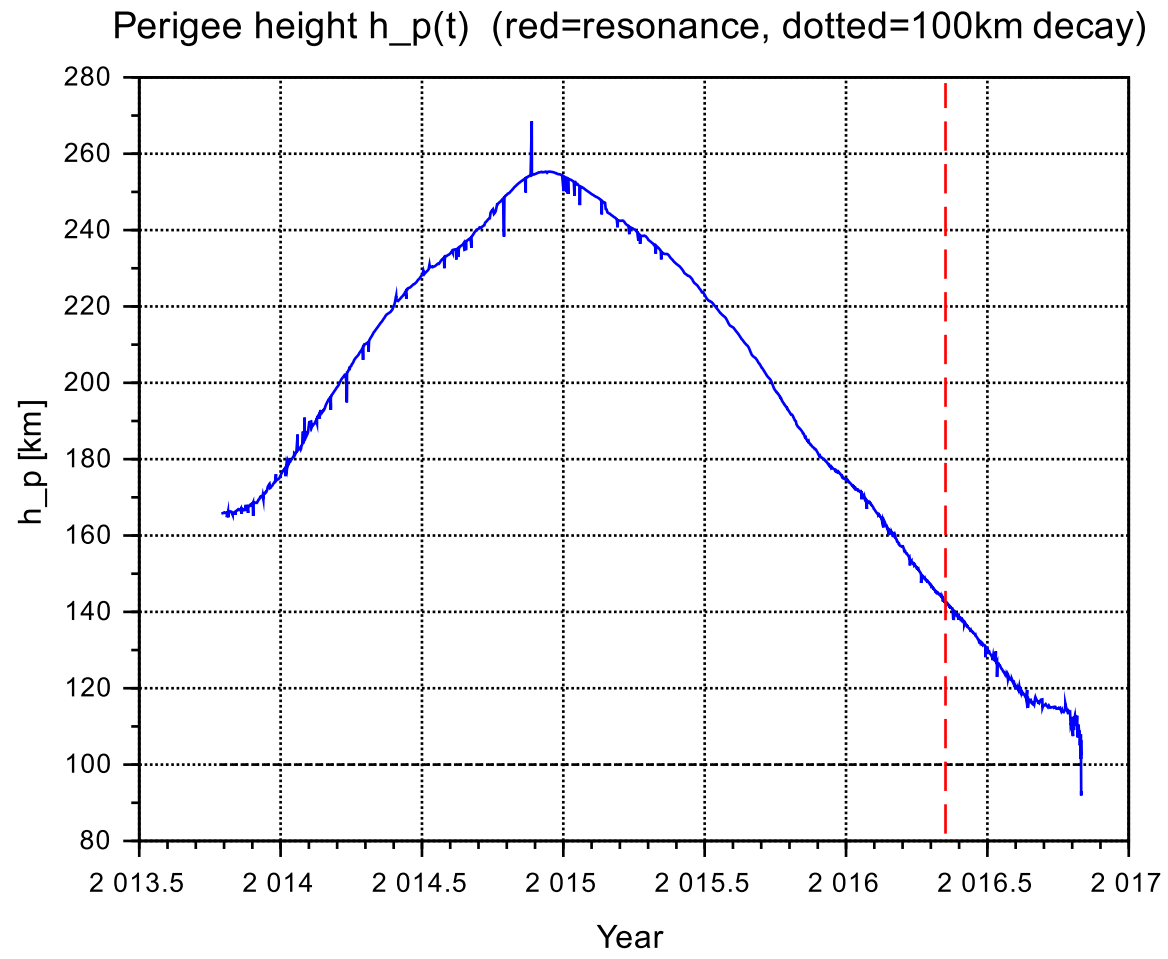
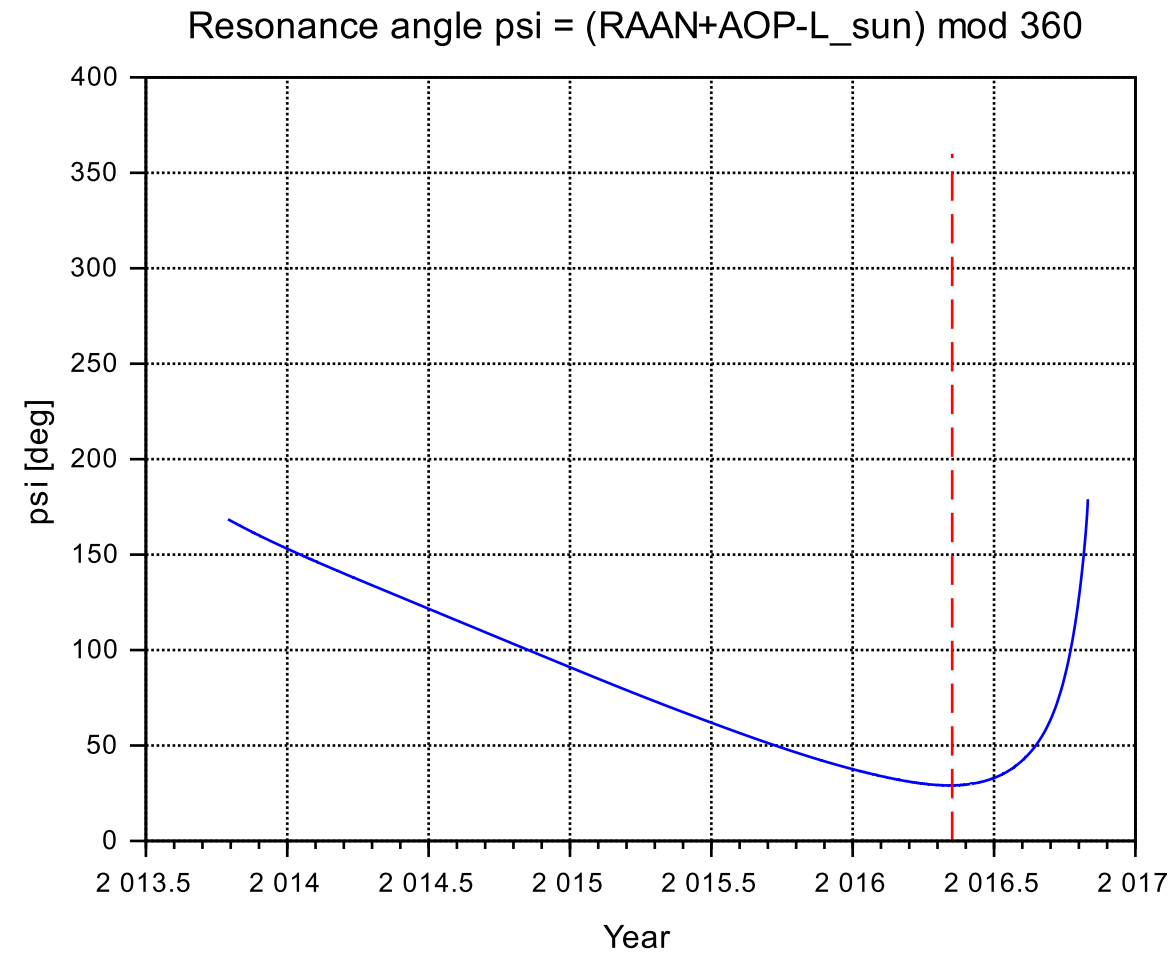
23 objects shown: 16 from OREMs original 97-object RPE campaign

+ 7 found after expanding the candidate pool.

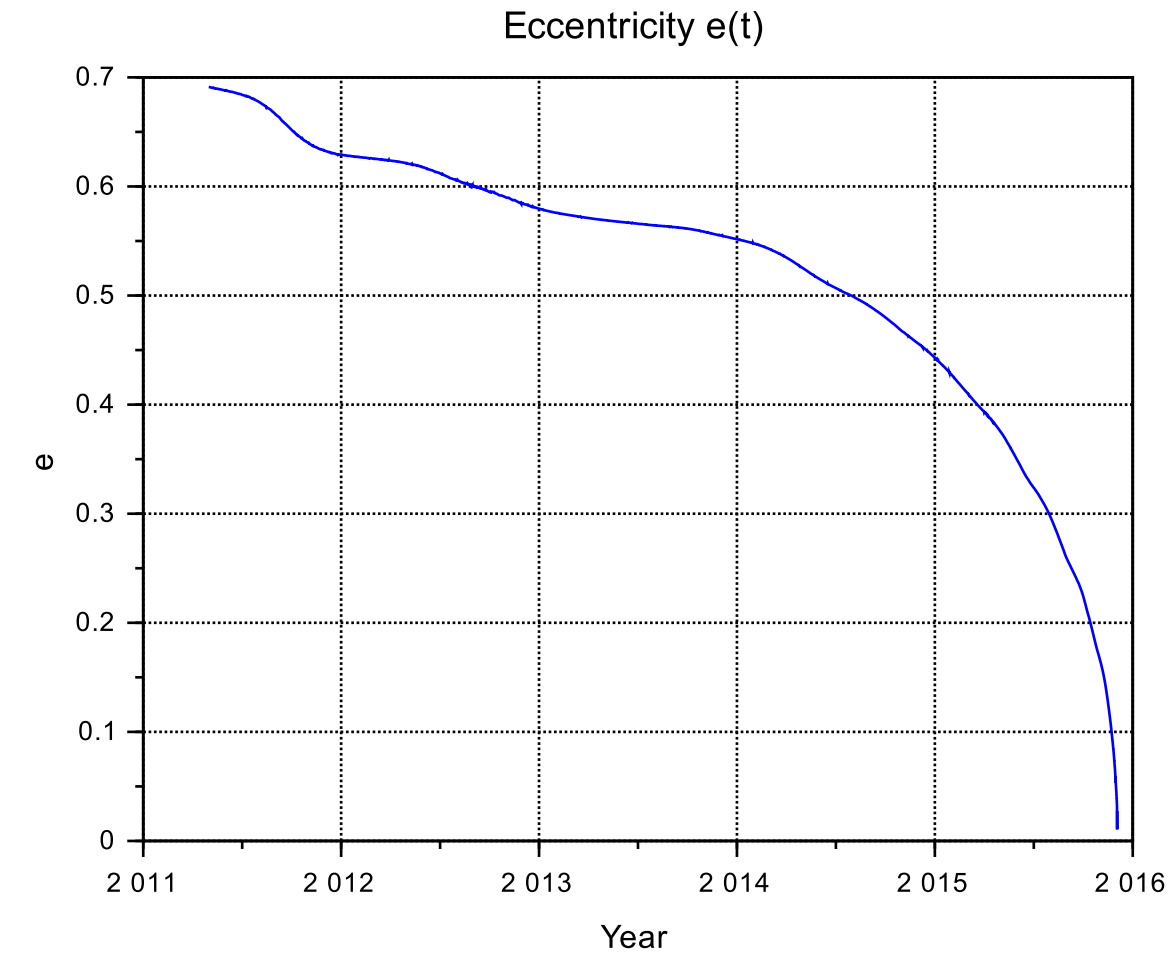
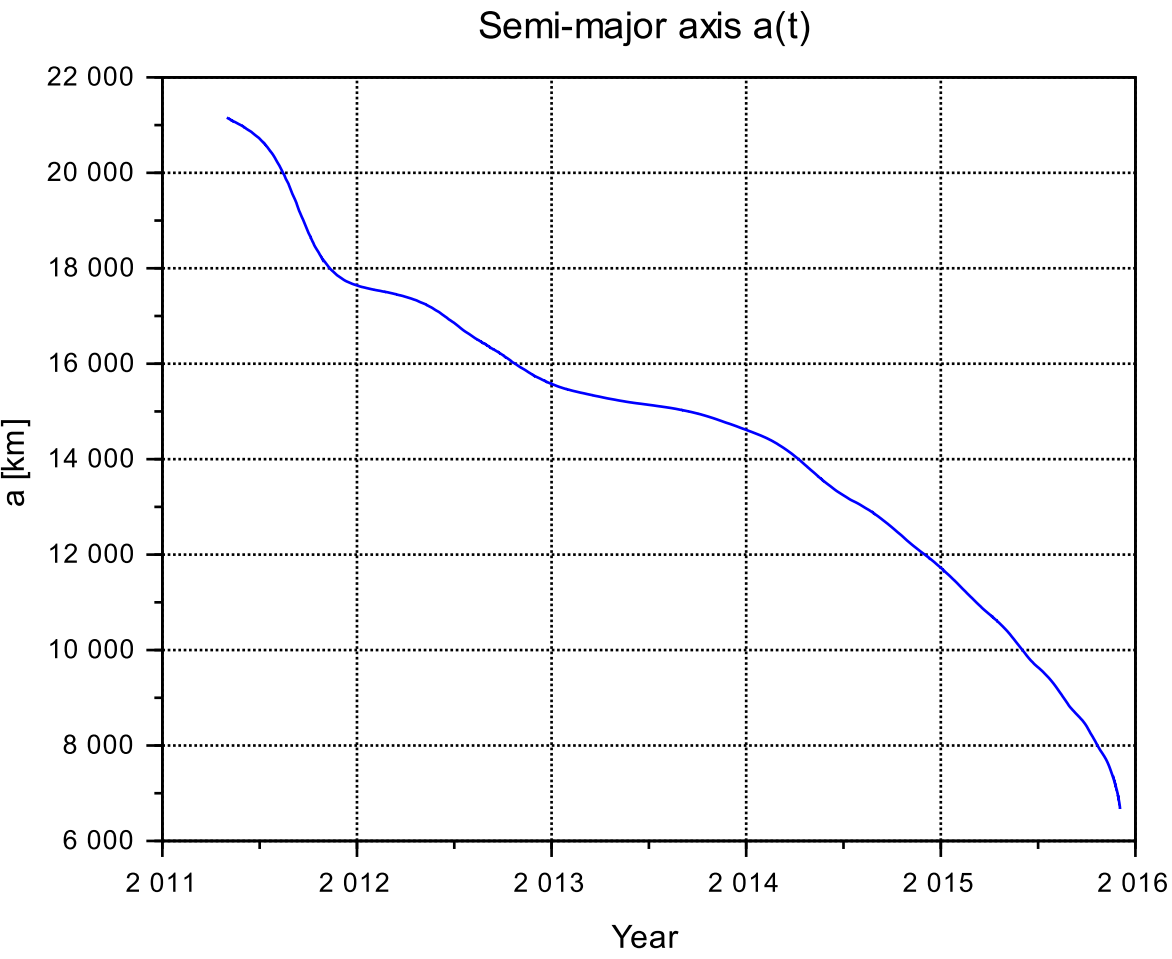
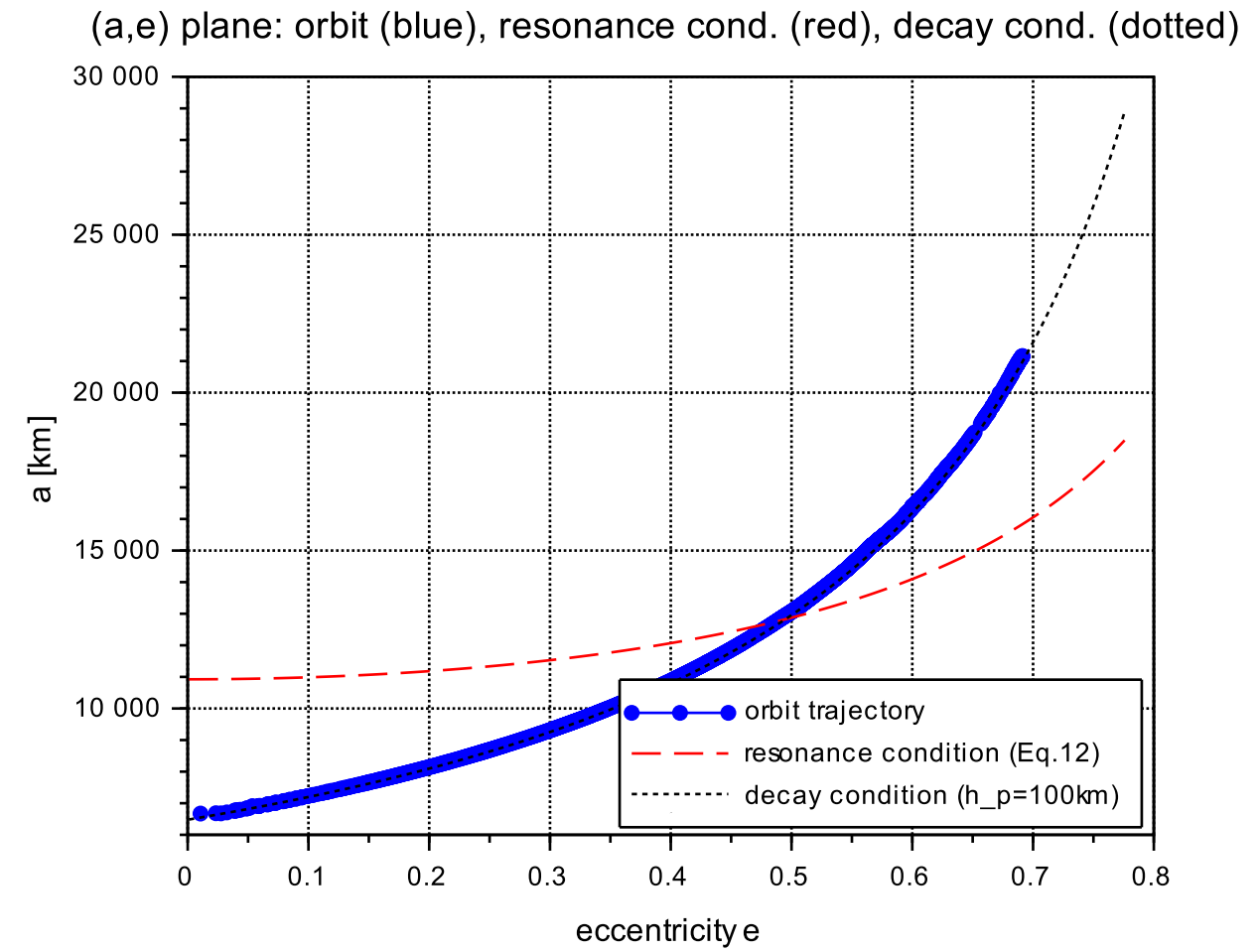
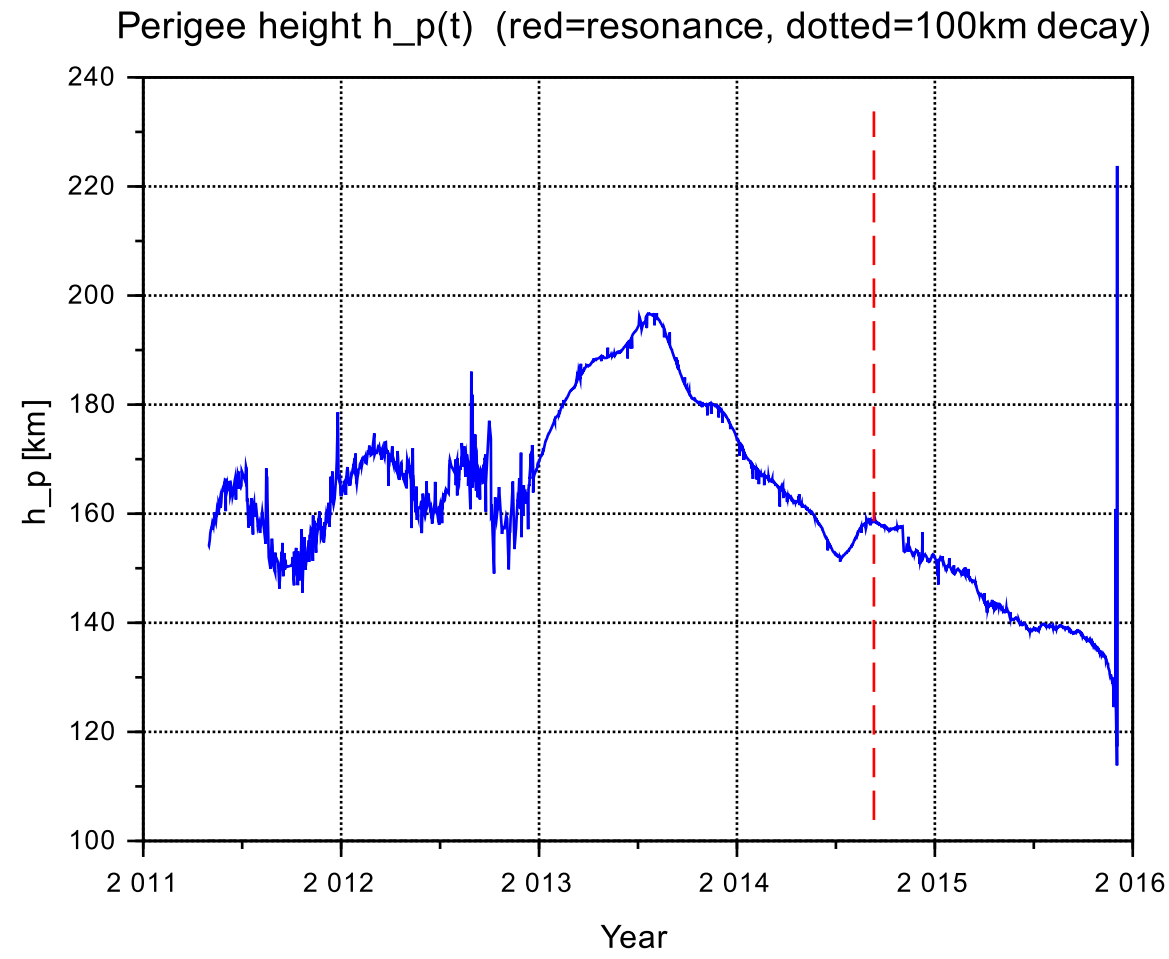
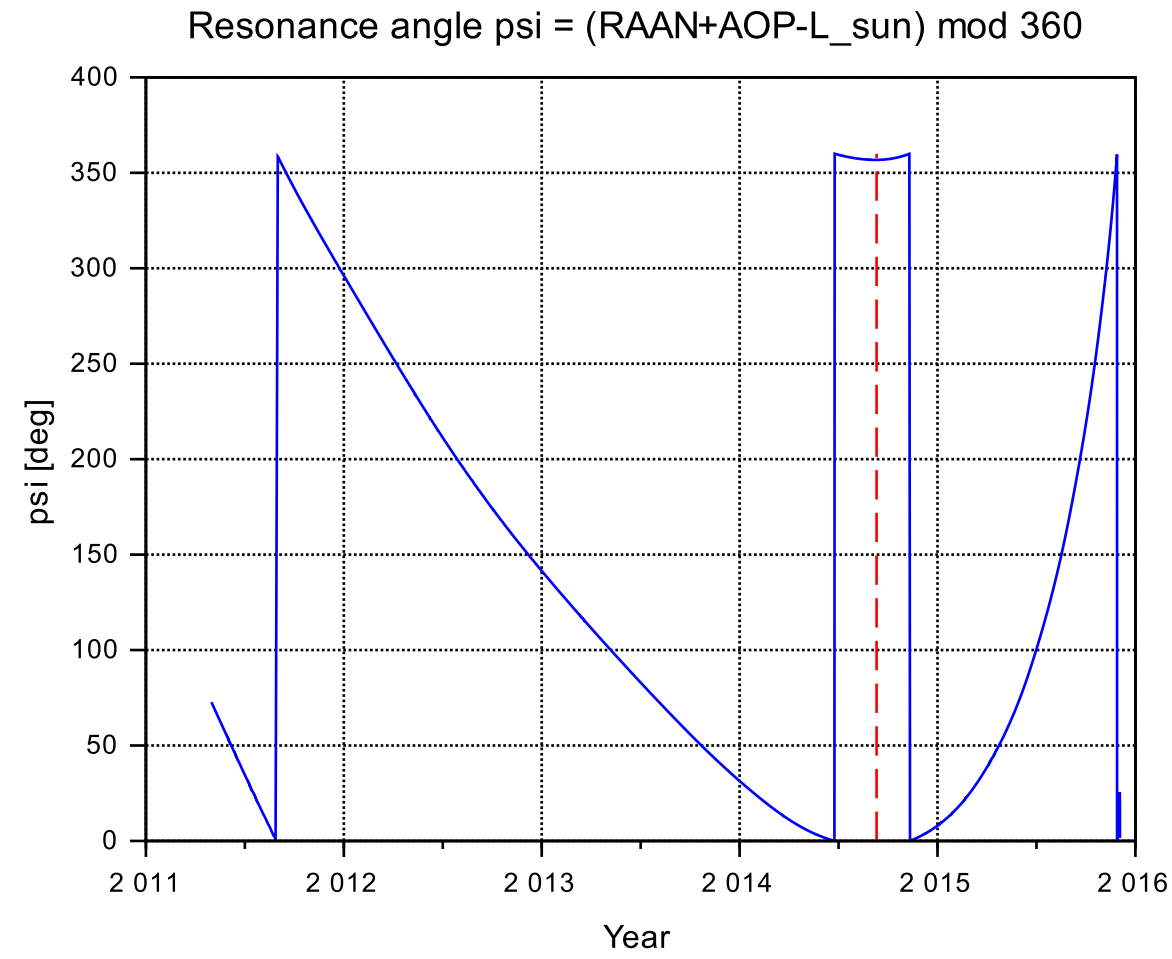
Source: OREM (github.com/hari251086/OREM), issue #56.

Index: 23 objects, in the order plotted

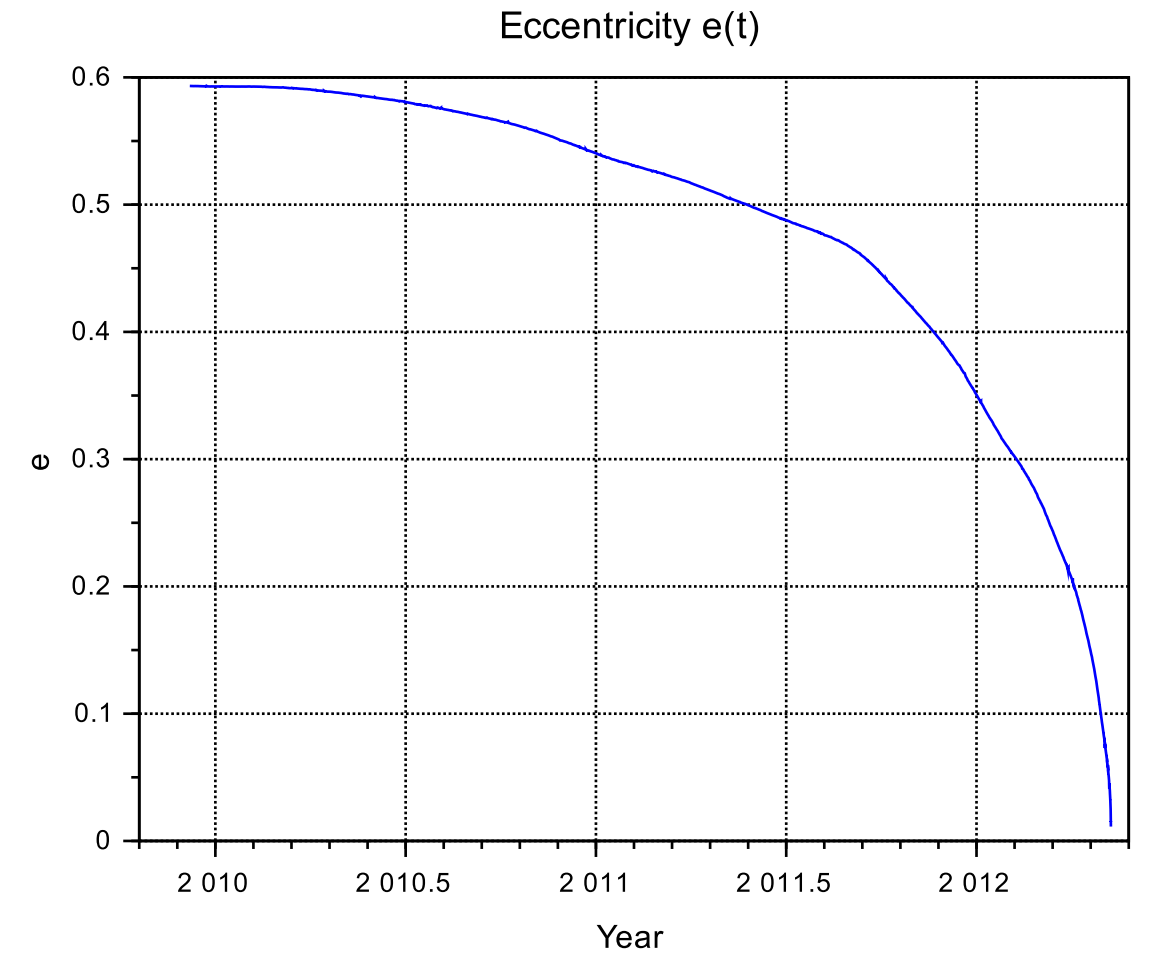
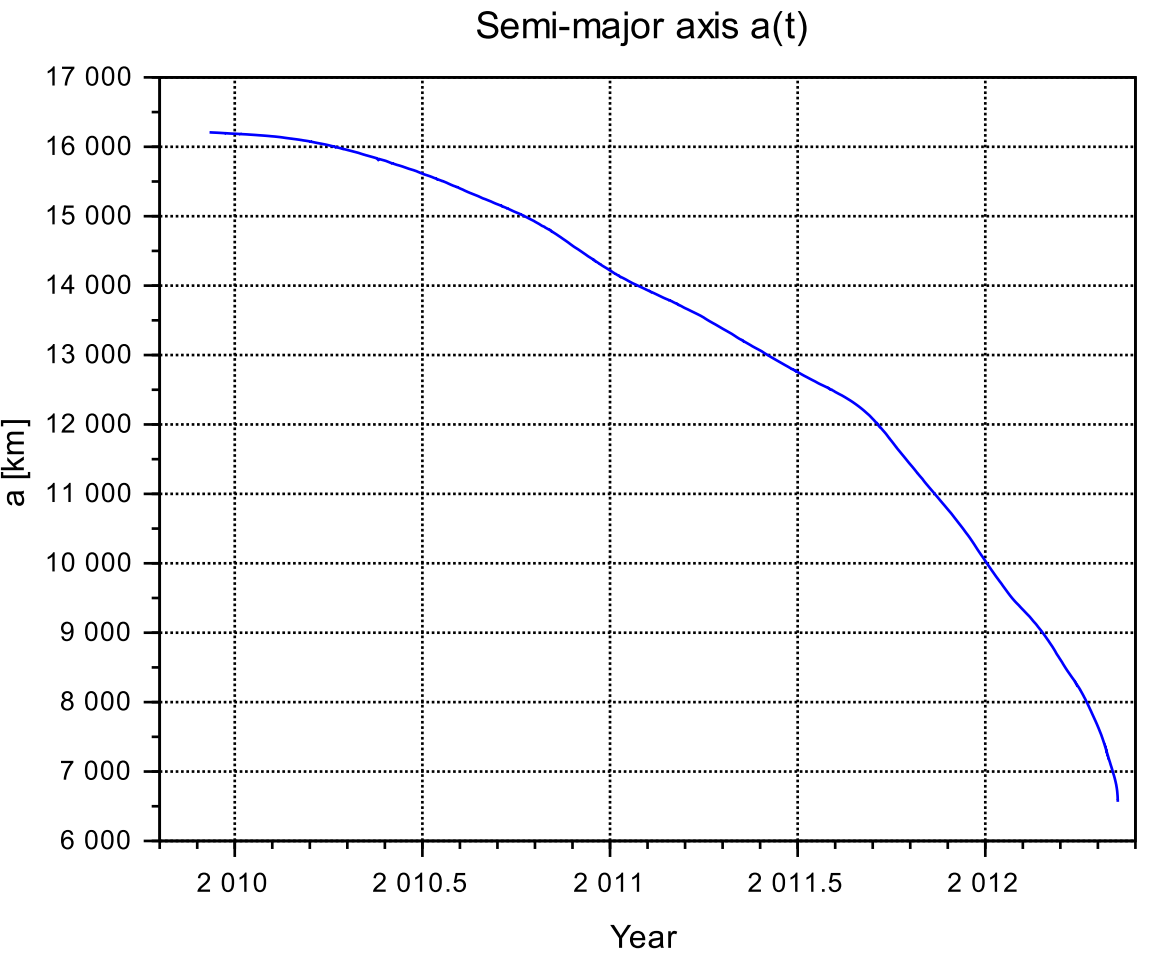
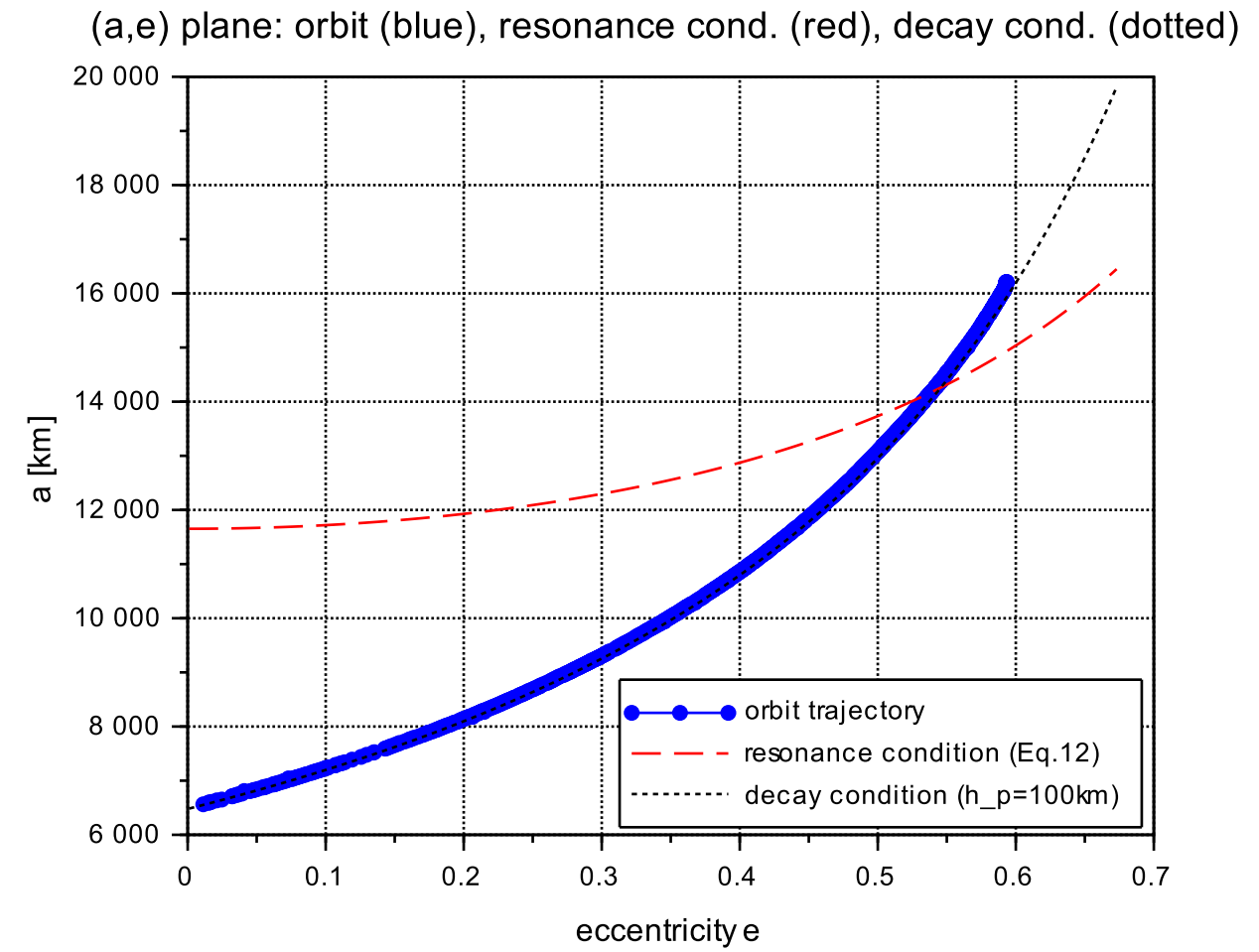
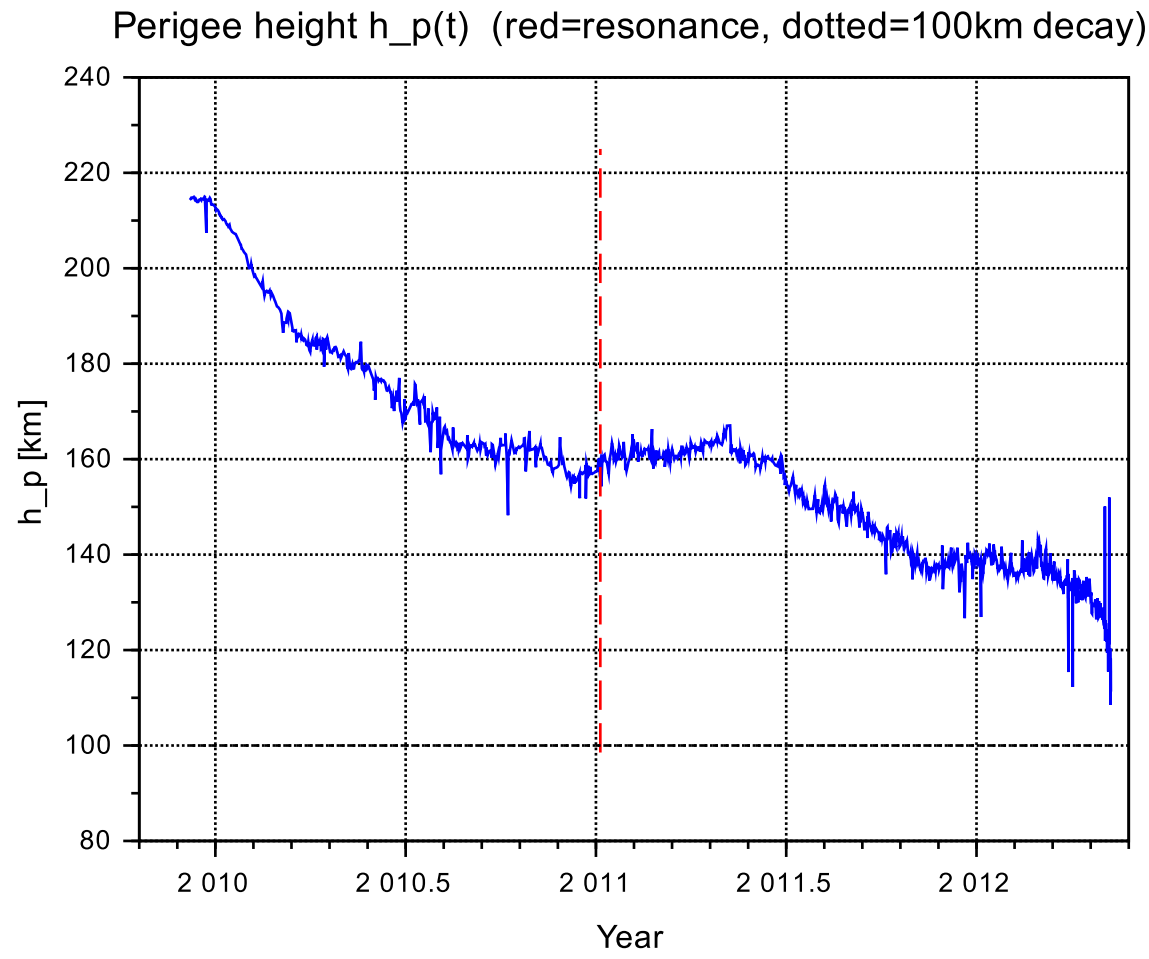
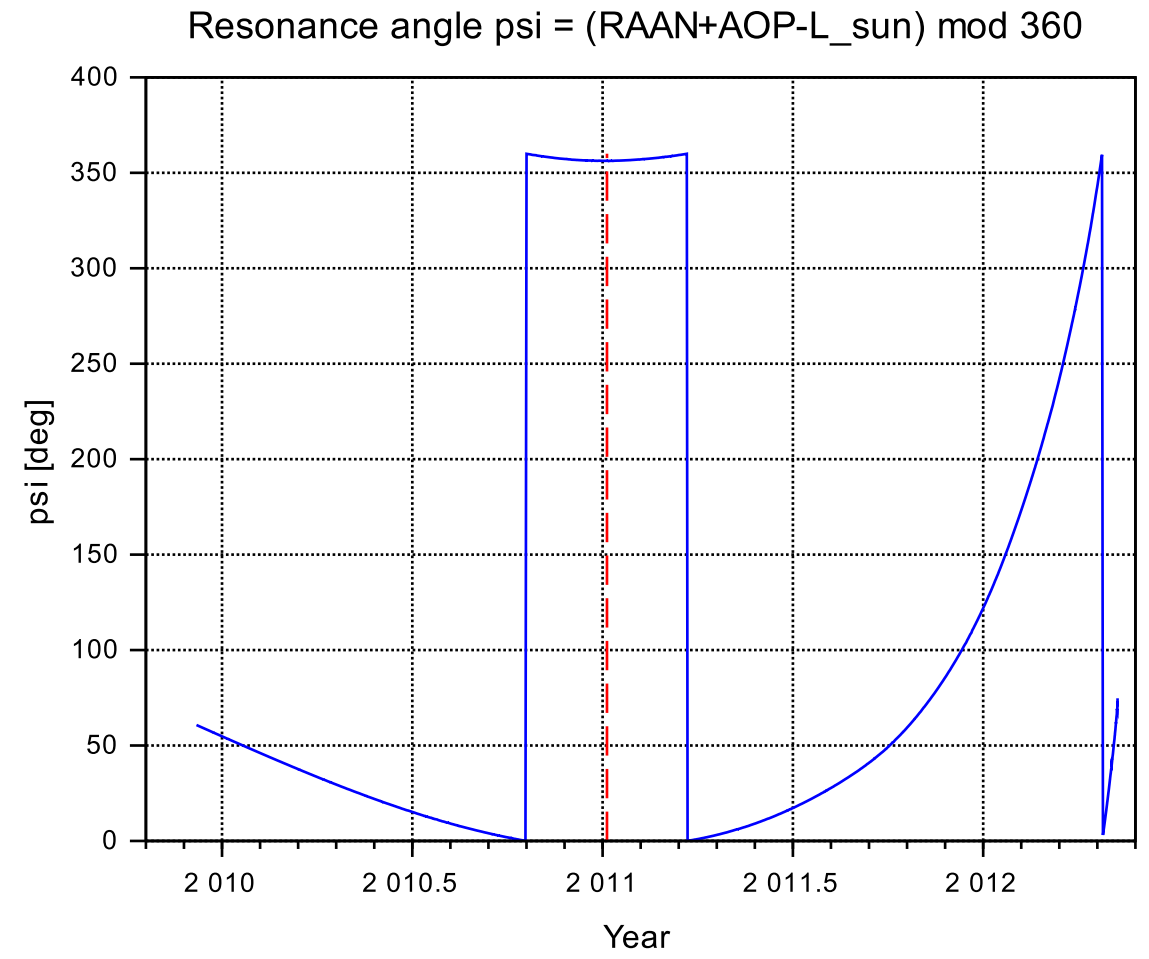
NORAD	Name	i(deg)	Resonance yr	Observed decay yr
35497	Ariane 5 ESC-A	5.77	2016.35	2016.8309
37151	Long March 3B	24.88	2014.69	2015.9192
27526	Ariane 5 R/B	17.75	2011.01	2012.3518
37819	Proton-M R/B	63.28	2012.68	2013.6961
59347	GTO R/B	25.38	2026.35	2026.4326
40943	GTO debris	6.34	2019.74	2020.1355
60328	CZ-3B R/B	26.84	2025.26	2025.2827
18983	SL-6 R/B(2)	63.7	1993.71	2002.0602
13112	SL-6 R/B(2)	64.52	1991.11	1992.8693
28141	CZ-2C R/B	25.03	2007.78	2007.7823
19881	Cosmos931 deb	59.51	1993.15	1993.1622
5980	Titan3C TS deb	26.36	1975.27	1976.2779
4570	SL-6 R/B(2)	65.77	1972.39	1972.4120
10724	AtlasCentaur deb	26.74	1980.67	1982.6626
28140	Doublestar TC-1	25.37	2007.78	2007.7823
44911	CZ-5 R/B	15.31	2022.33	2022.3368
7096	SKYNET 2A	35.36	1974.05	1974.0657
68538	ARTEMIS 2 (INTEGRITY)	33.62	2026.25	2026.2710
6294	MOLNIYA 1-22	65.06	1976.08	1976.1109
18954	ARIANE 3 DEB (SYLDA)	6.9	1989.04	1989.0856
29462	SL-12 DEB	65.31	2010.43	2012.7543
5941	PROGNOZ 1	64.93	1975.65	1980.0479
6231	MOLNIYA 1-21	65.41	1972.82	1977.8330



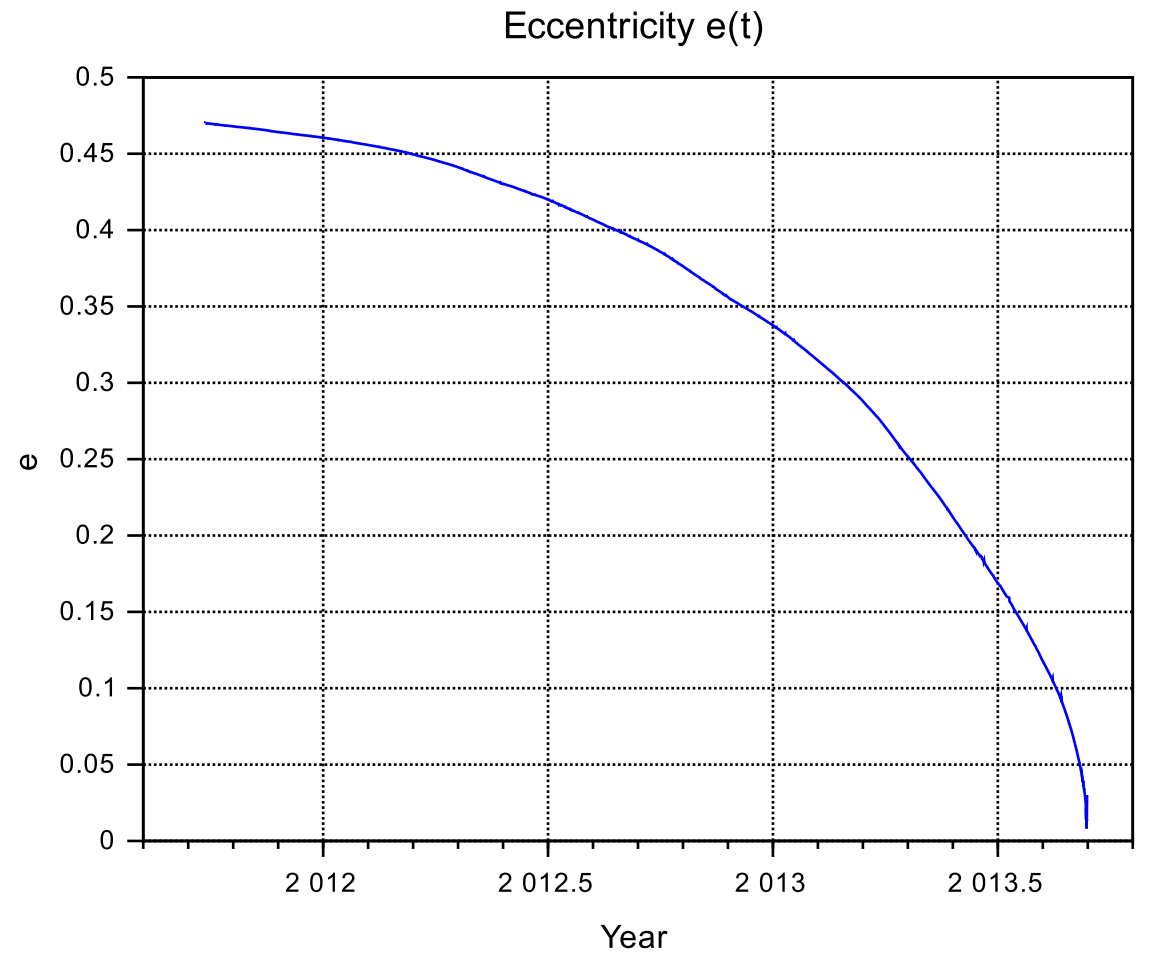
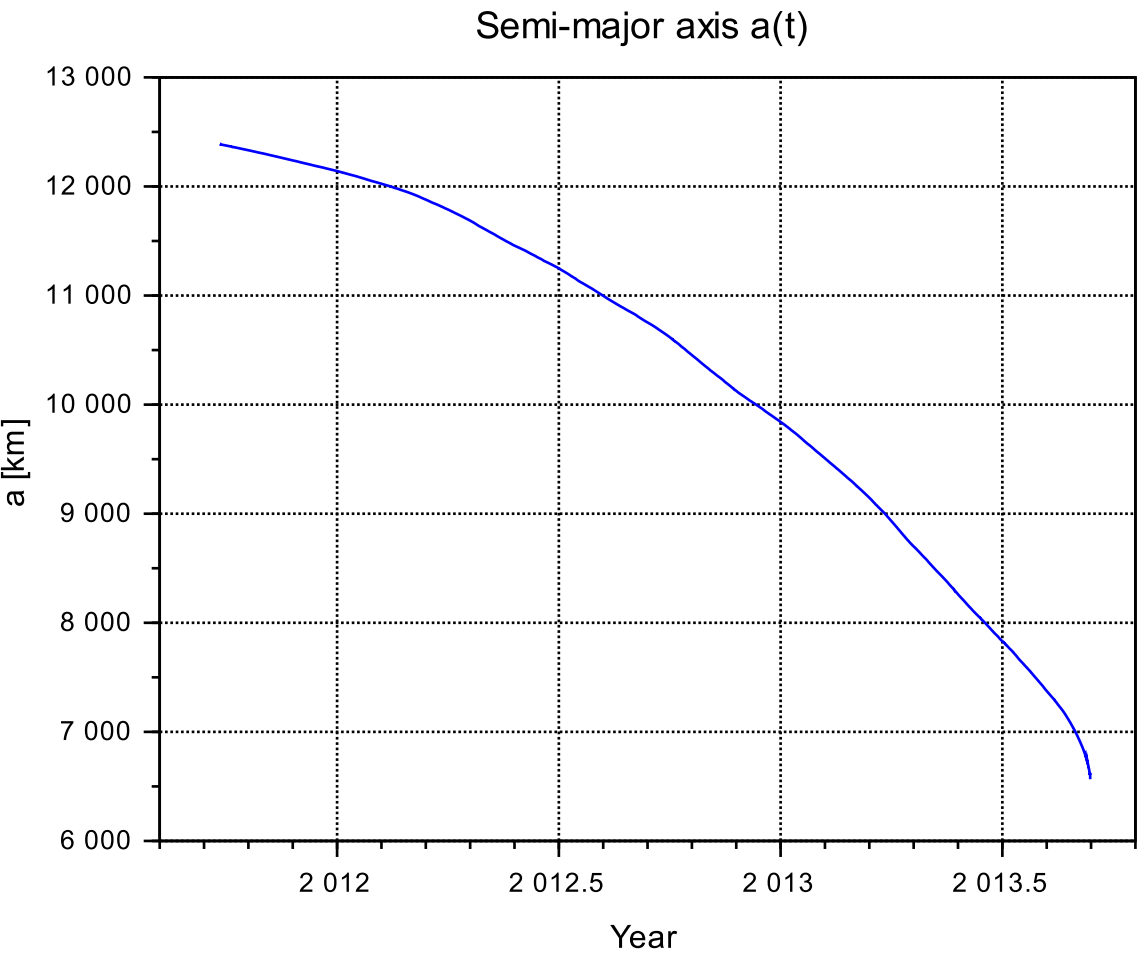
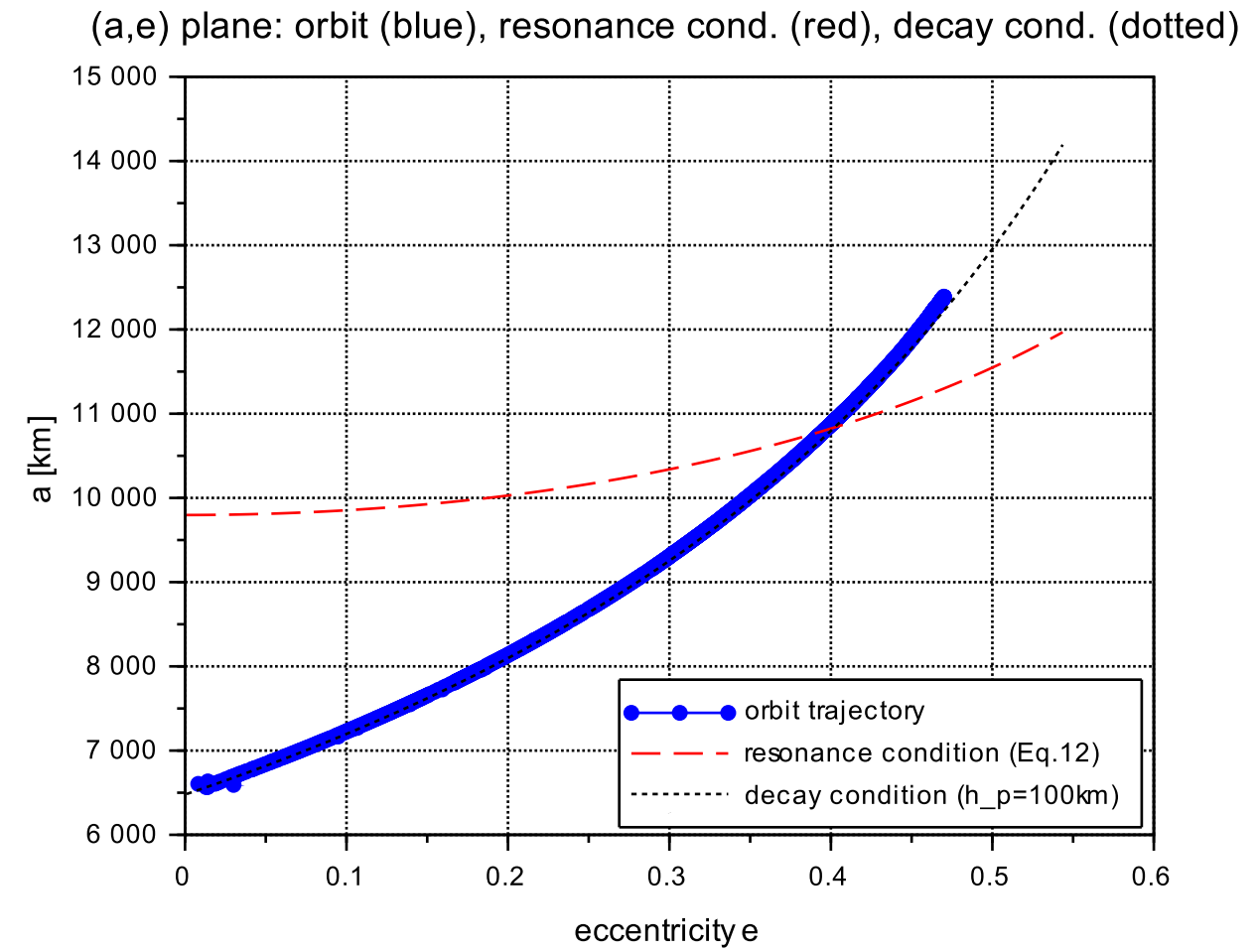
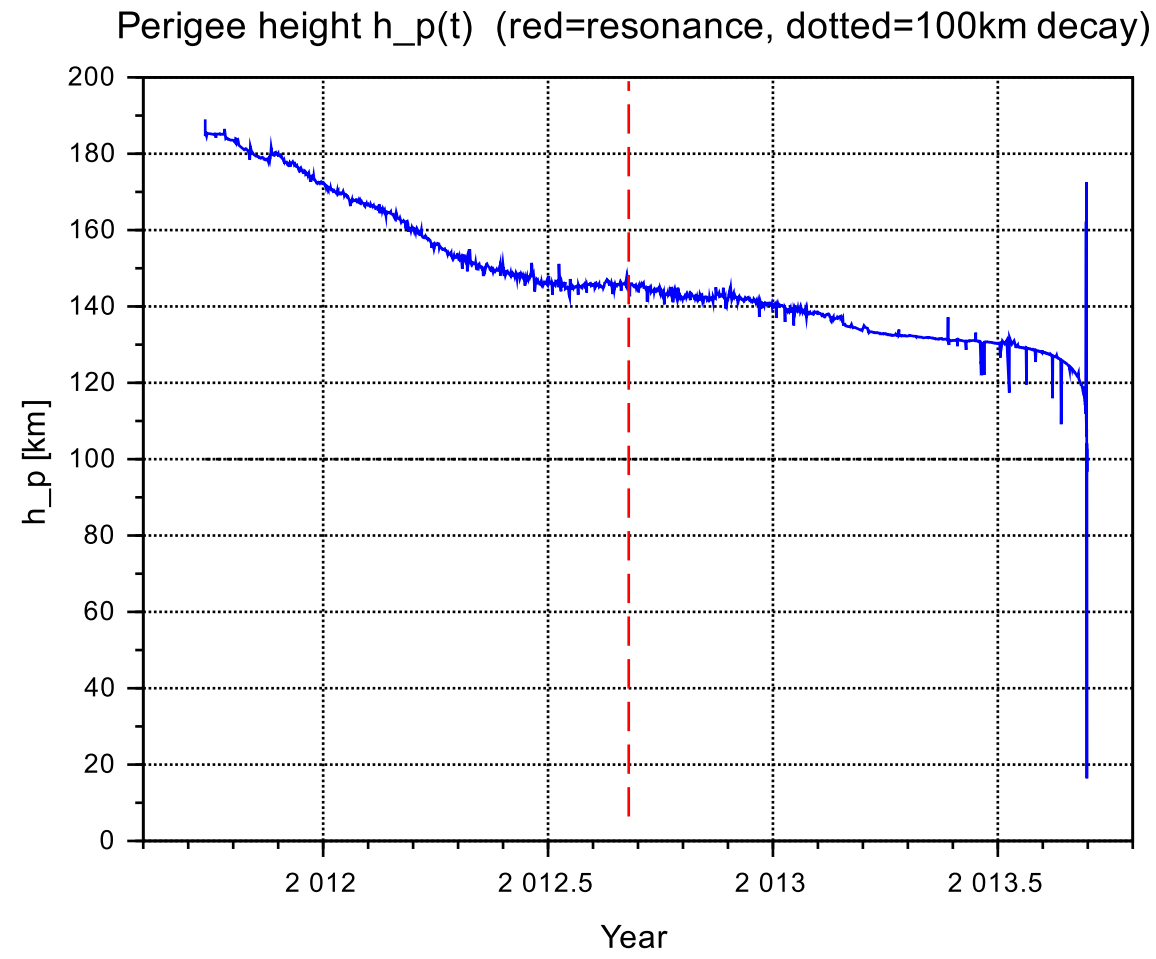
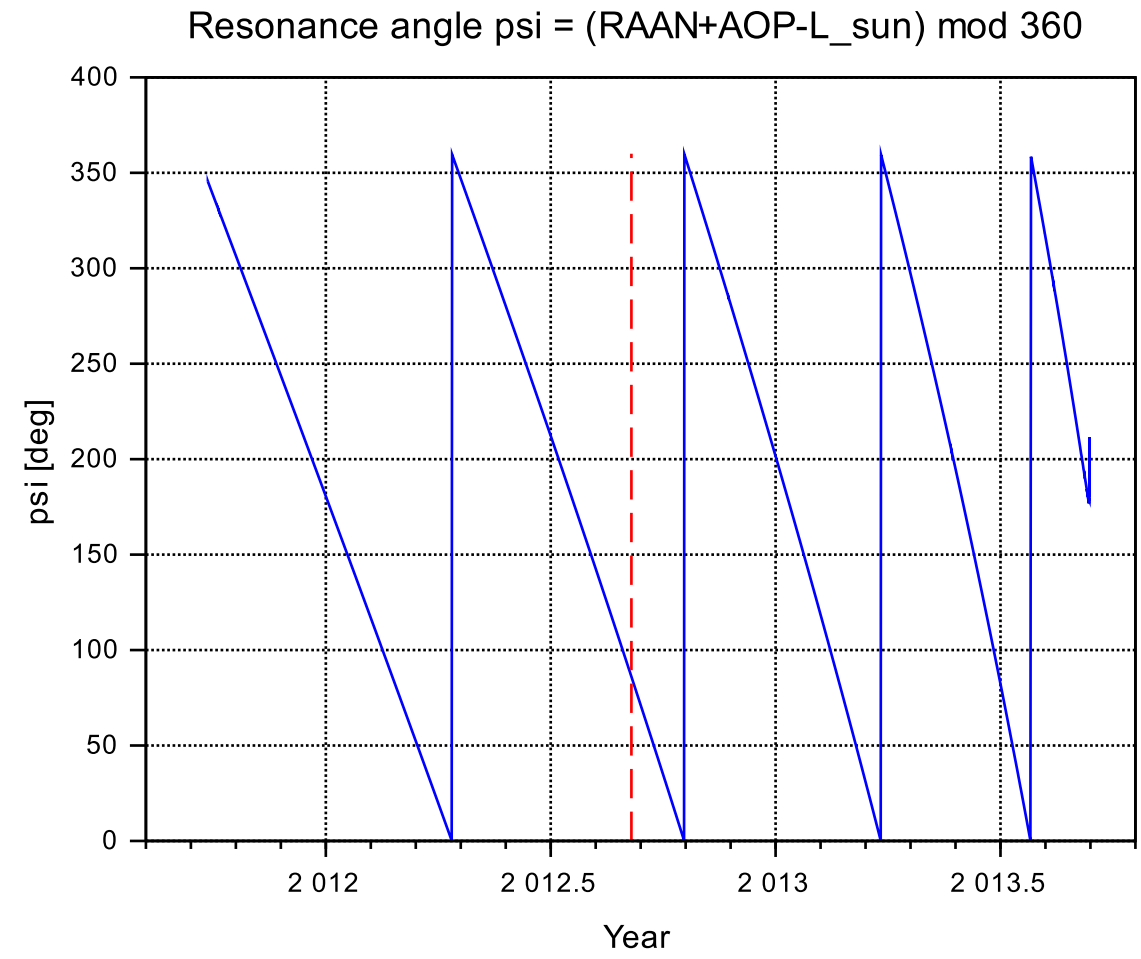
NORAD 35497 Ariane 5 ESC-A
Mean inclination: 5.77 deg
 λ_{crit} (Eq.12): 12280.7
Resonance epoch: year 2016.35
TLE record: 2013.79 to 2016.83
Observed decay: year 2016.8309



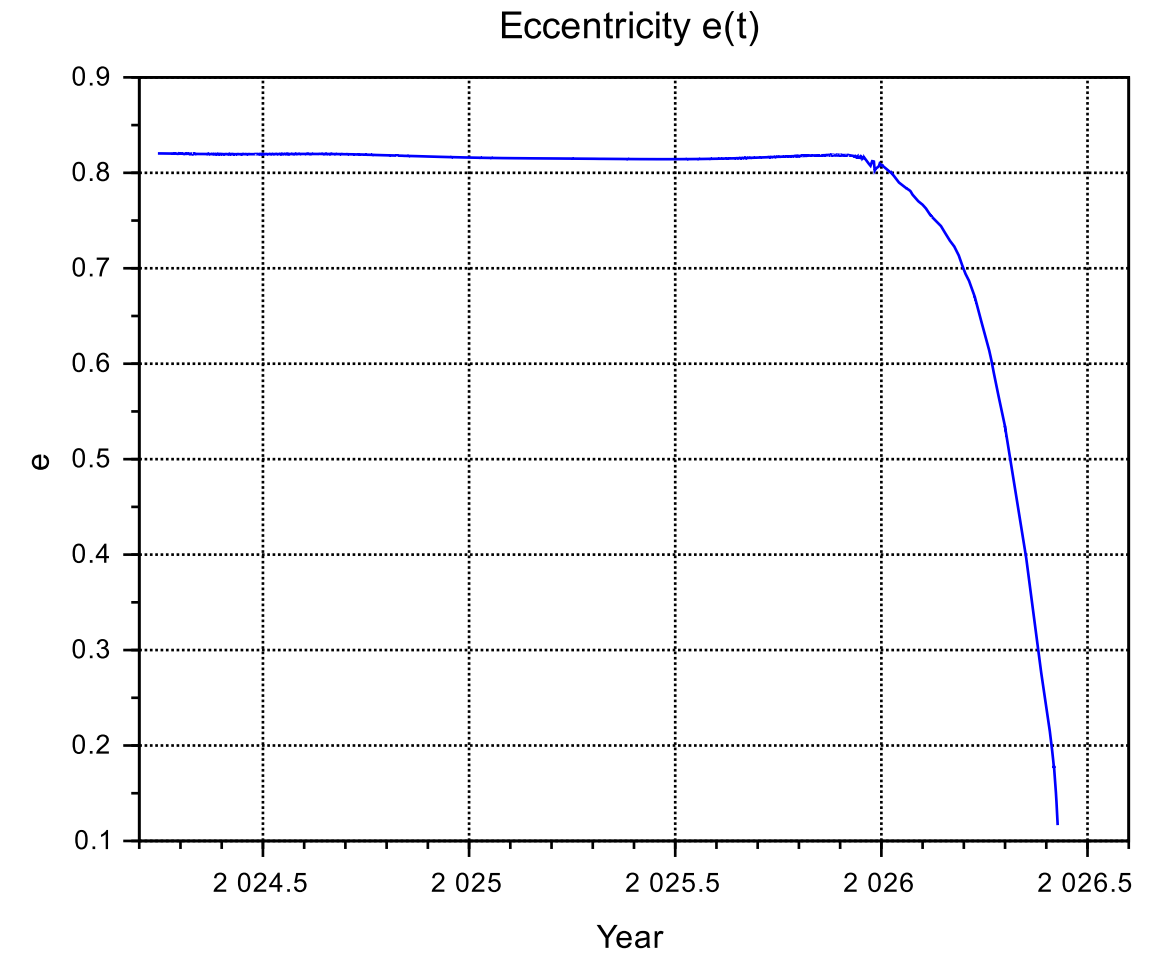
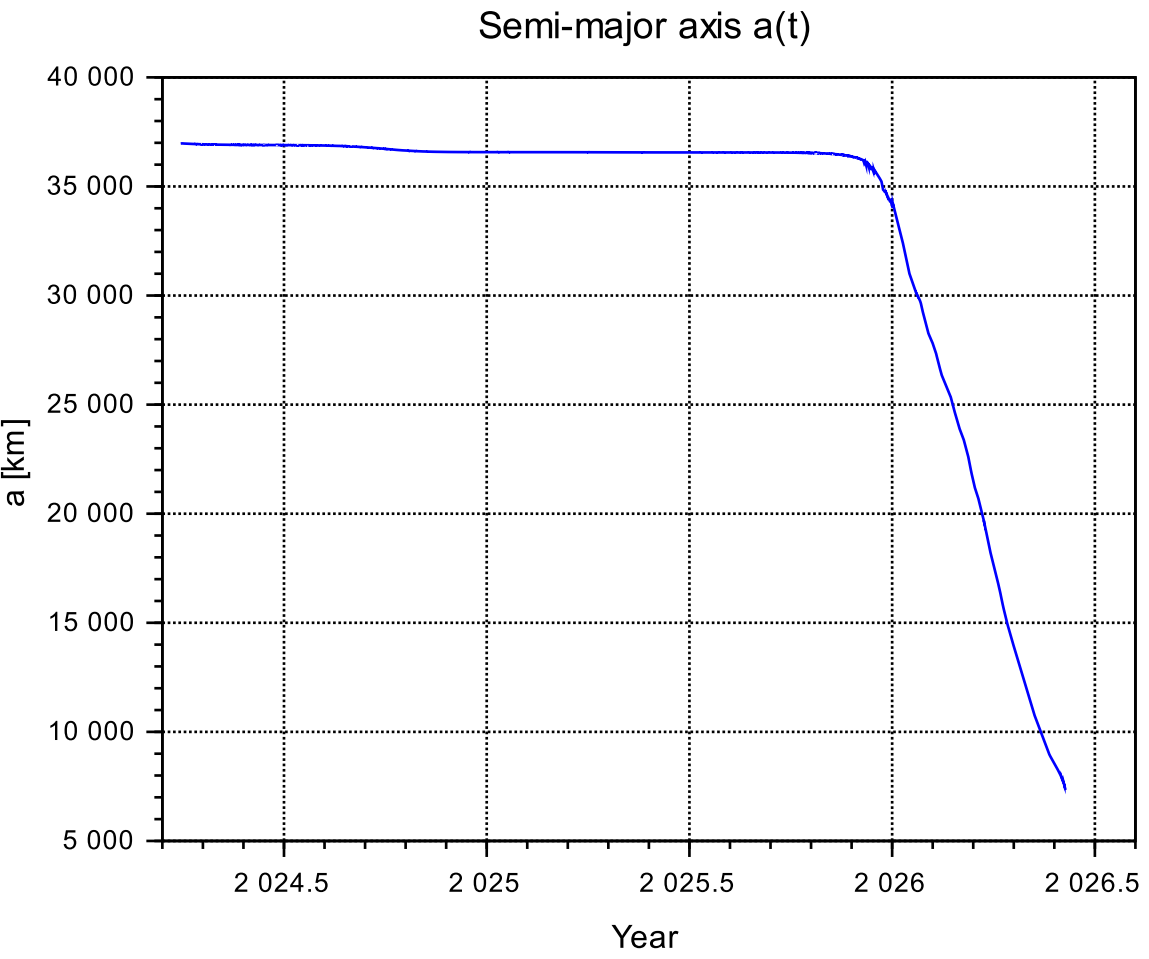
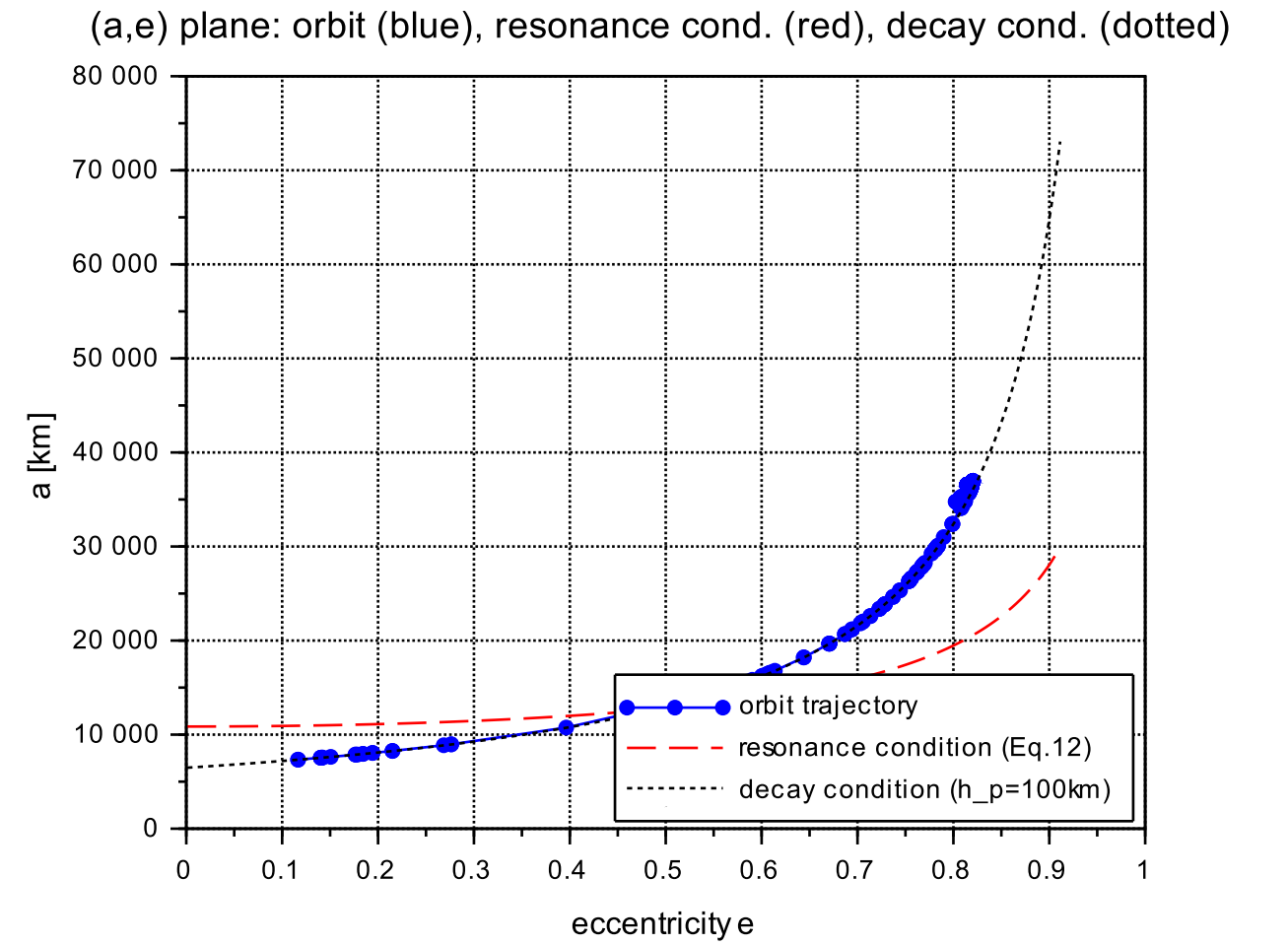
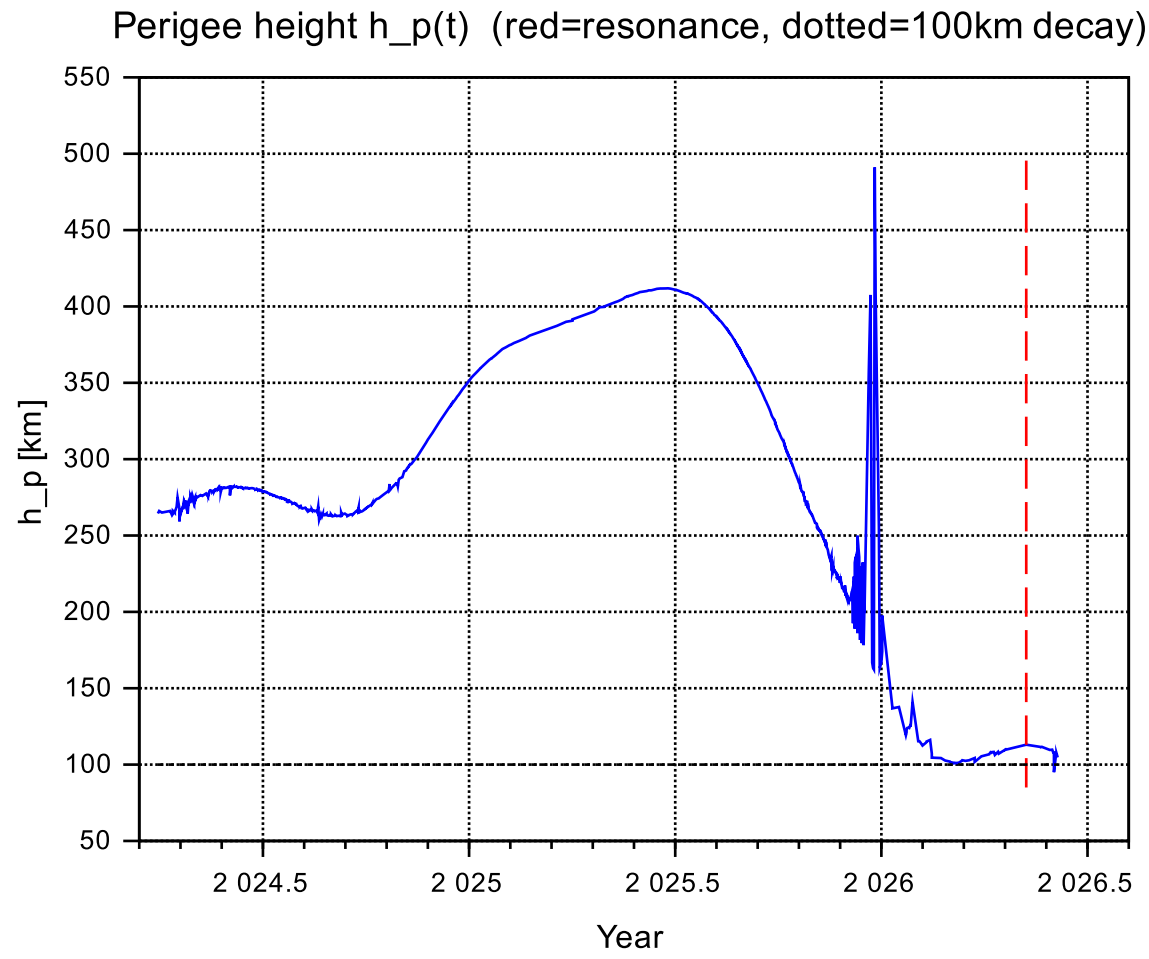
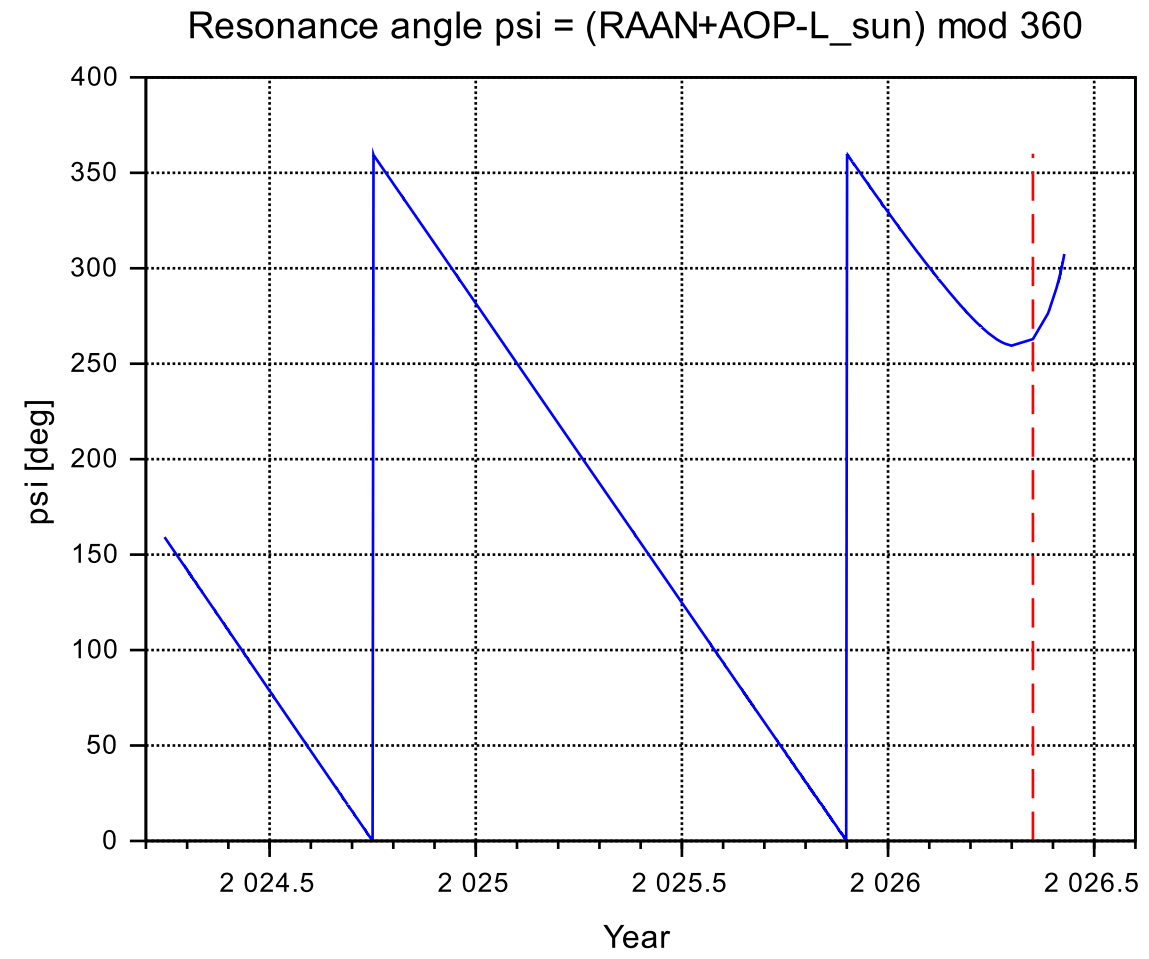
NORAD 37151 Long March 3B
Mean inclination: 24.88 deg
 λ_{crit} (Eq.12): 10924.05
Resonance epoch: year 2014.69
TLE record: 2011.33 to 2015.92
Observed decay: year 2015.9192



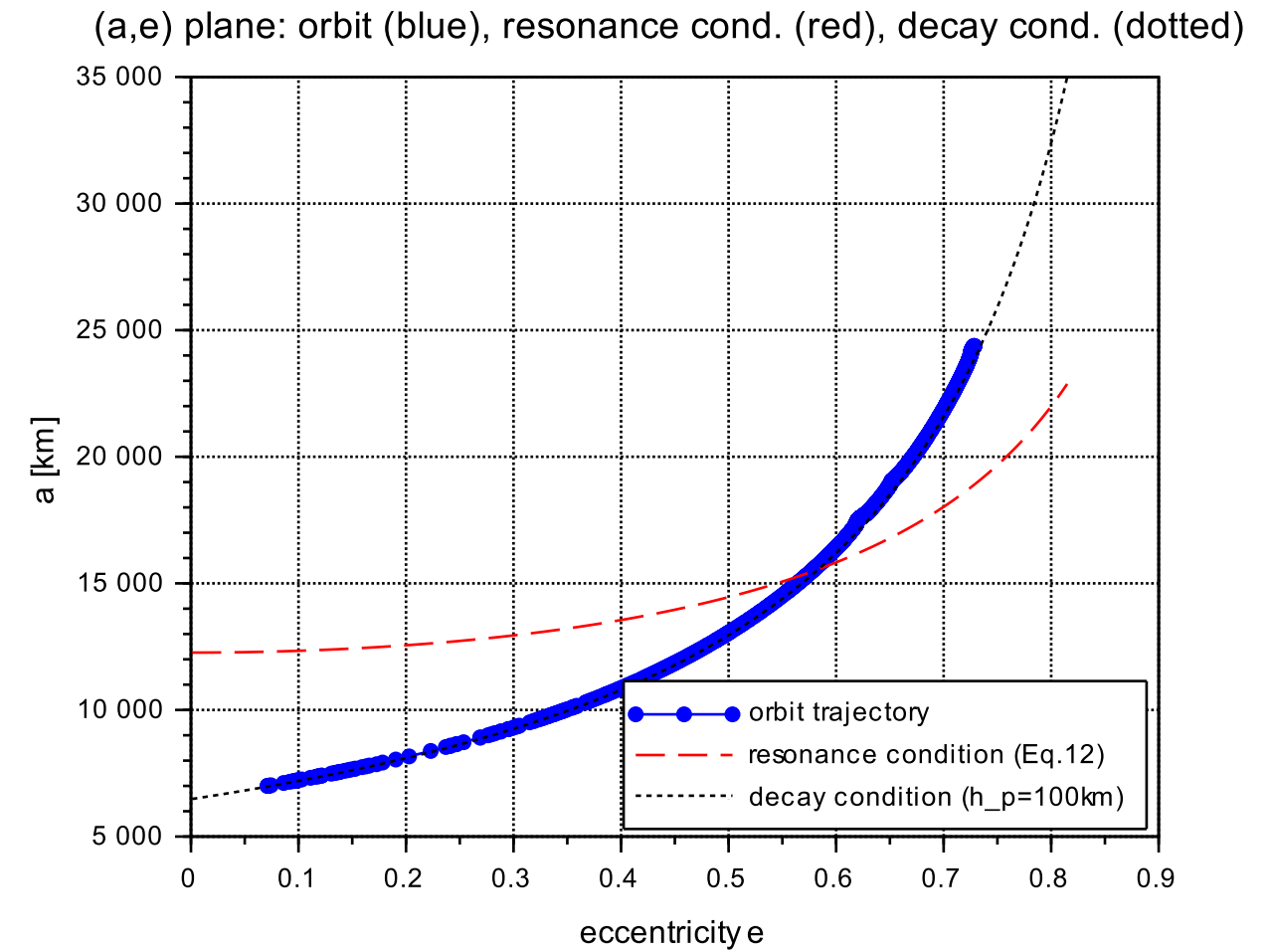
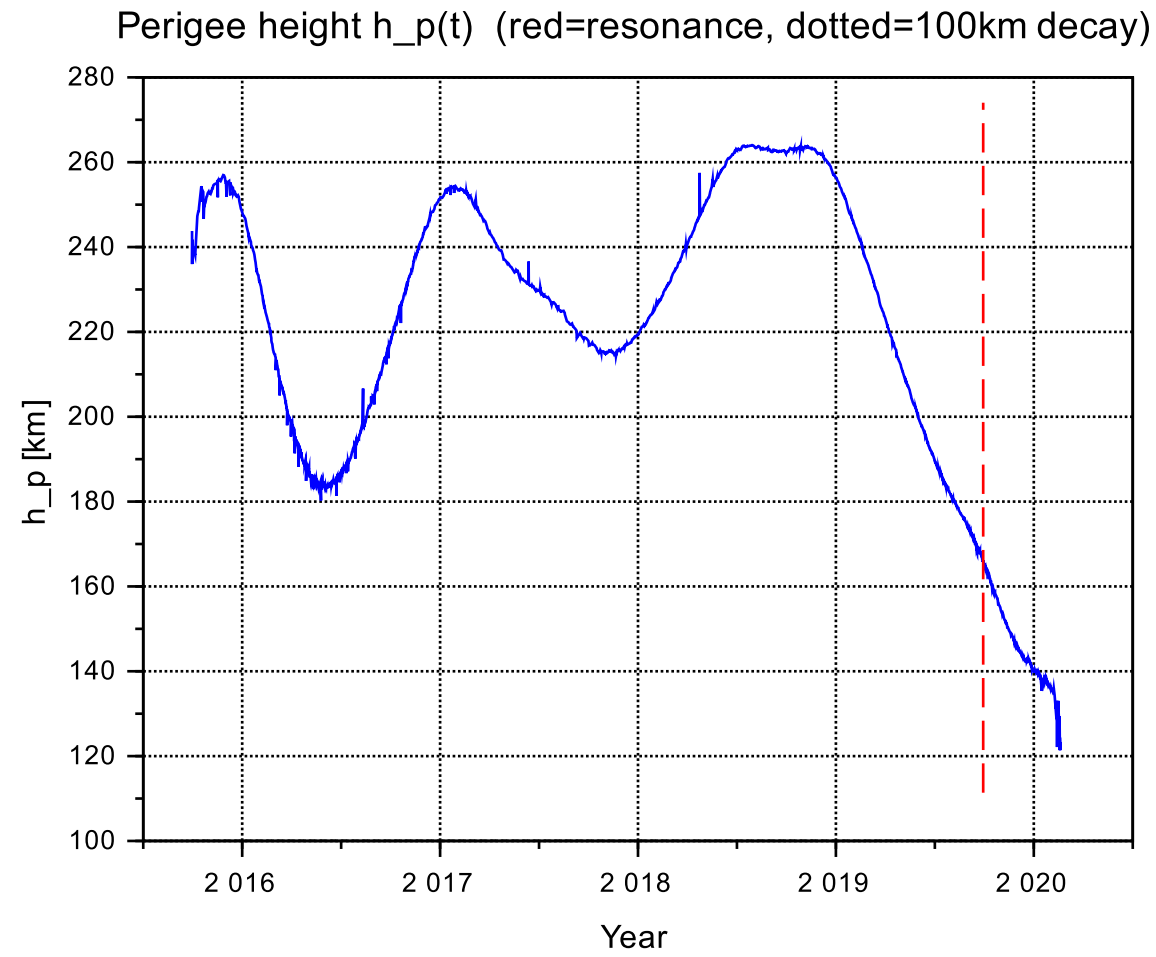
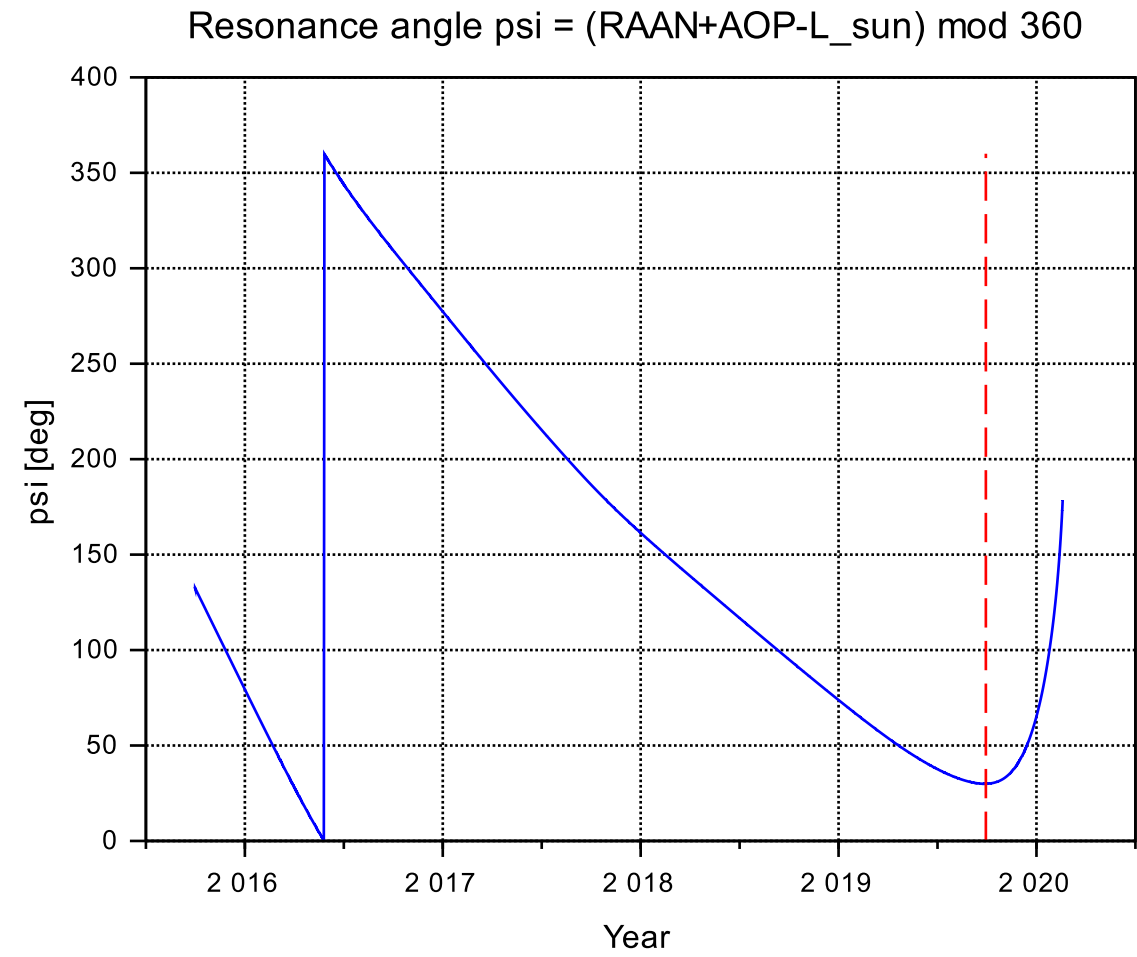
NORAD 27526 Ariane 5 R/B
Mean inclination: 17.75 deg
 λ_{crit} (Eq.12): 11652.26
Resonance epoch: year 2011.01
TLE record: 2009.93 to 2012.35
Observed decay: year 2012.3518



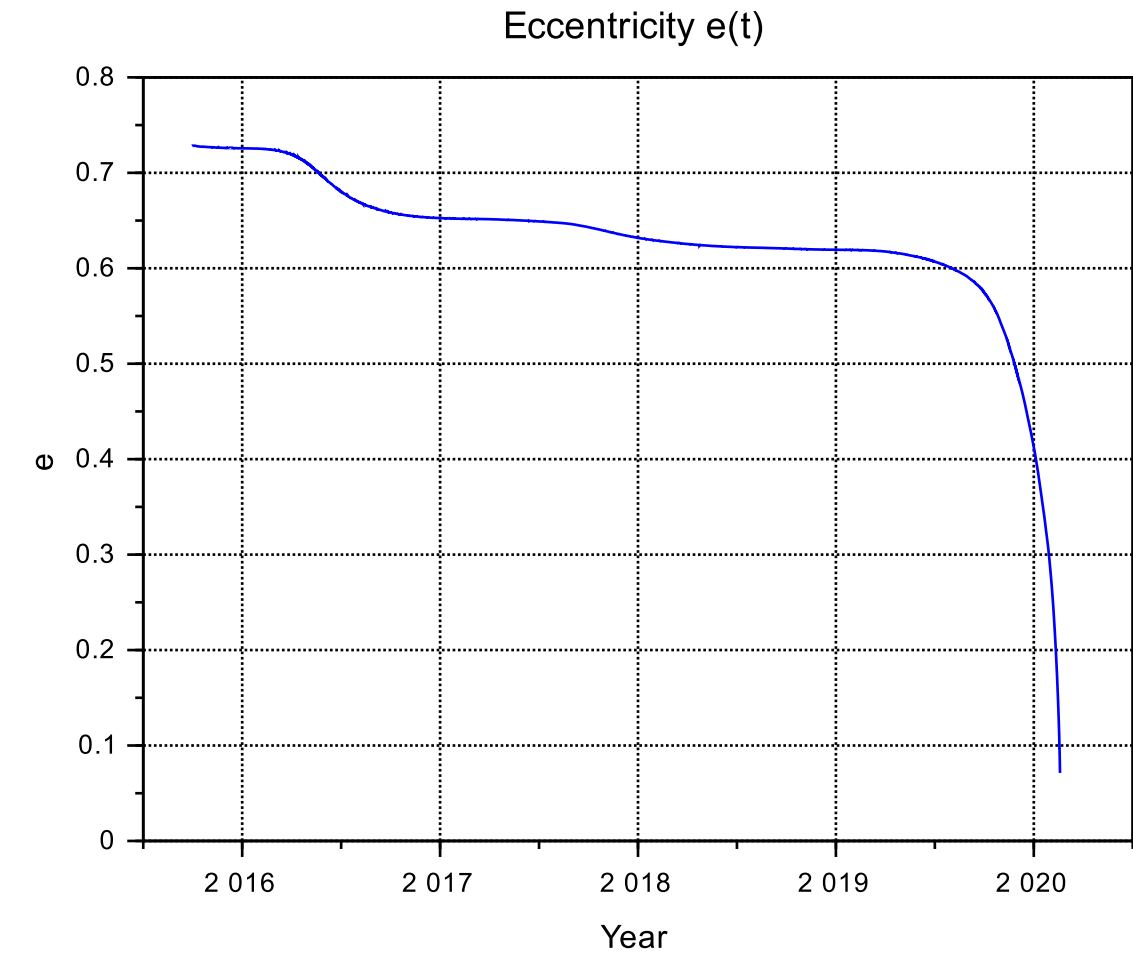
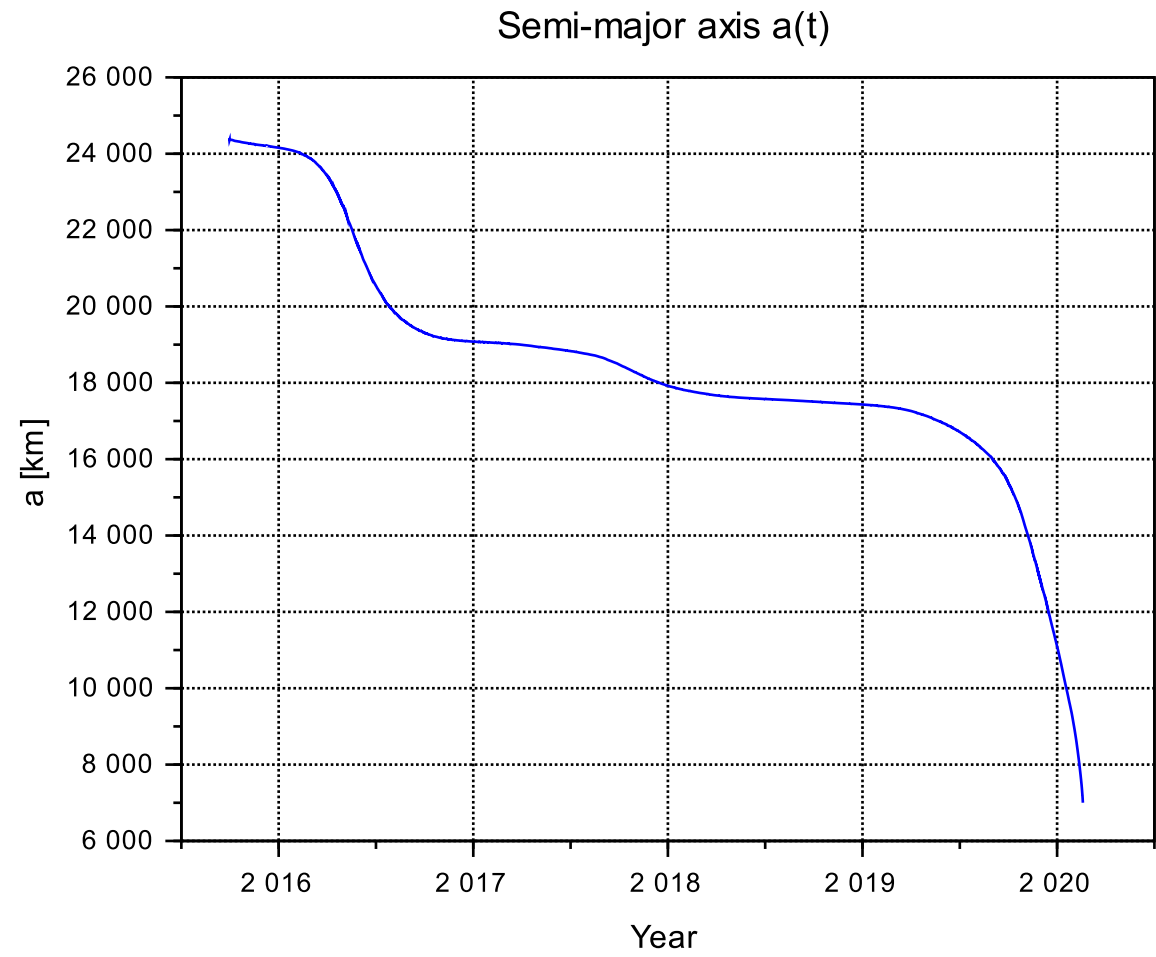
NORAD 37819 Proton-M R/B
Mean inclination: 63.28 deg
 λ_{crit} (Eq.12): 9797.04
Resonance epoch: year 2012.68
TLE record: 2011.74 to 2013.7
Observed decay: year 2013.6961

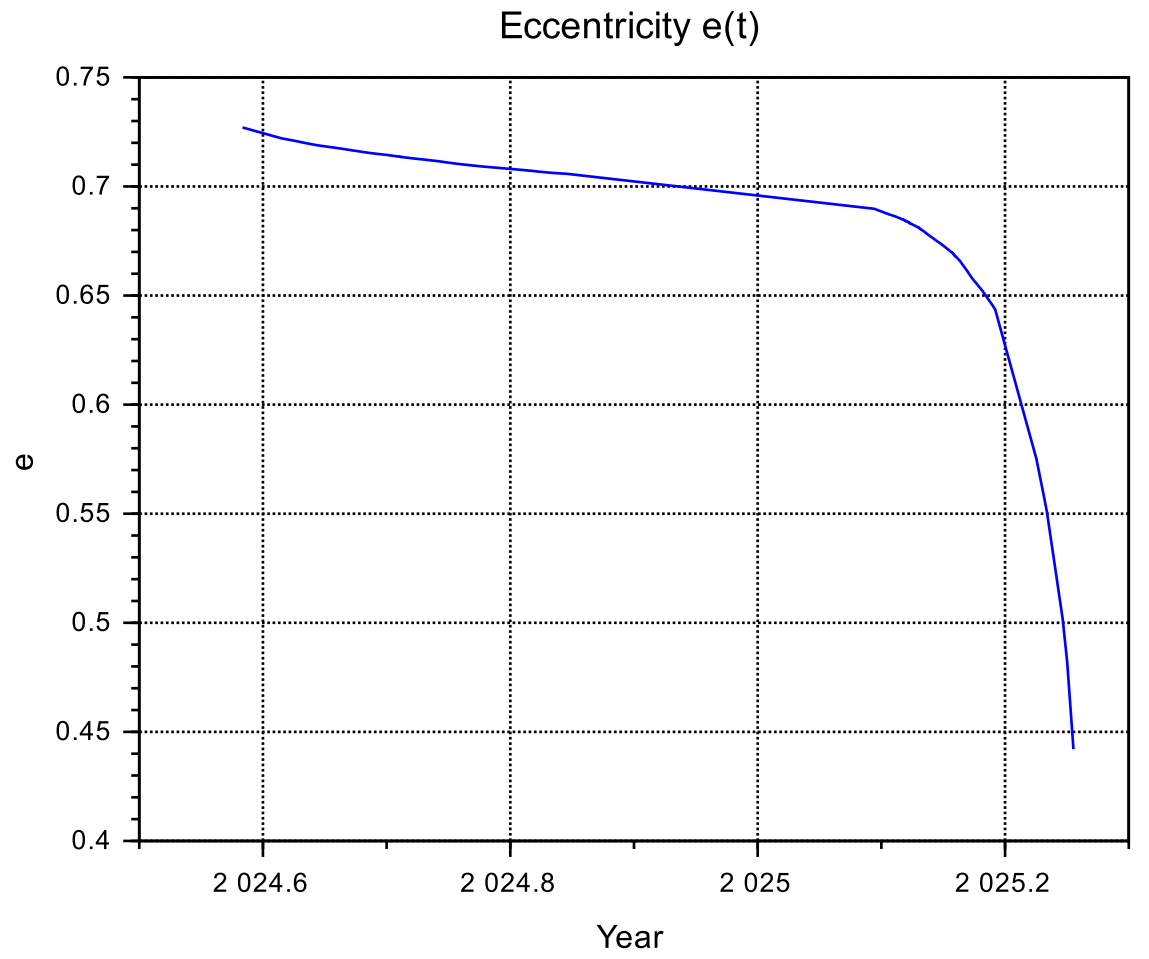
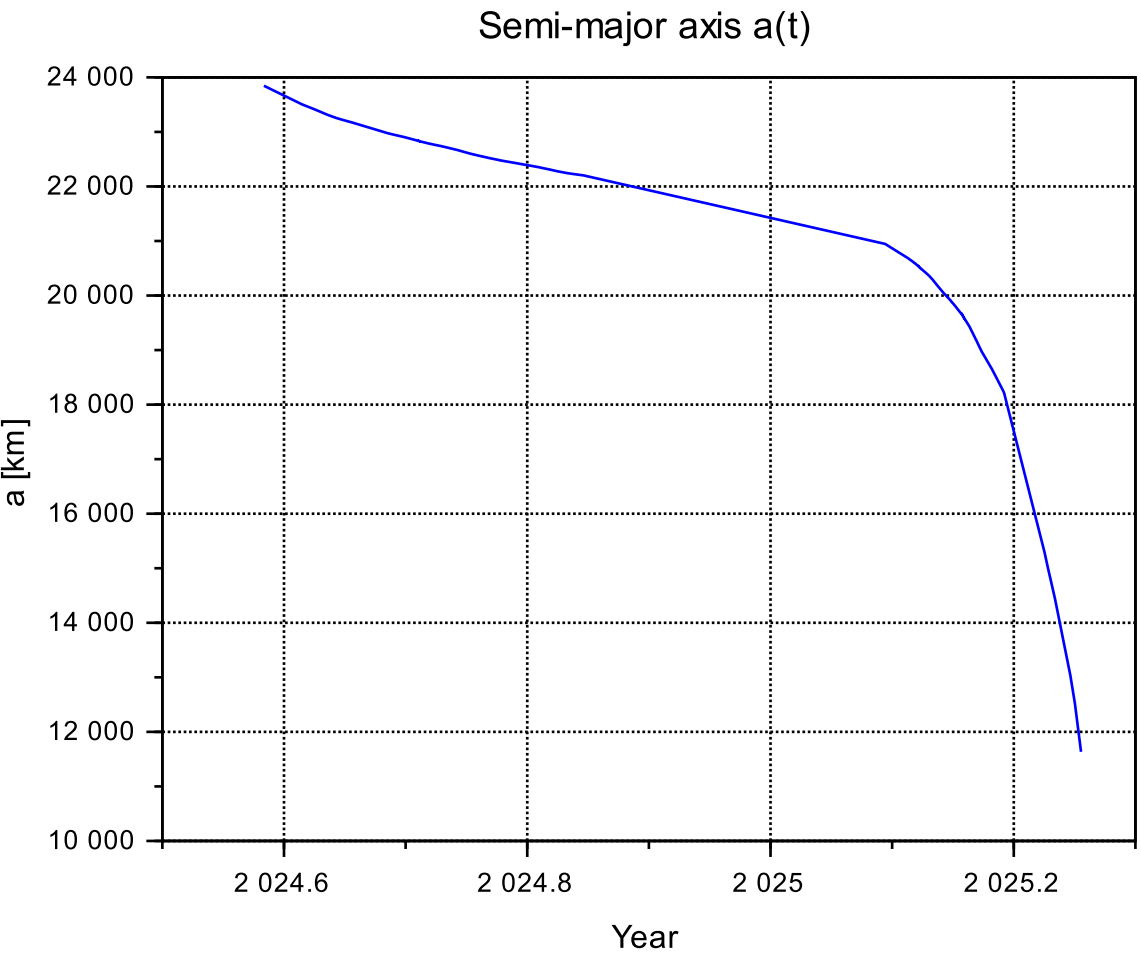
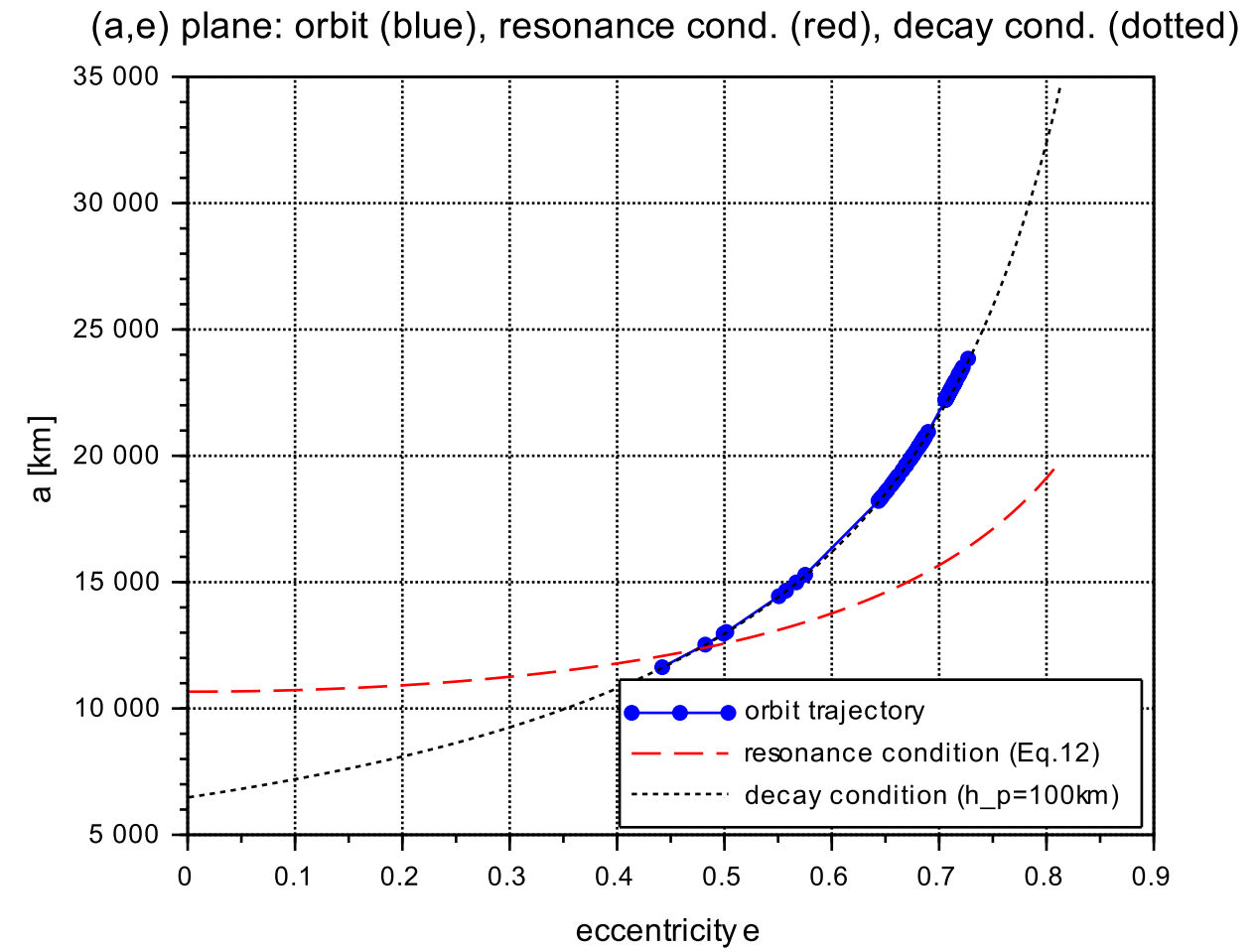
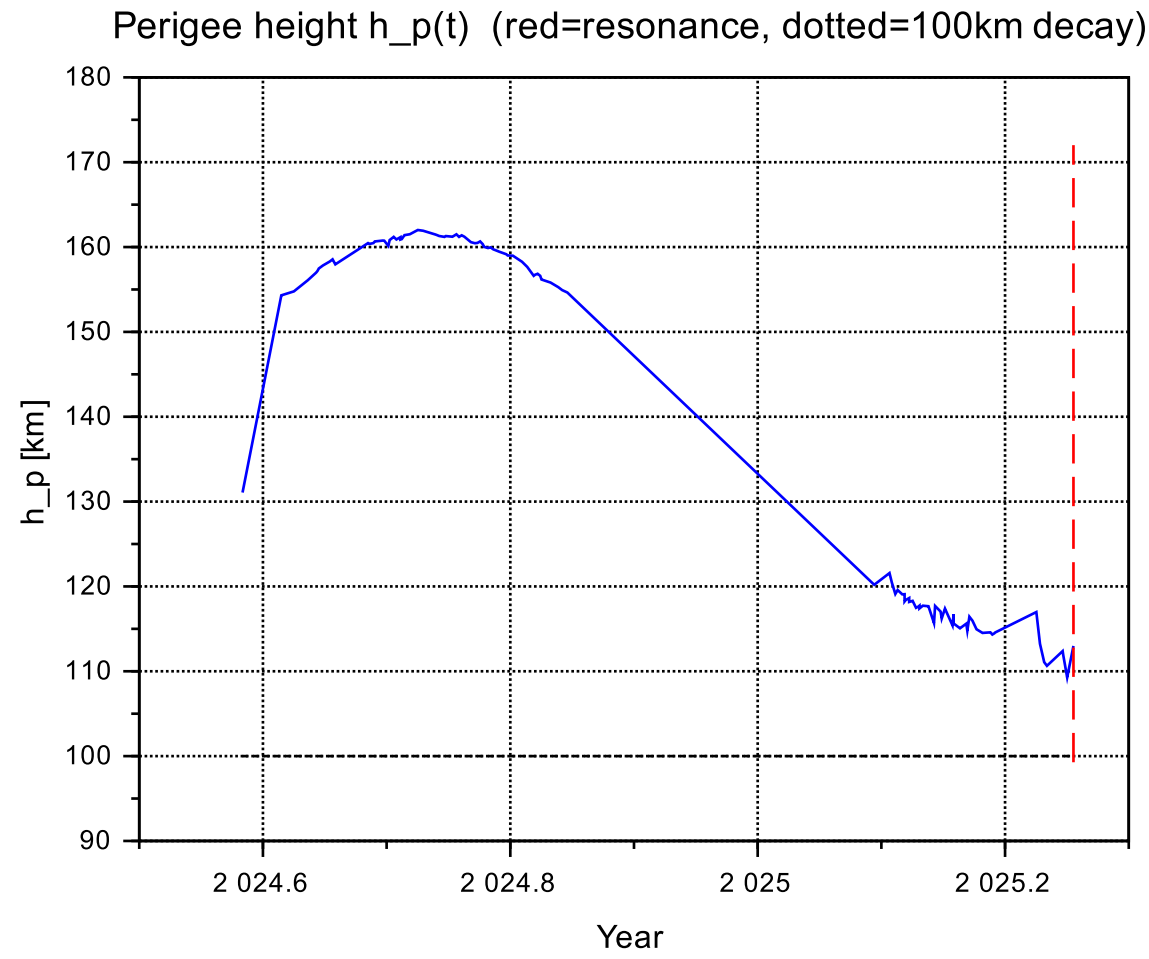
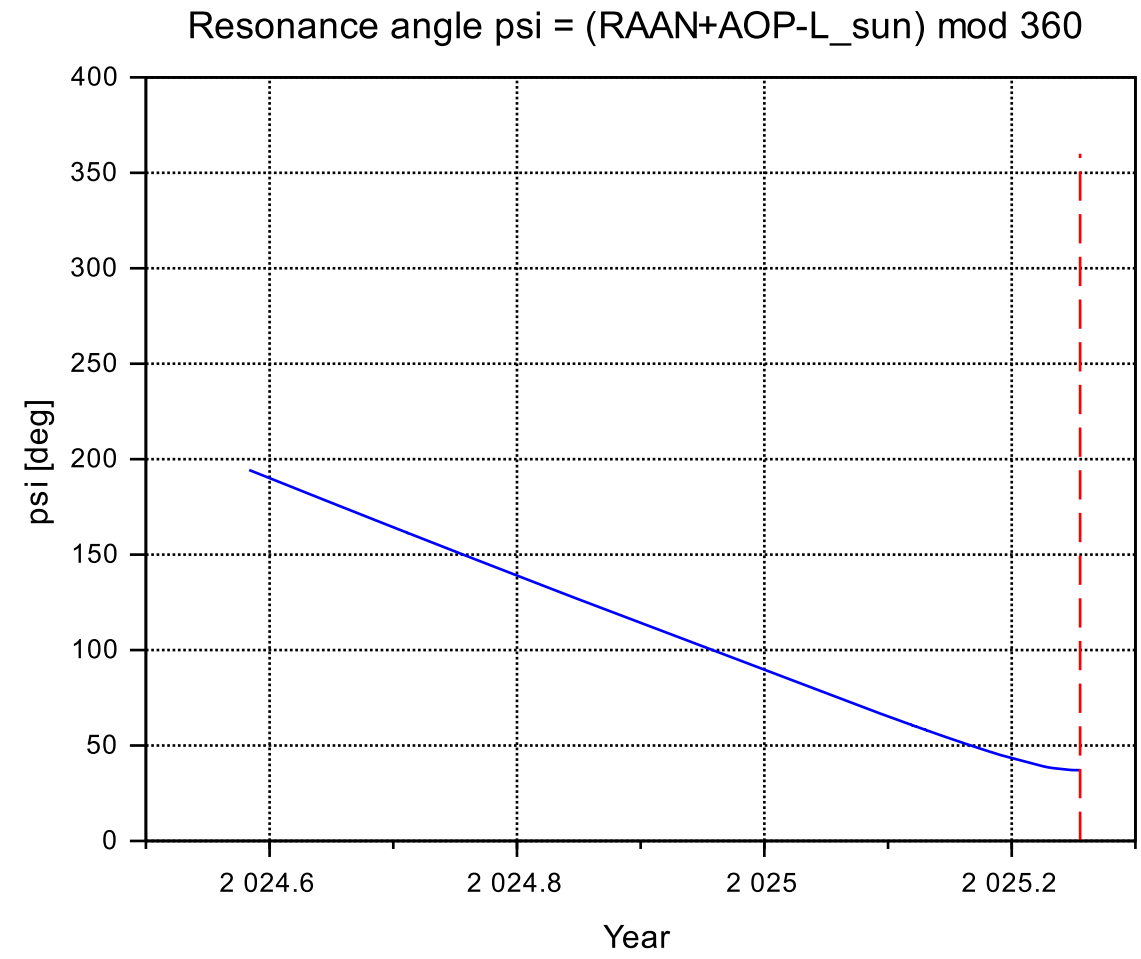


NORAD 59347 GTO R/B
Mean inclination: 25.38 deg
 λ_{crit} (Eq.12): 10860.06
Resonance epoch: year 2026.35
TLE record: 2024.25 to 2026.43
Observed decay: year 2026.4326

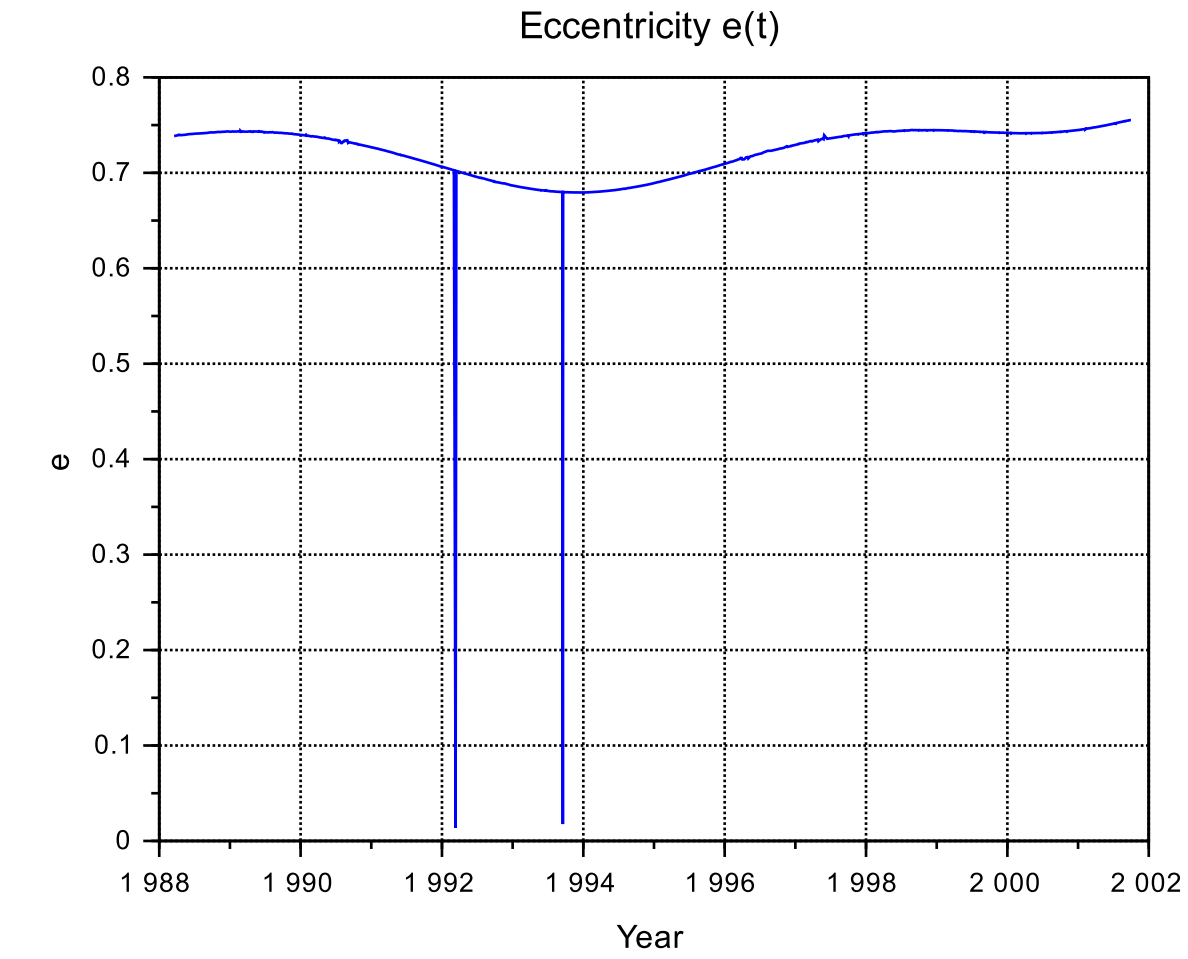
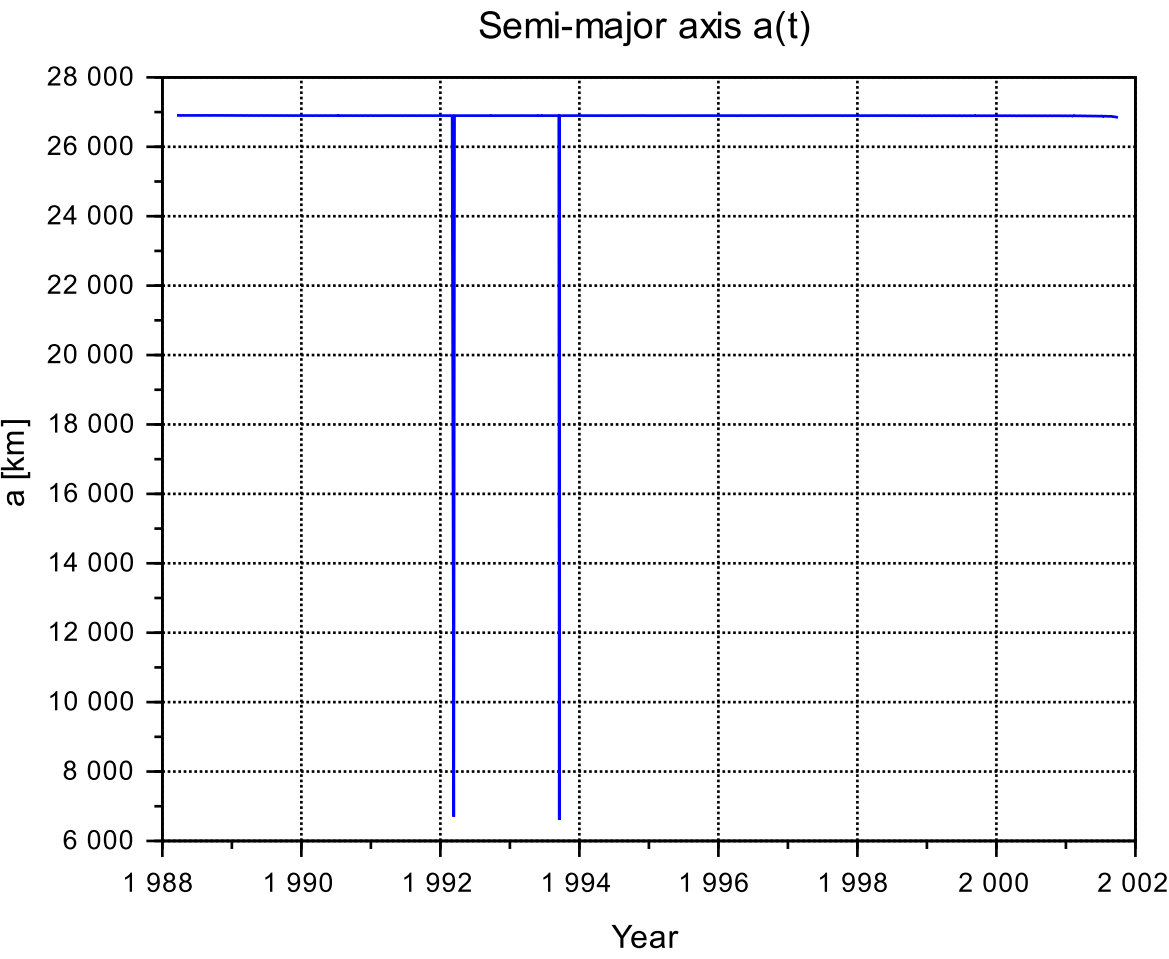
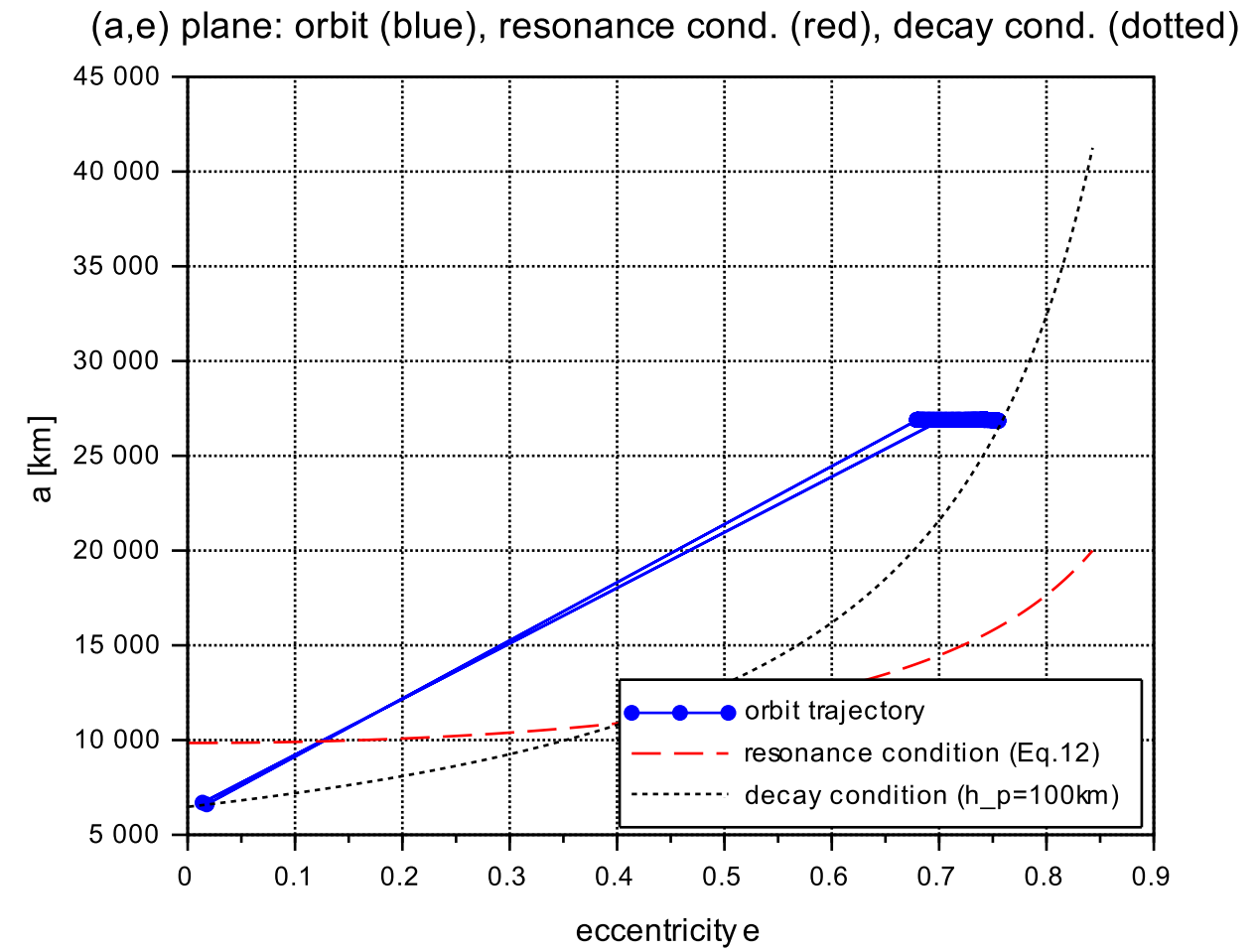
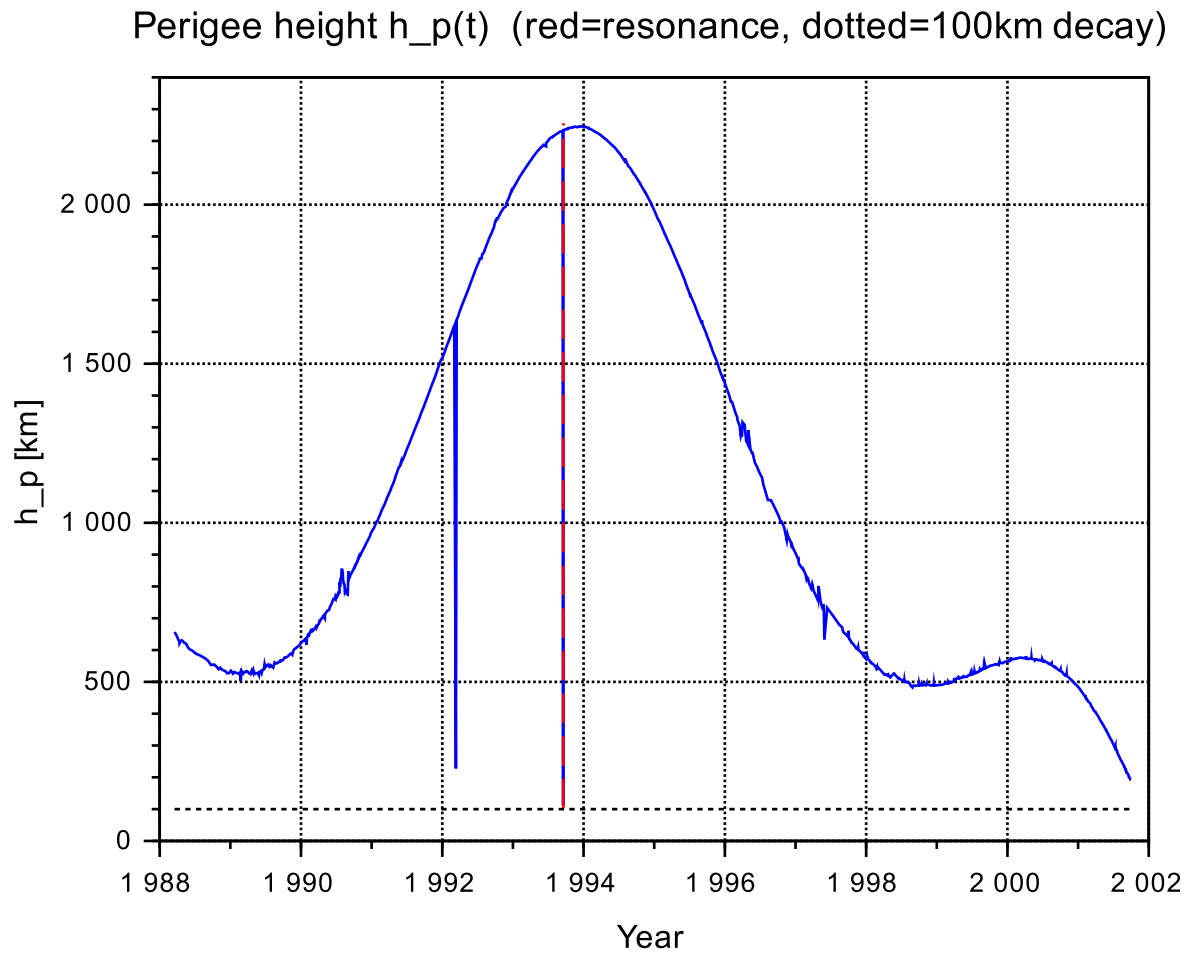
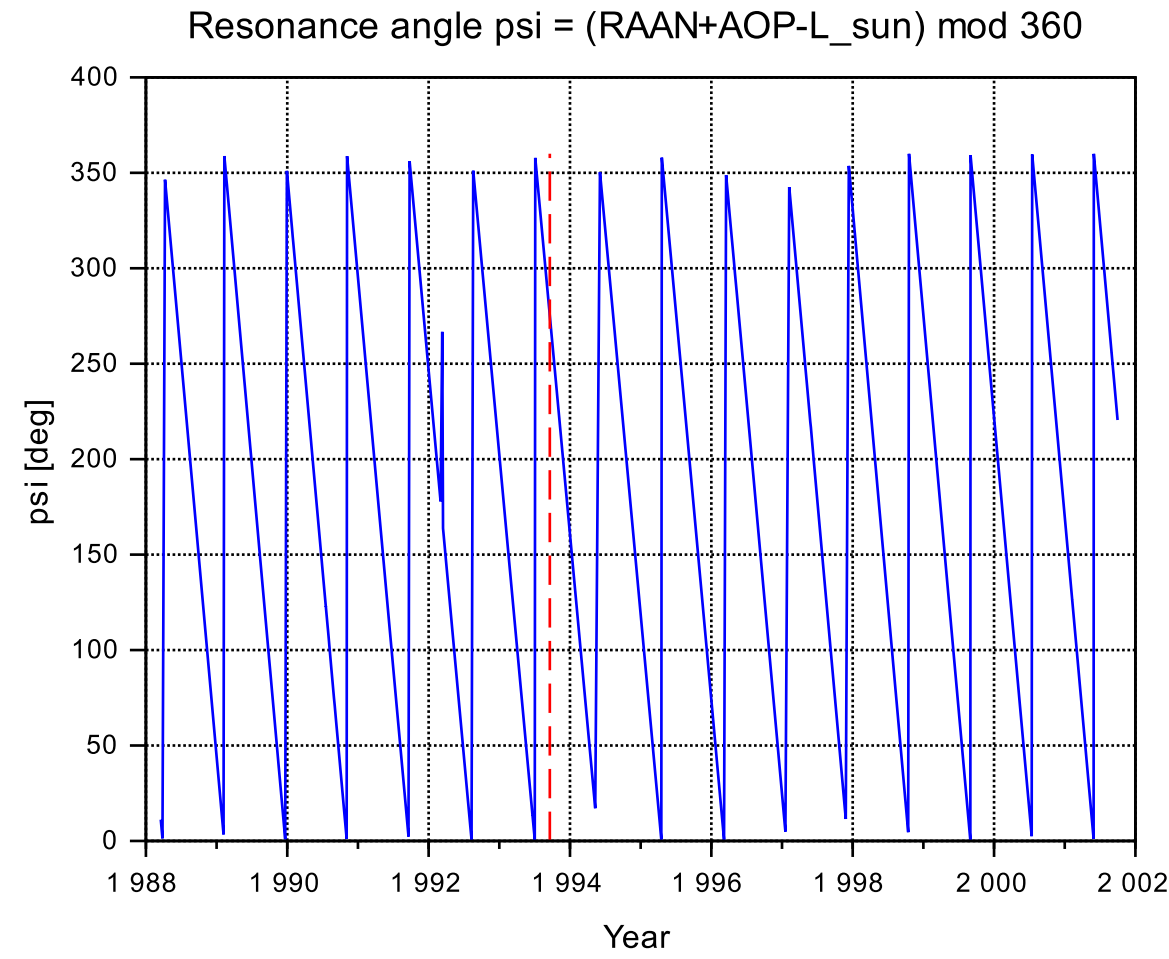


NORAD 40943 GTO debris
Mean inclination: 6.34 deg
 λ_{crit} (Eq.12): 12265.89
Resonance epoch: year 2019.74
TLE record: 2015.75 to 2020.13
Observed decay: year 2020.1355

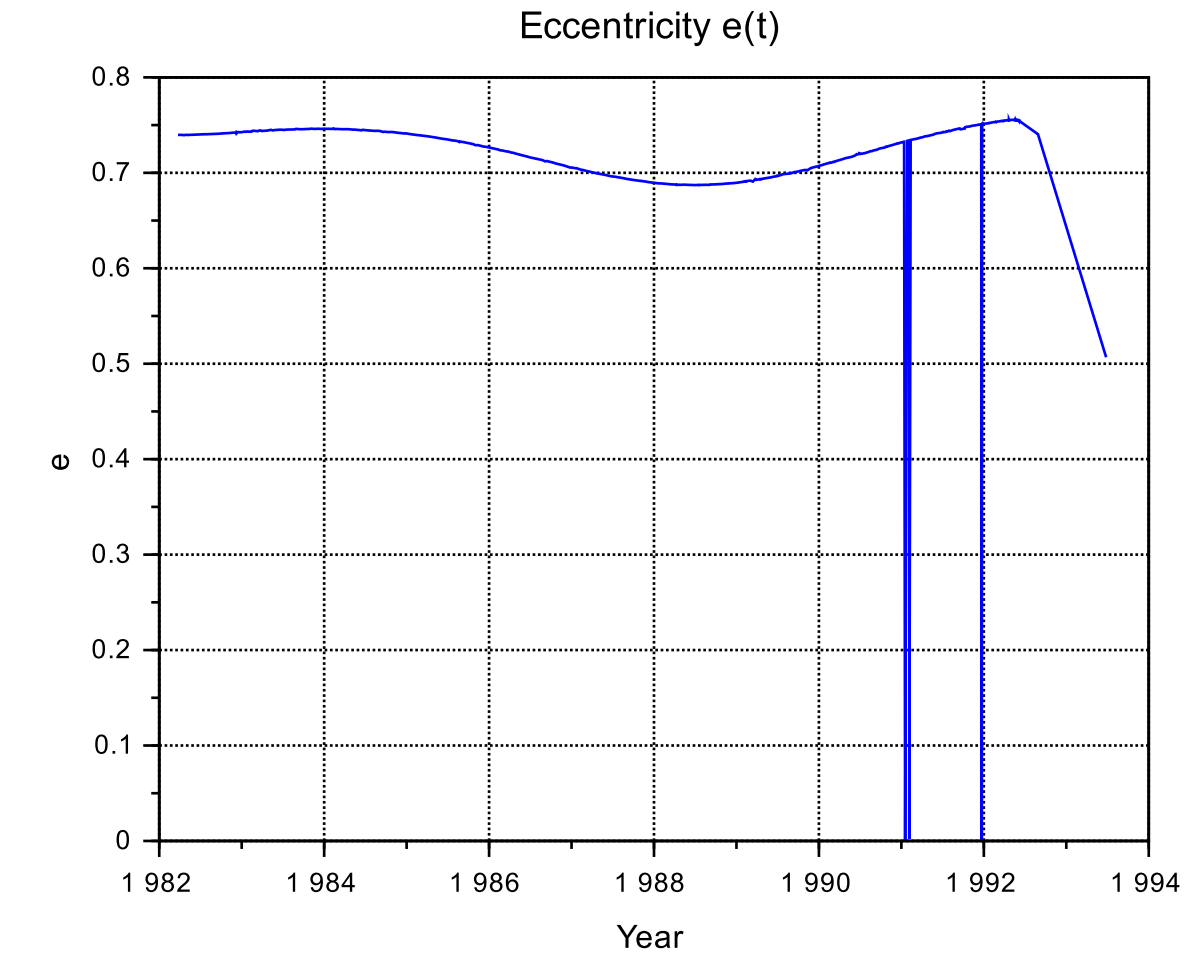
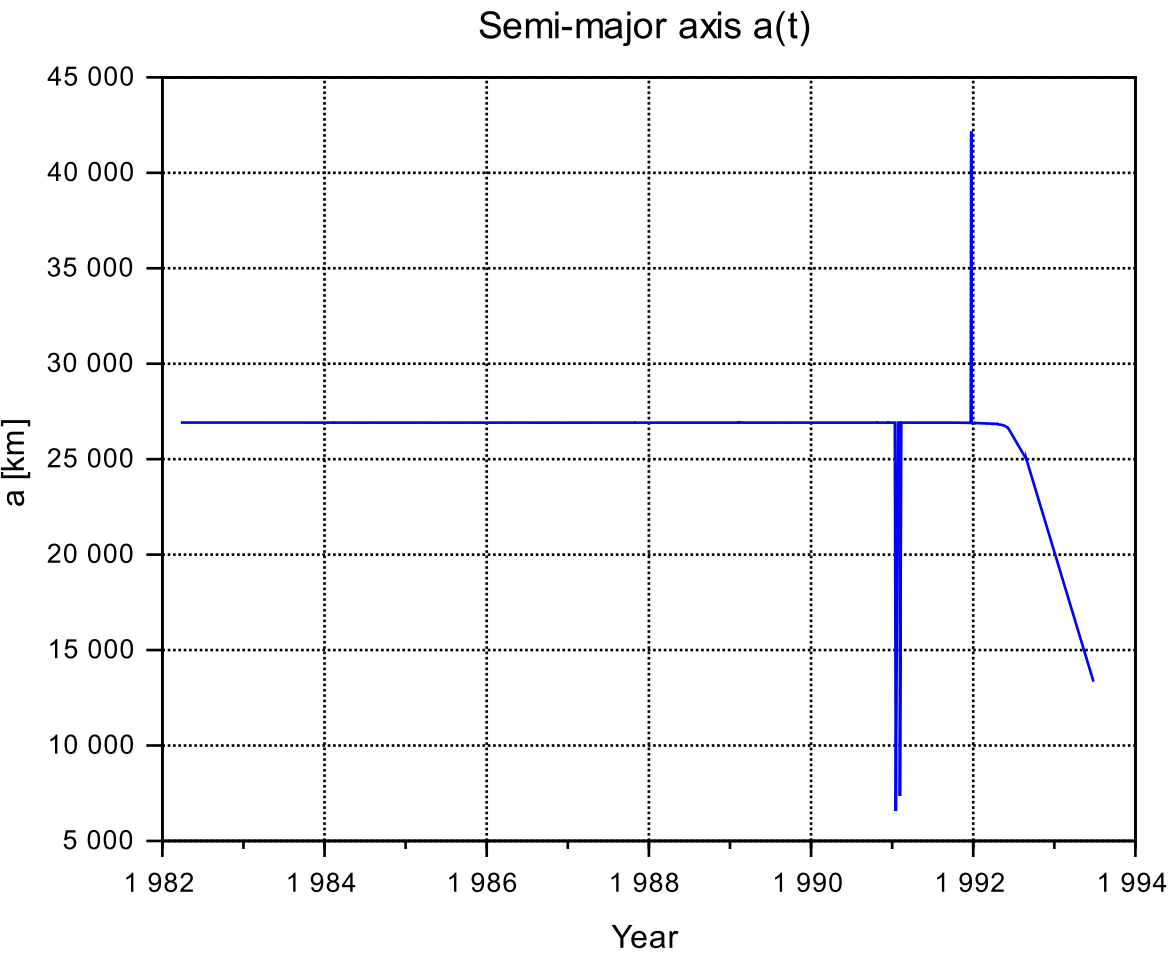
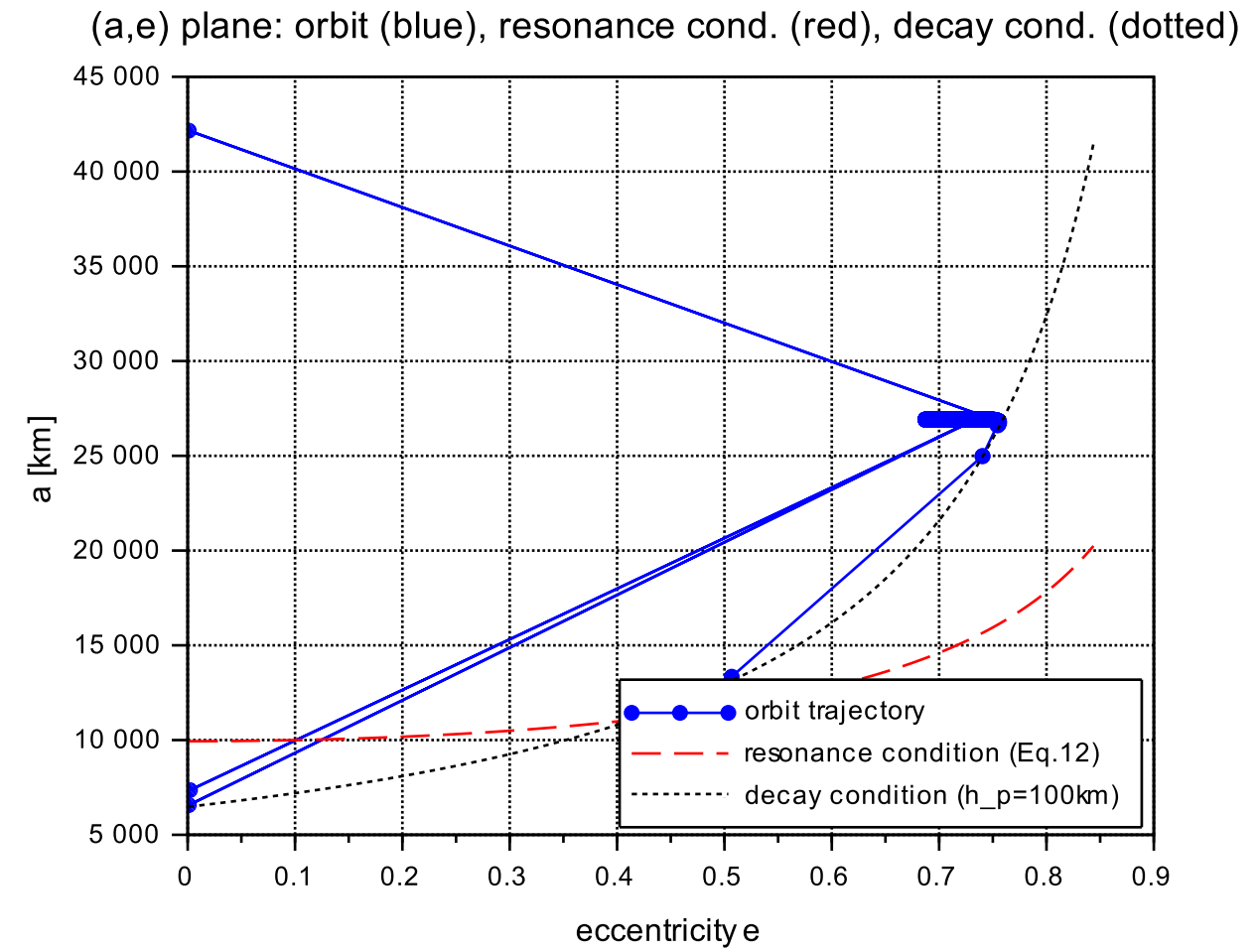
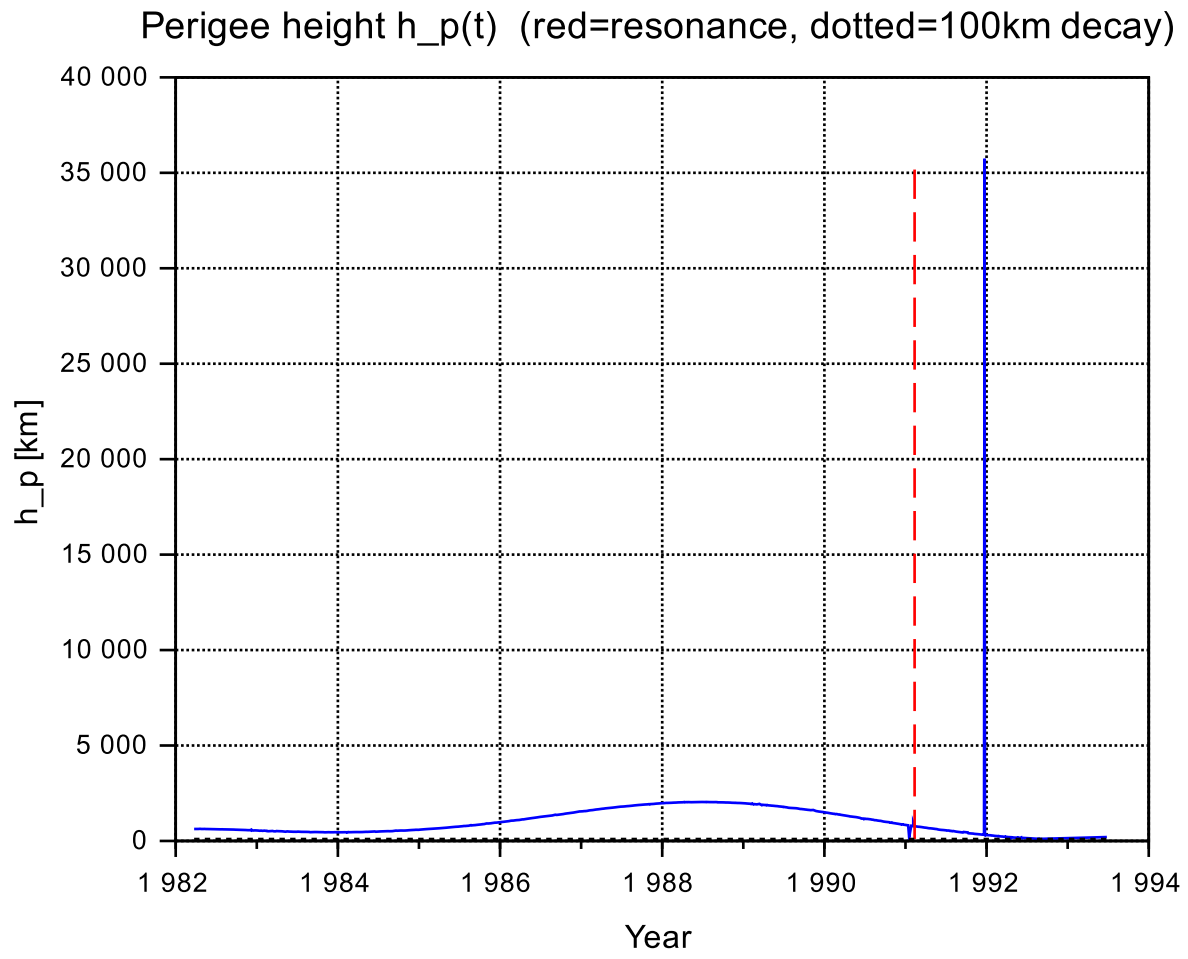
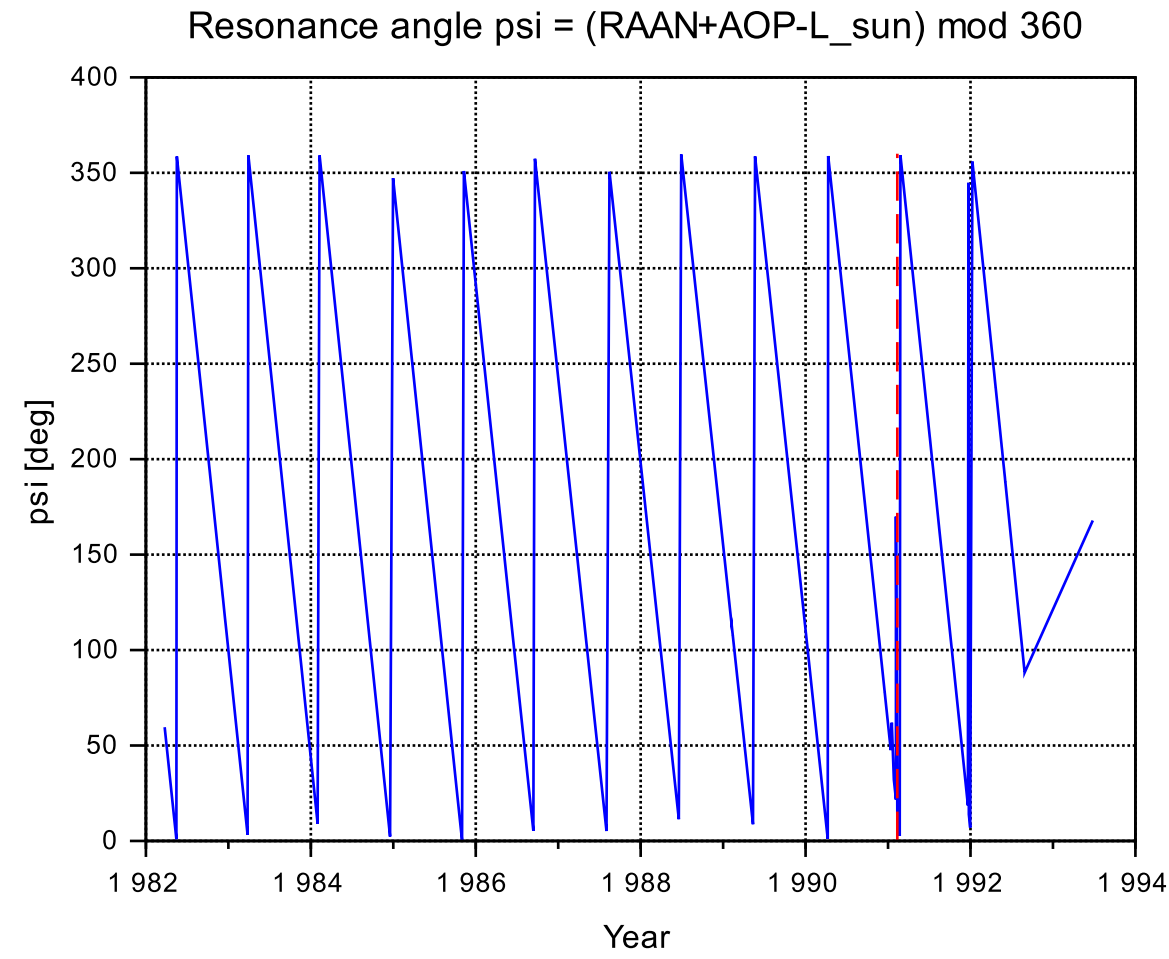




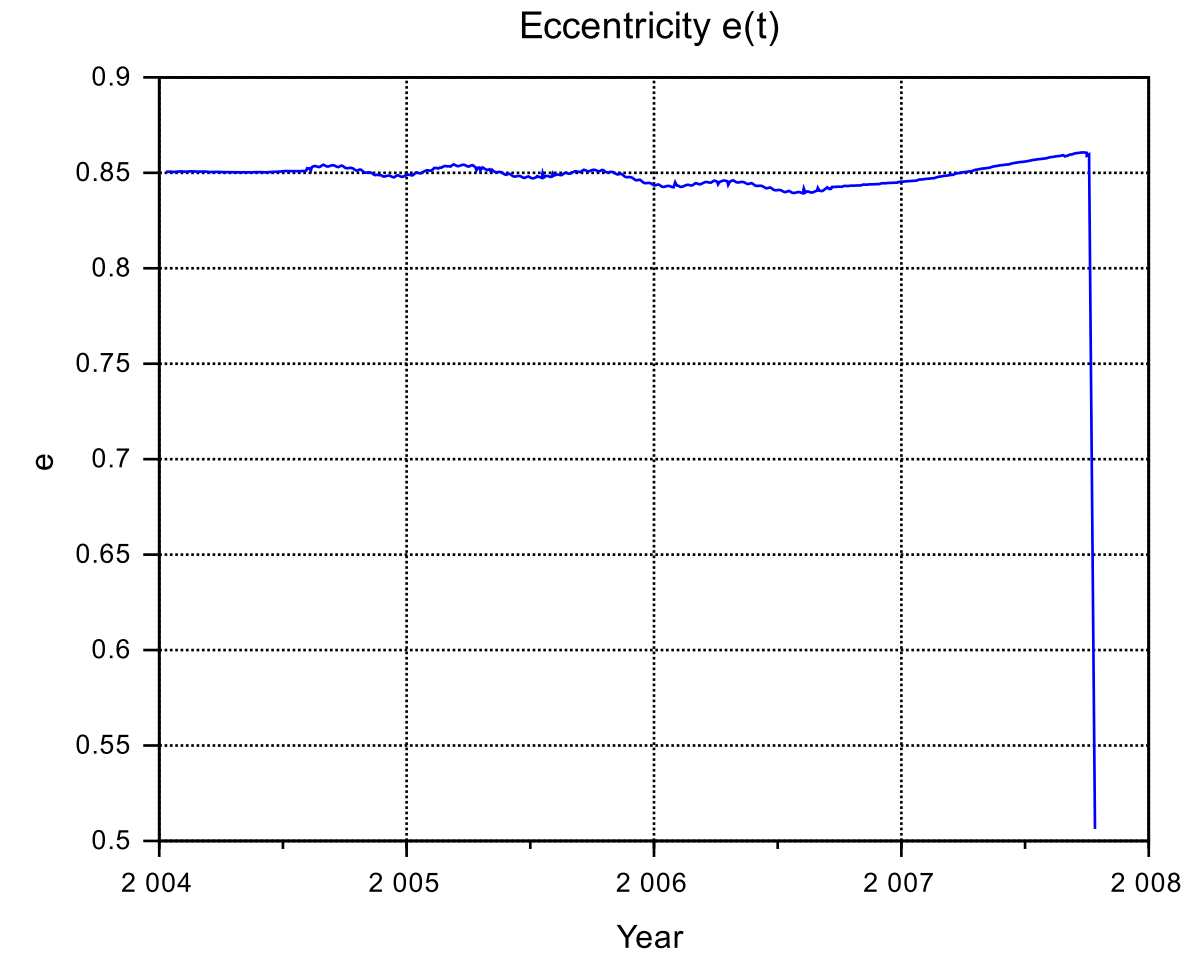
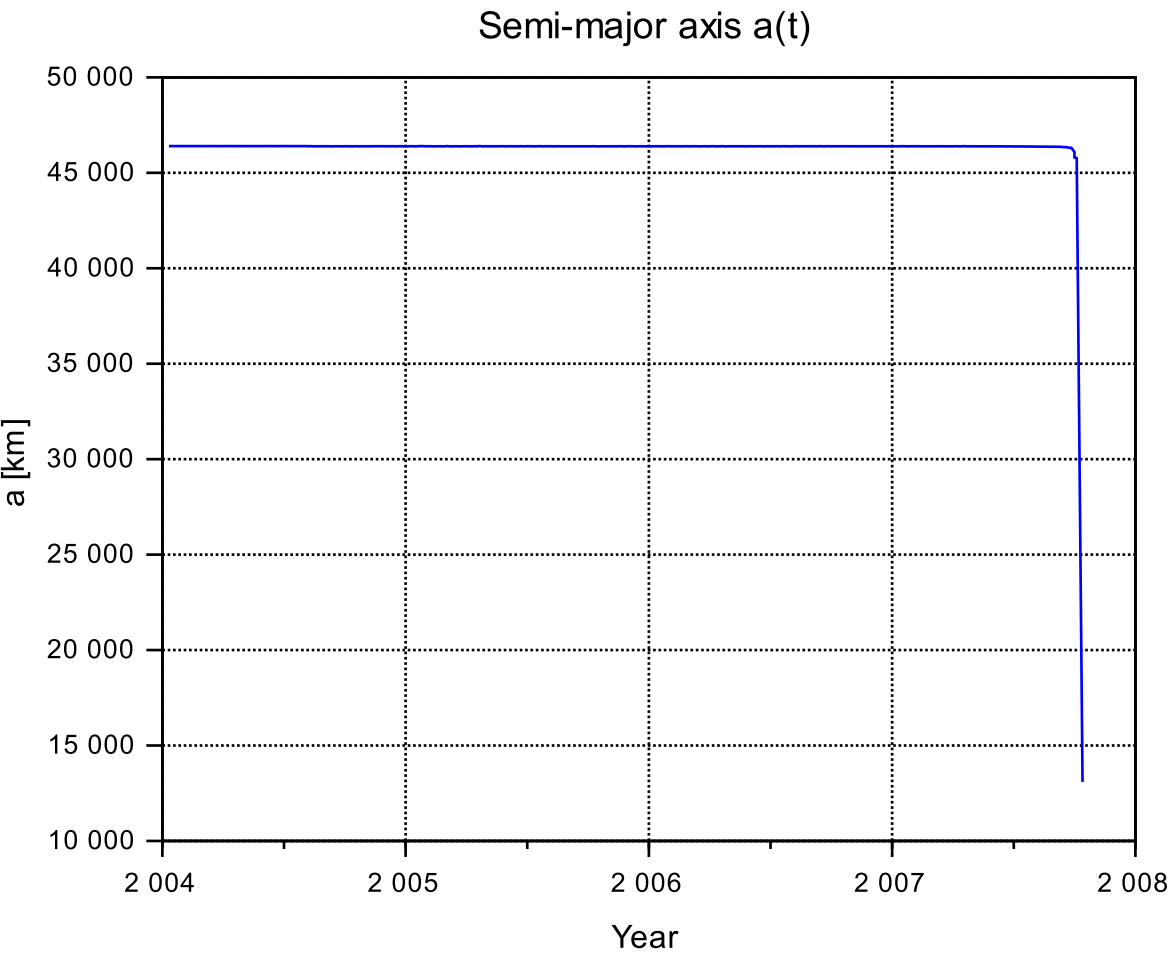
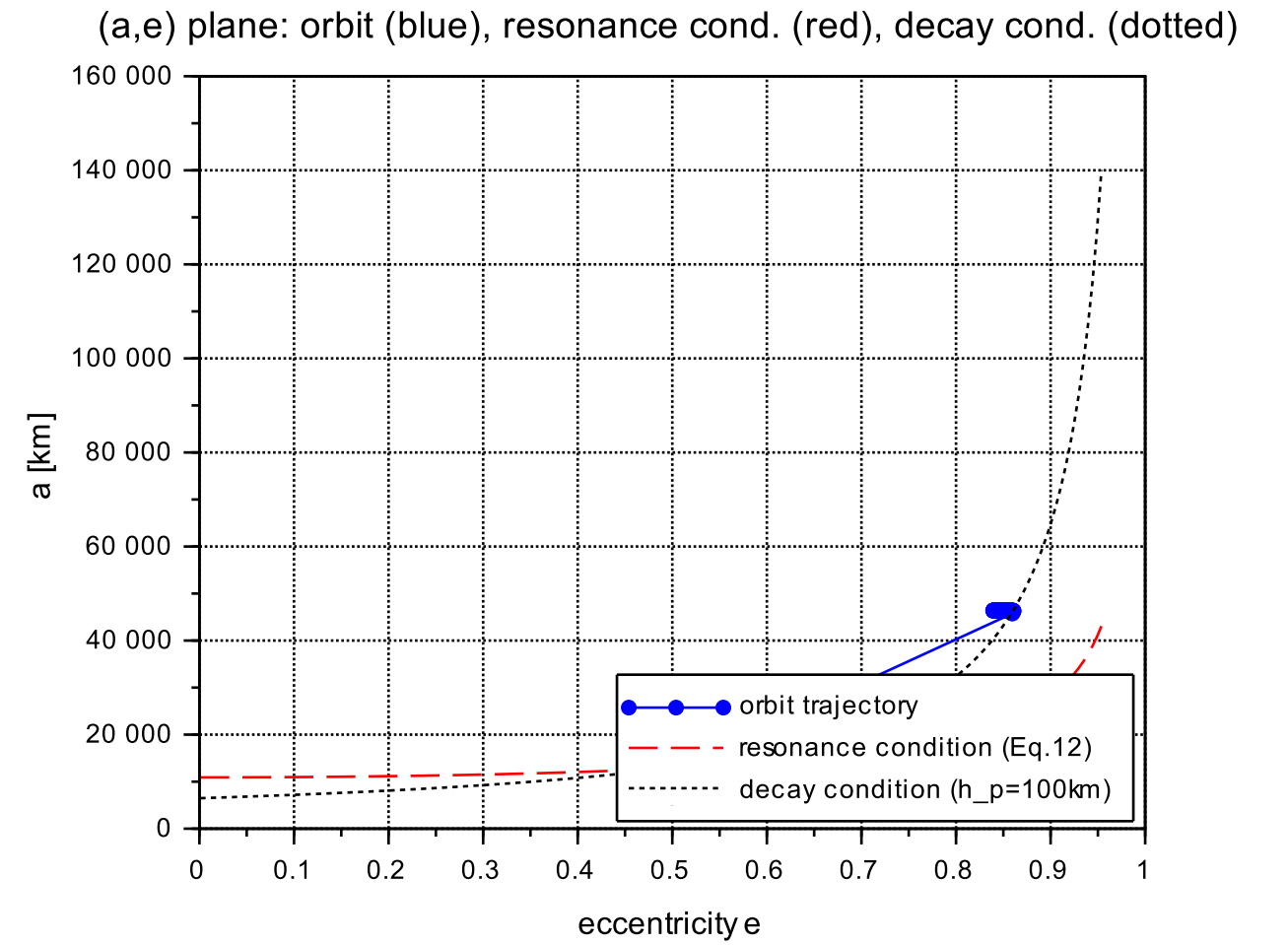
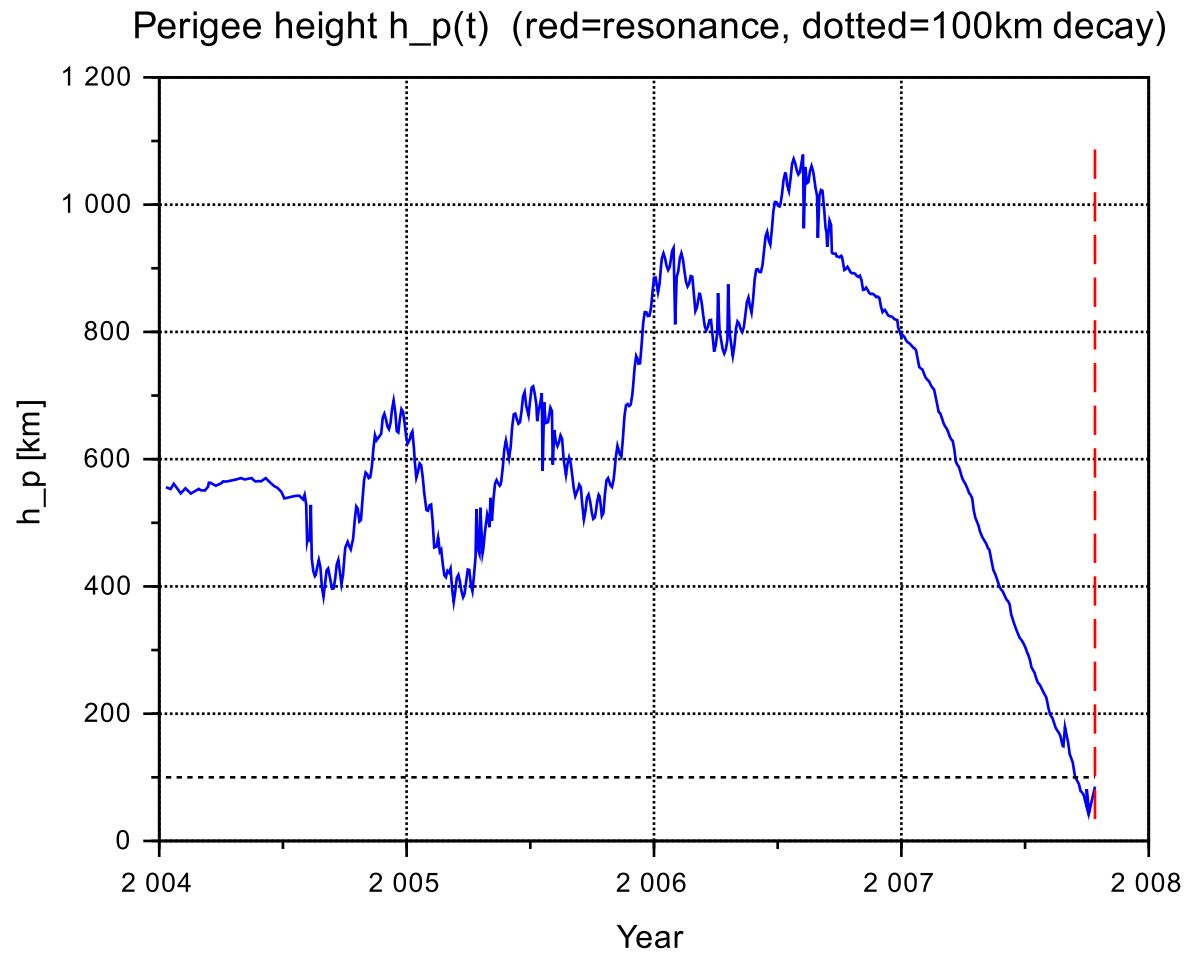
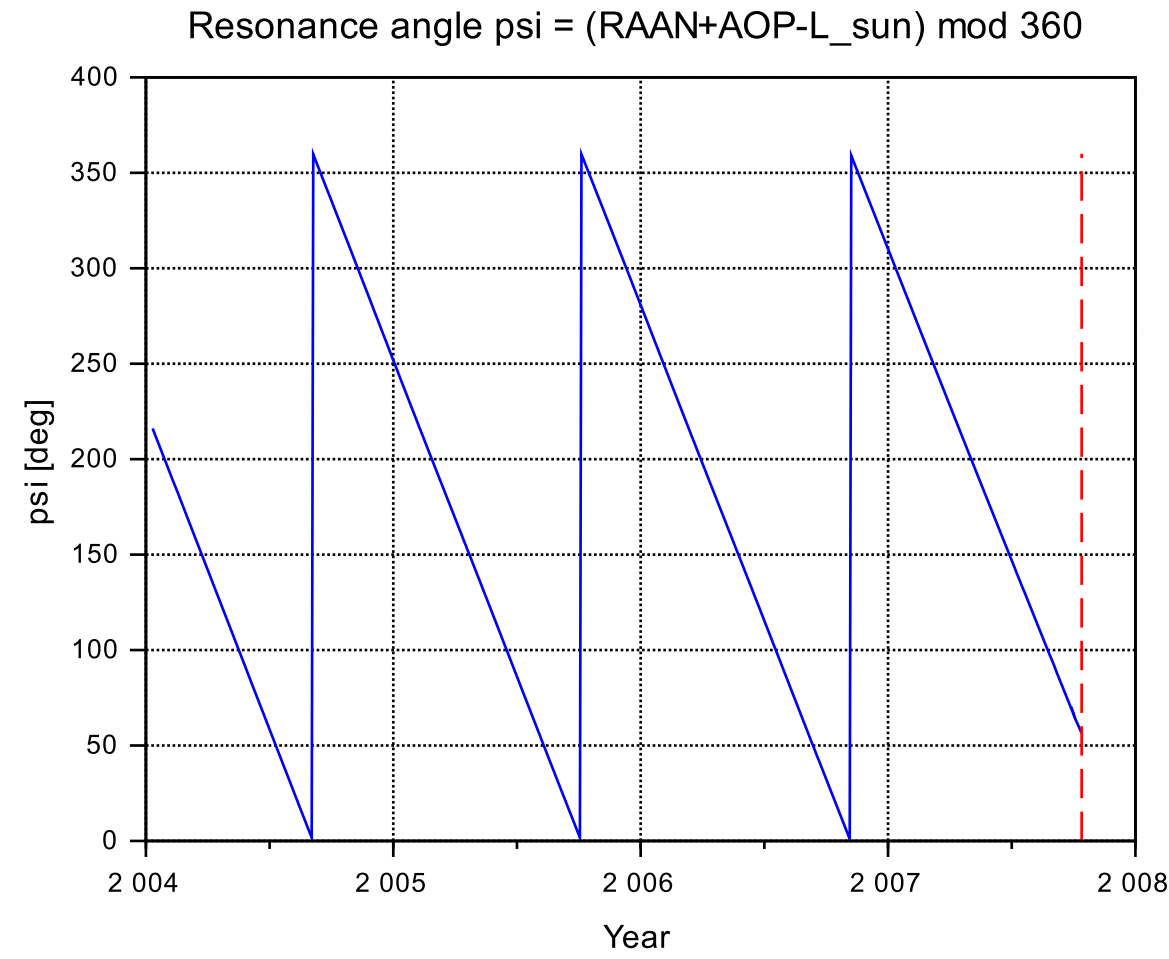
NORAD 60328 CZ-3B R/B
Mean inclination: 26.84 deg
 λ_{crit} (Eq.12): 10665.13
Resonance epoch: year 2025.26
TLE record: 2024.58 to 2025.26
Observed decay: year 2025.2827



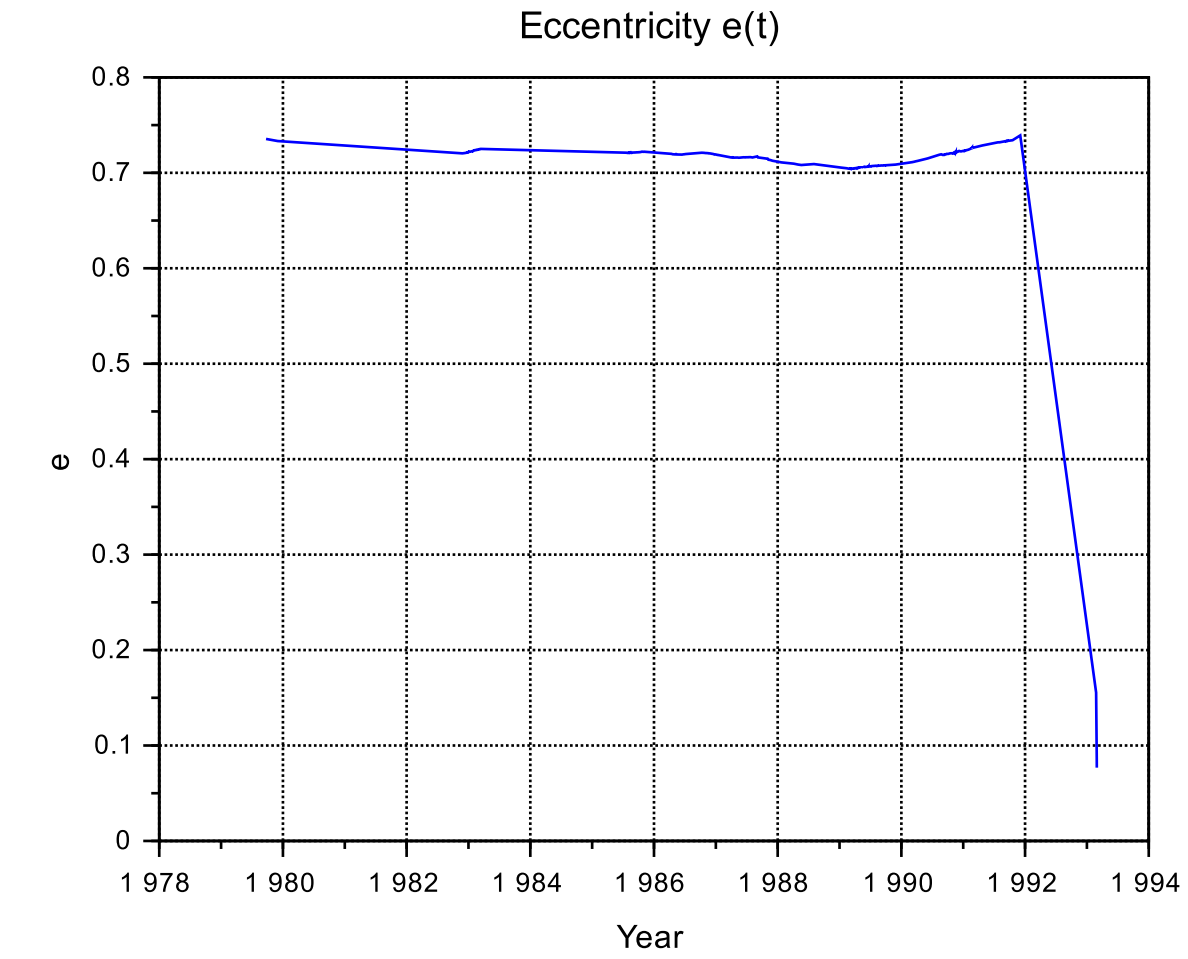
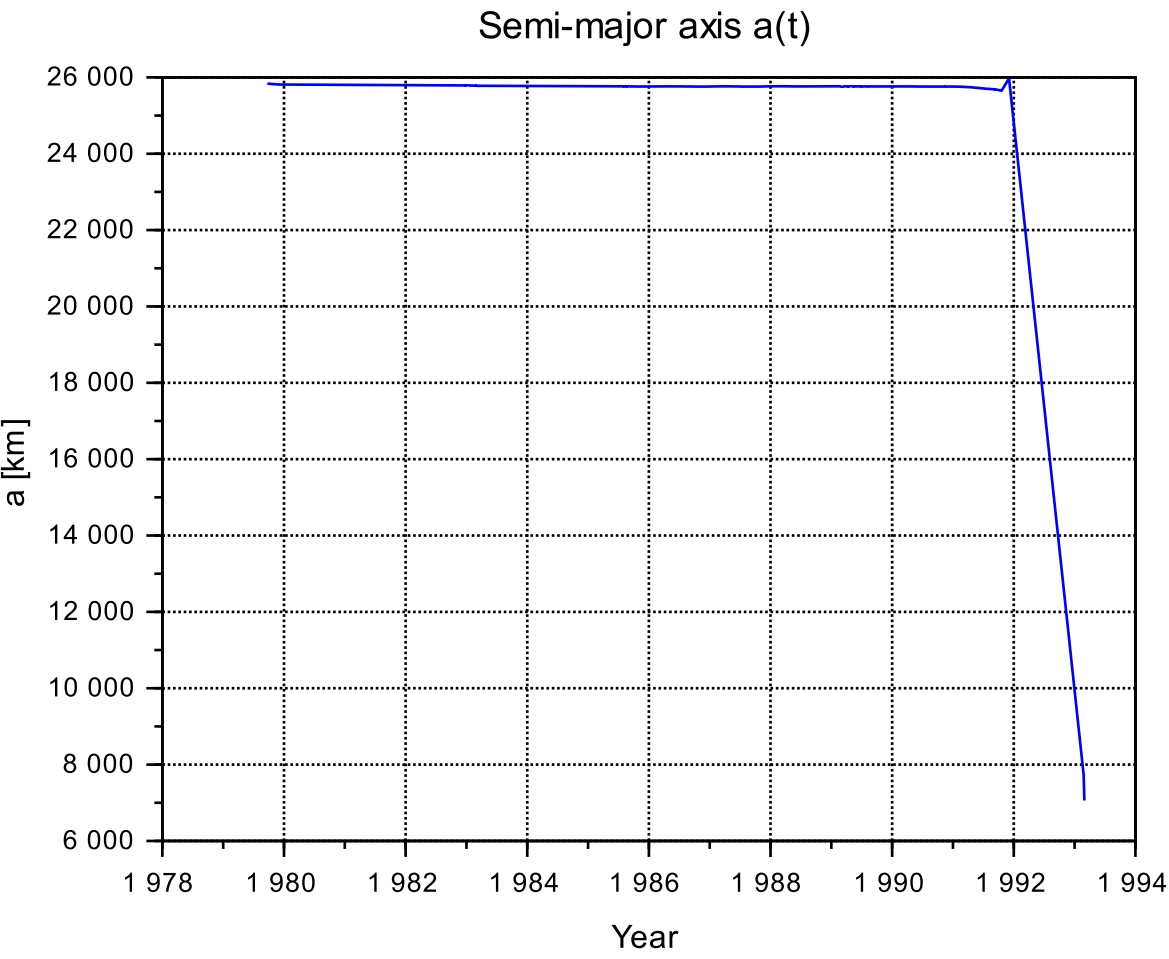
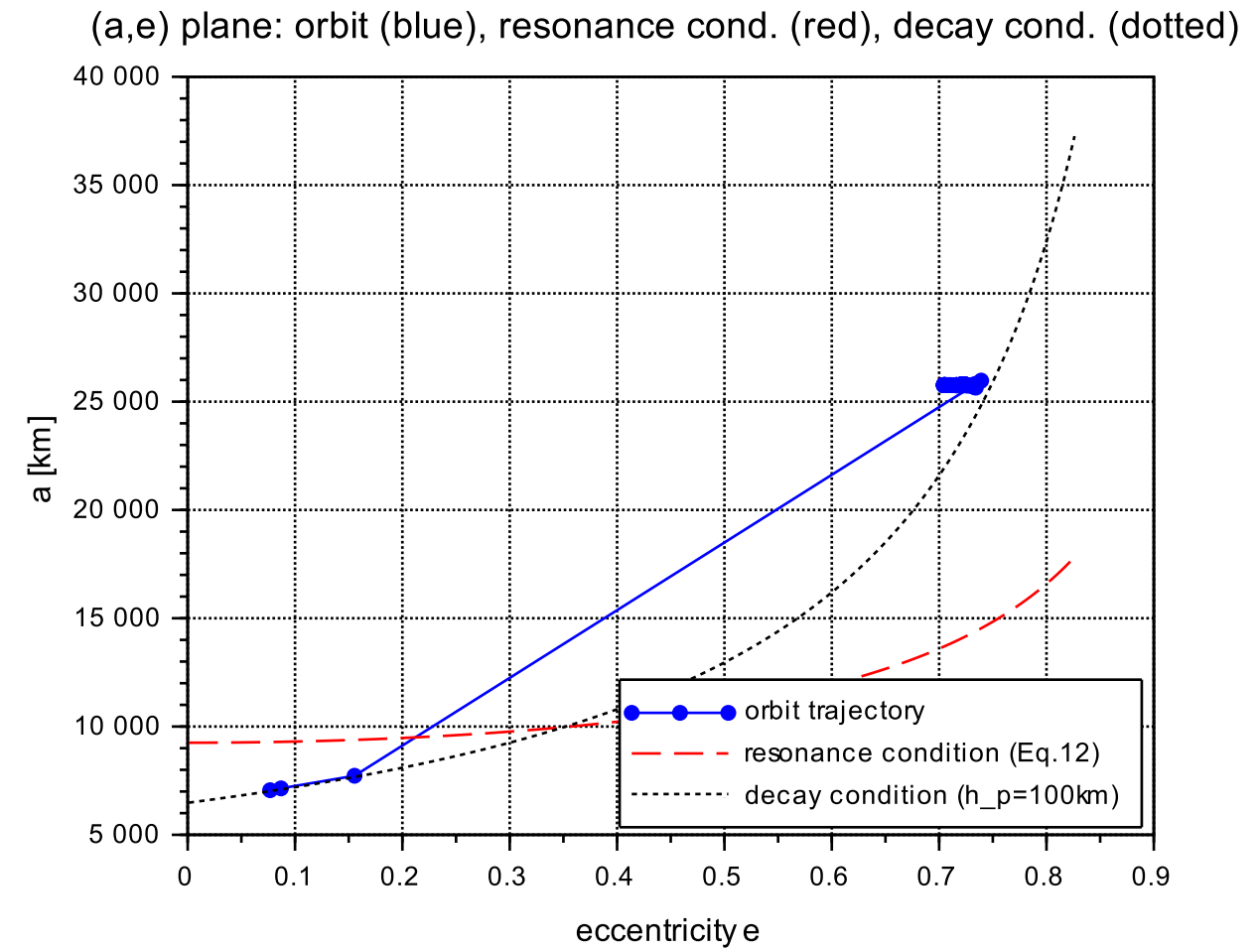
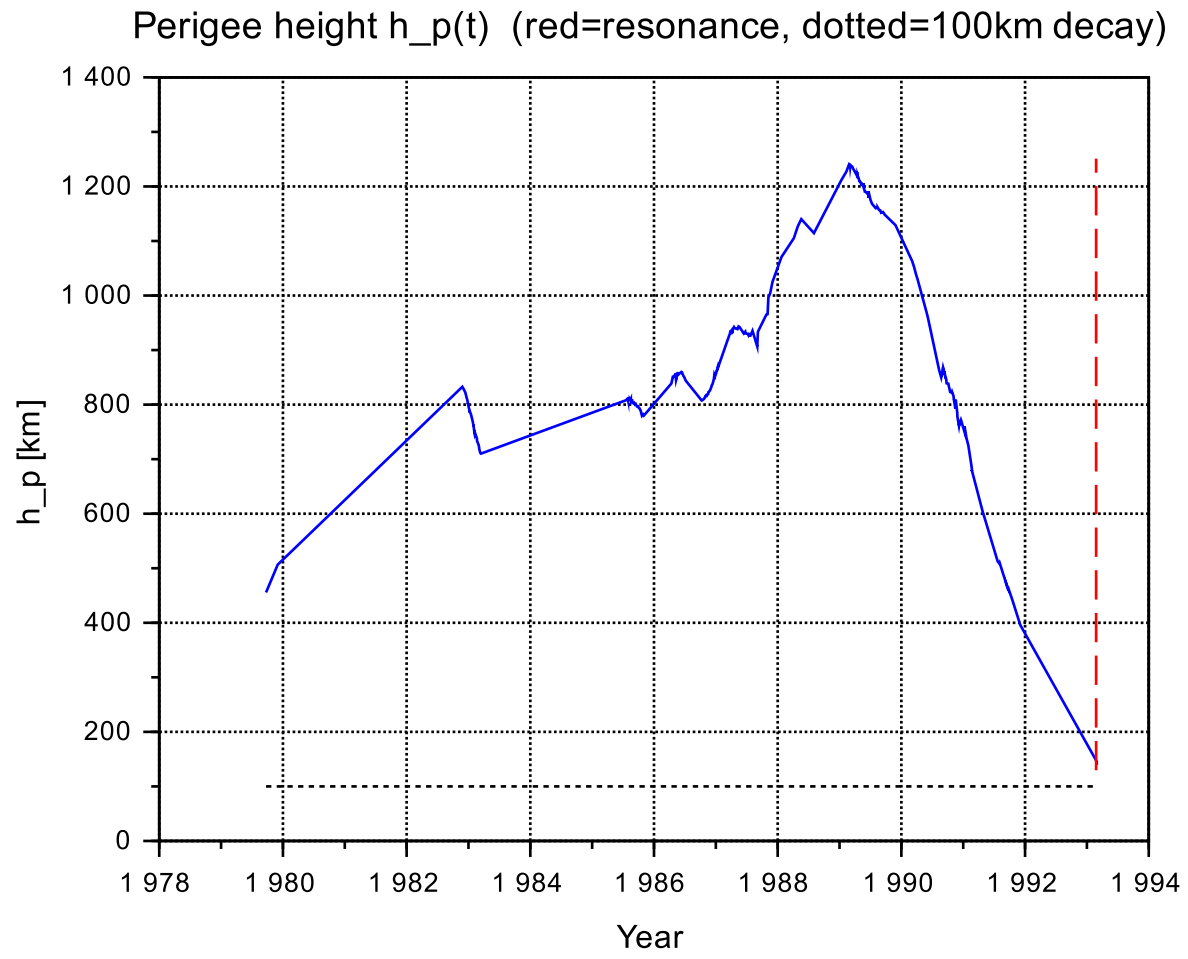
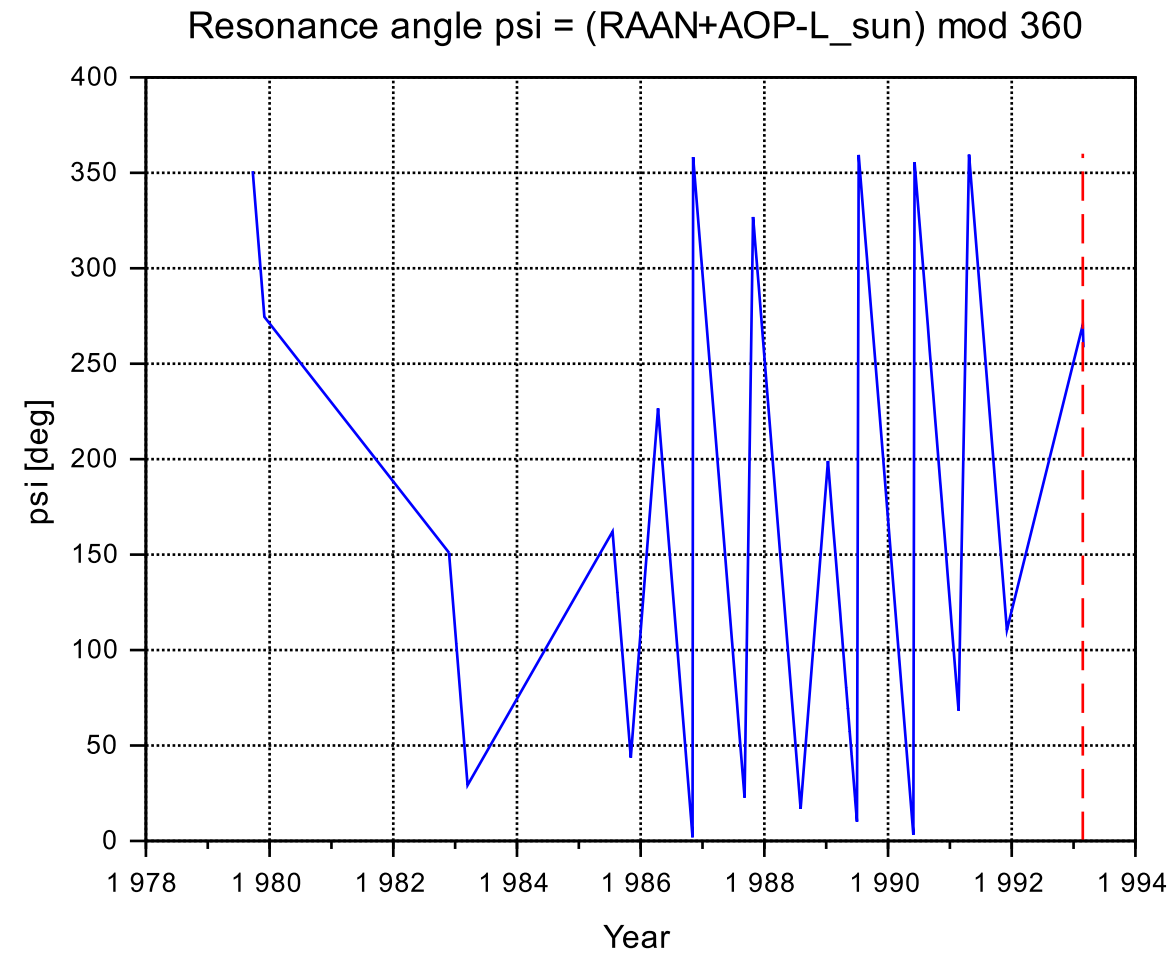
NORAD 18983 SL-6 R/B(2)
Mean inclination: 63.7 deg
 λ_{crit} (Eq.12): 9846.84
Resonance epoch: year 1993.71
TLE record: 1988.21 to 2001.75
Observed decay: year 2002.0602



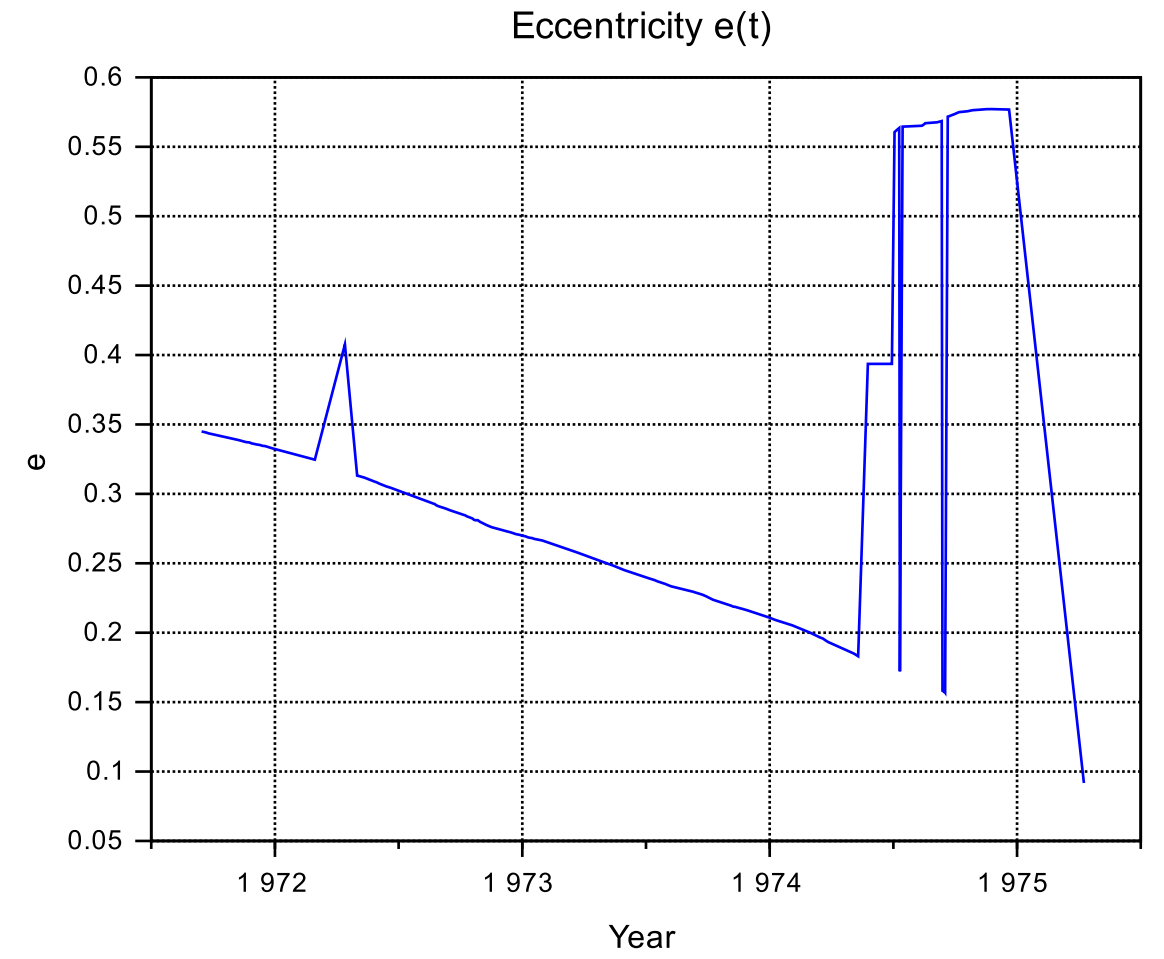
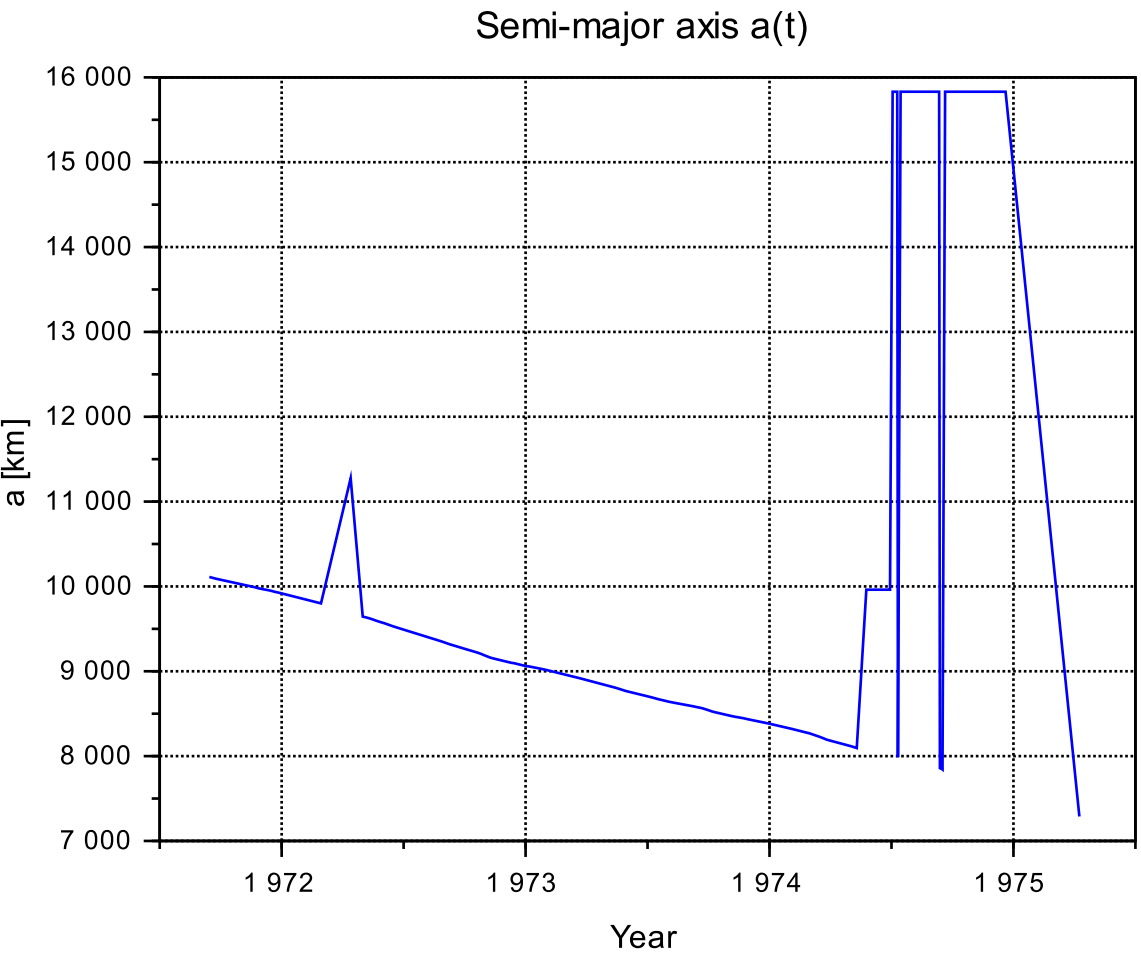
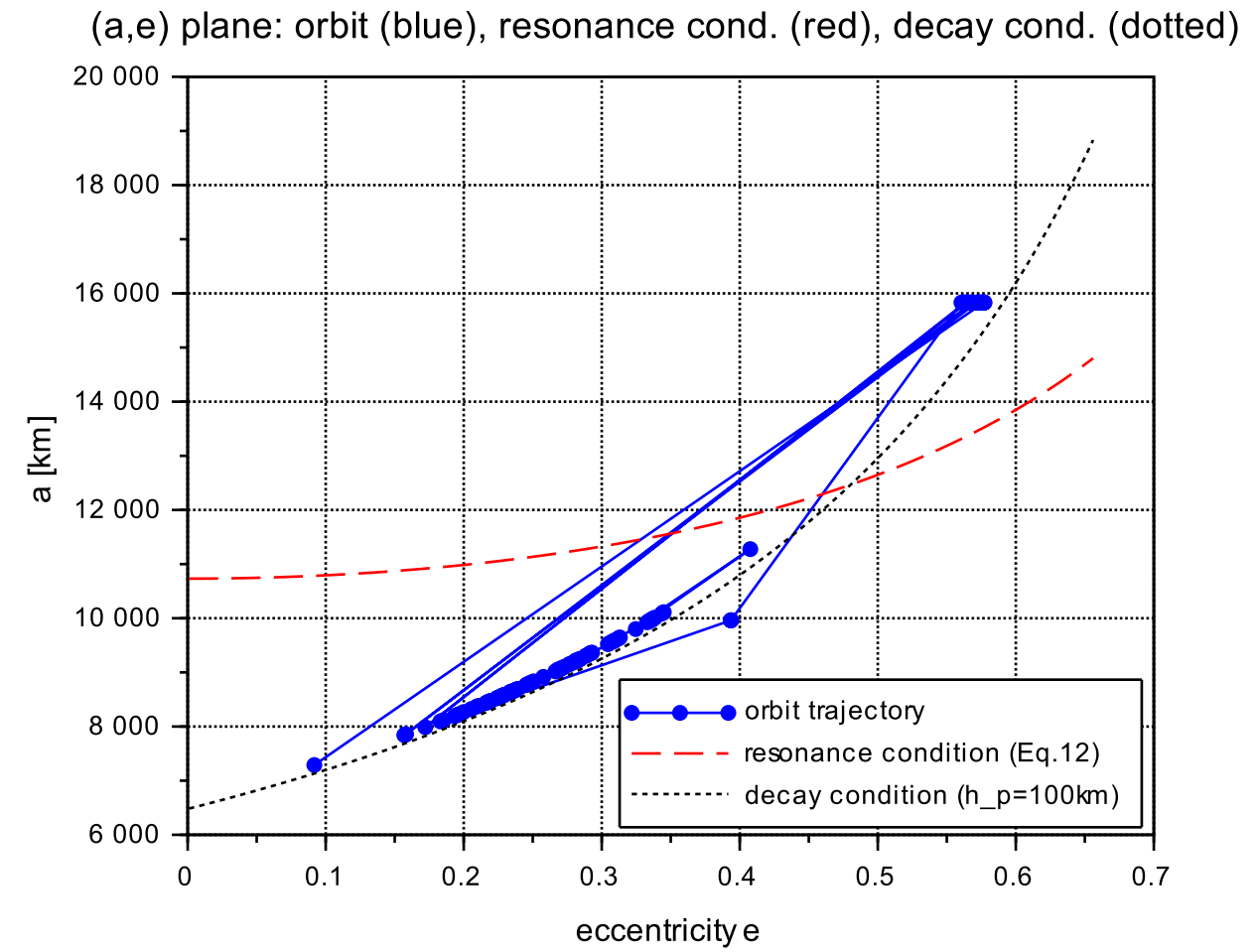
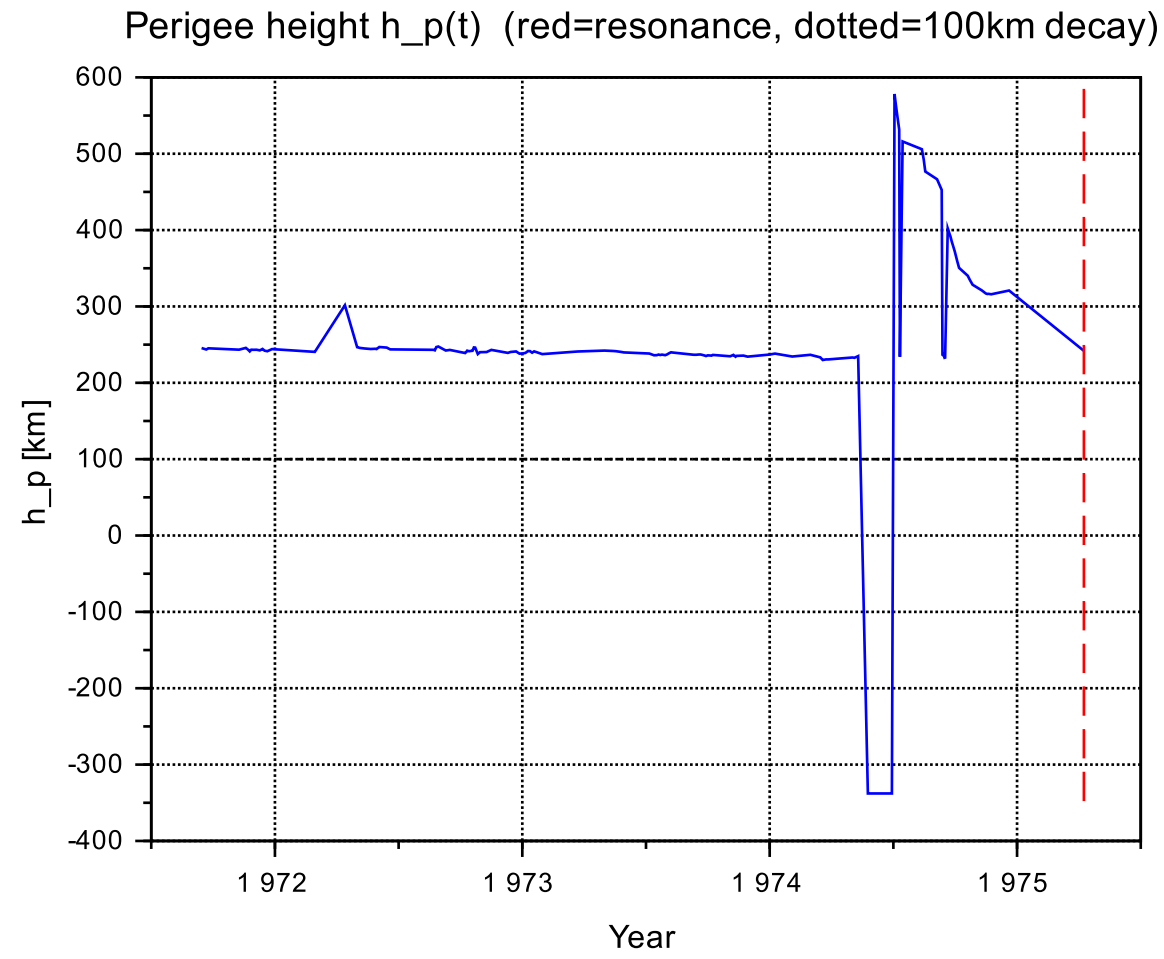
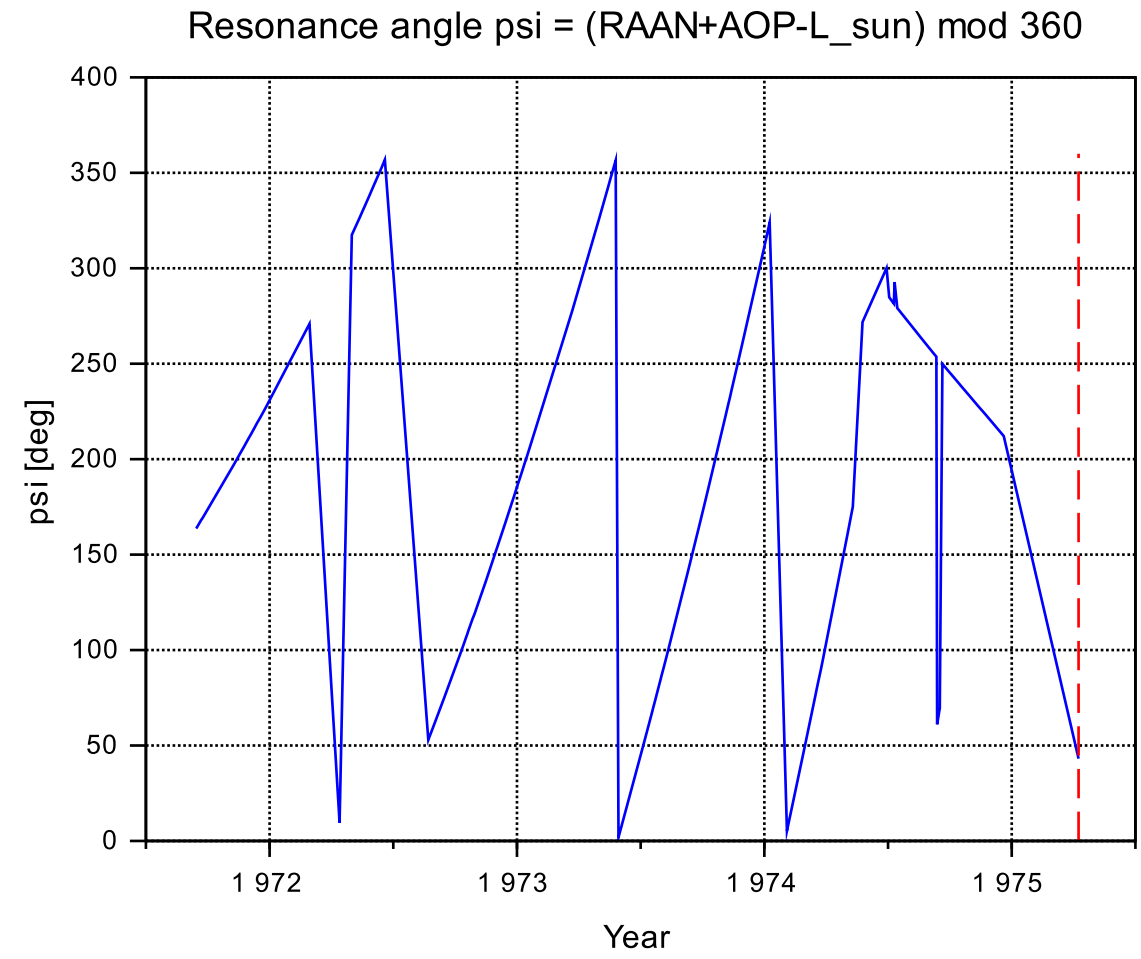
NORAD 13112 SL-6 R/B(2)
Mean inclination: 64.52 deg
 λ_{crit} (Eq.12): 9940.79
Resonance epoch: year 1991.11
TLE record: 1982.23 to 1993.48
Observed decay: year 1992.8693



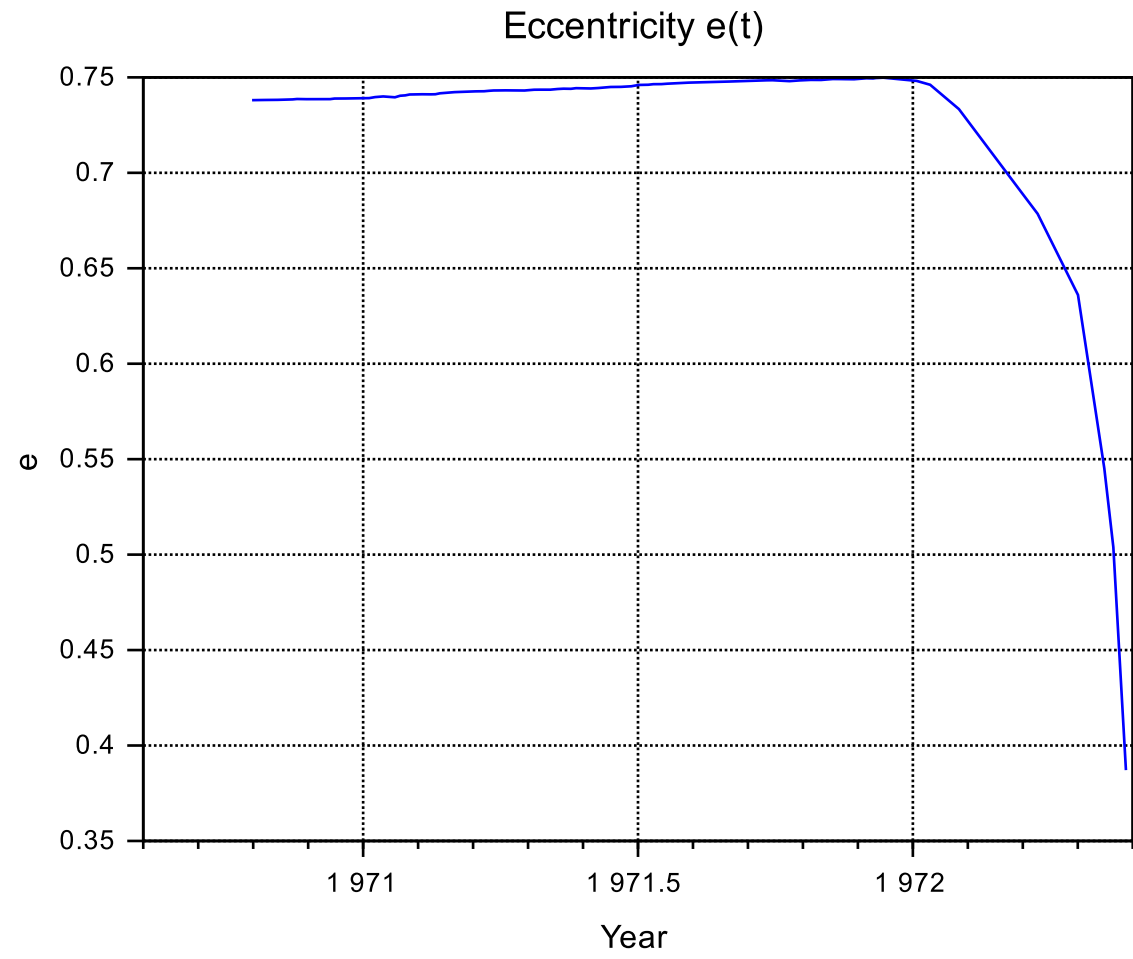
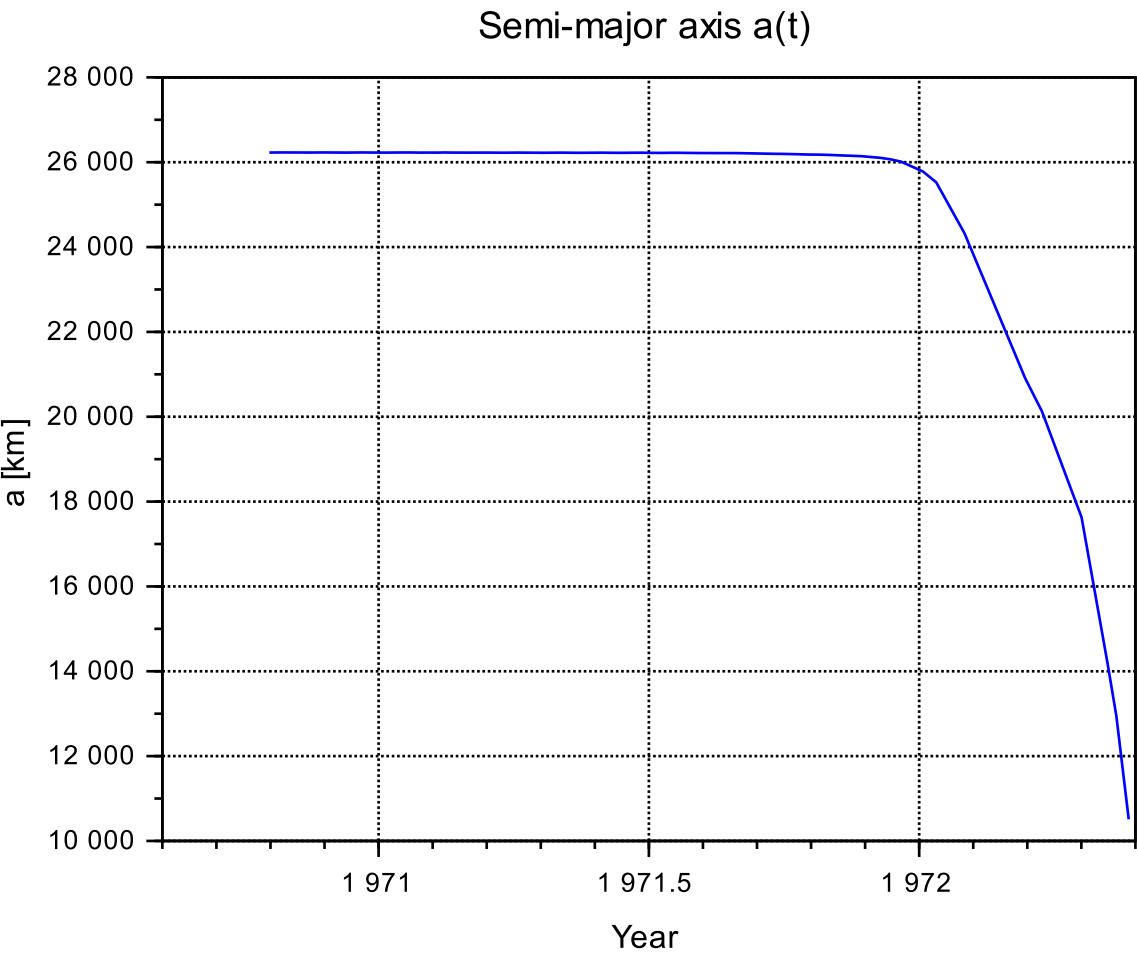
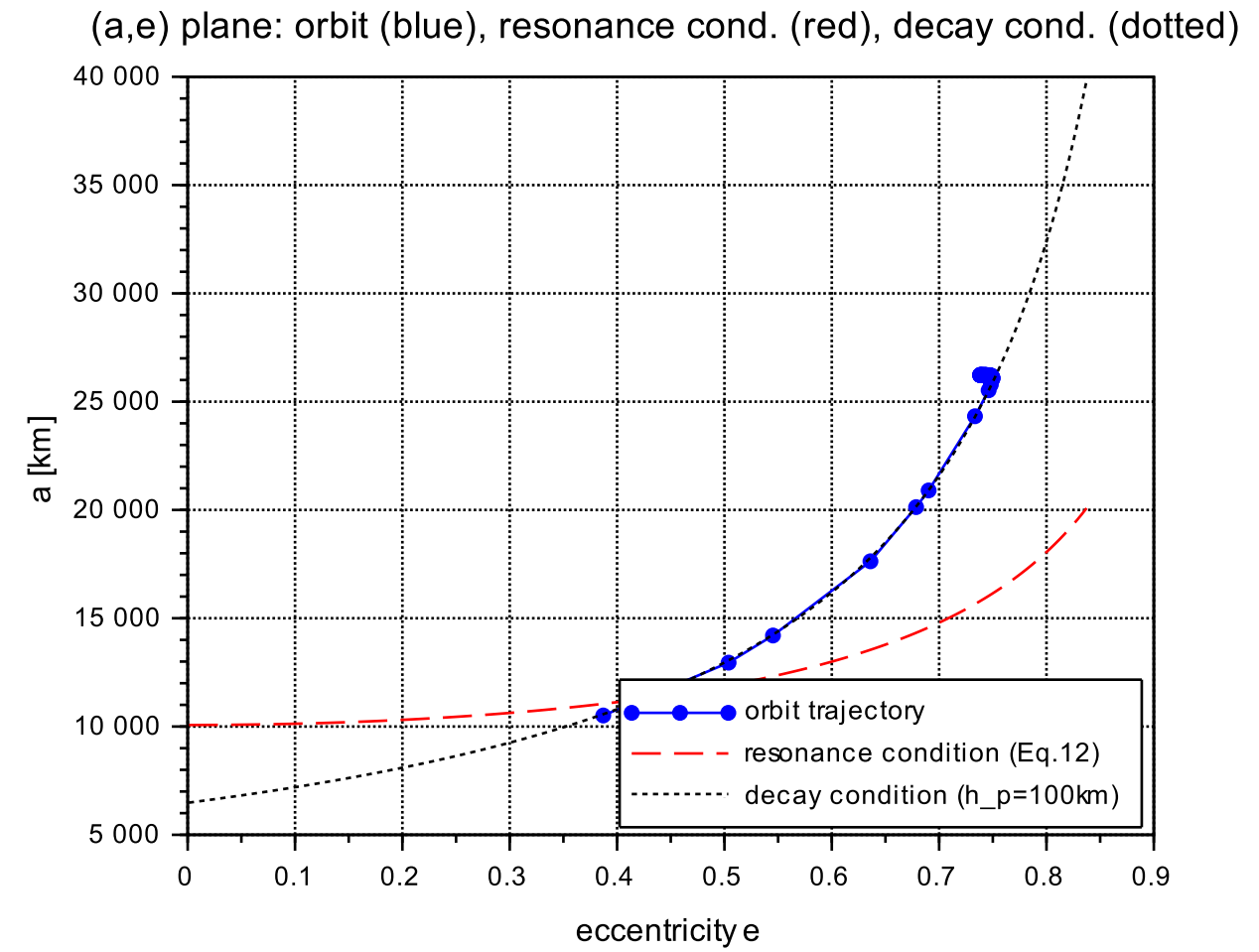
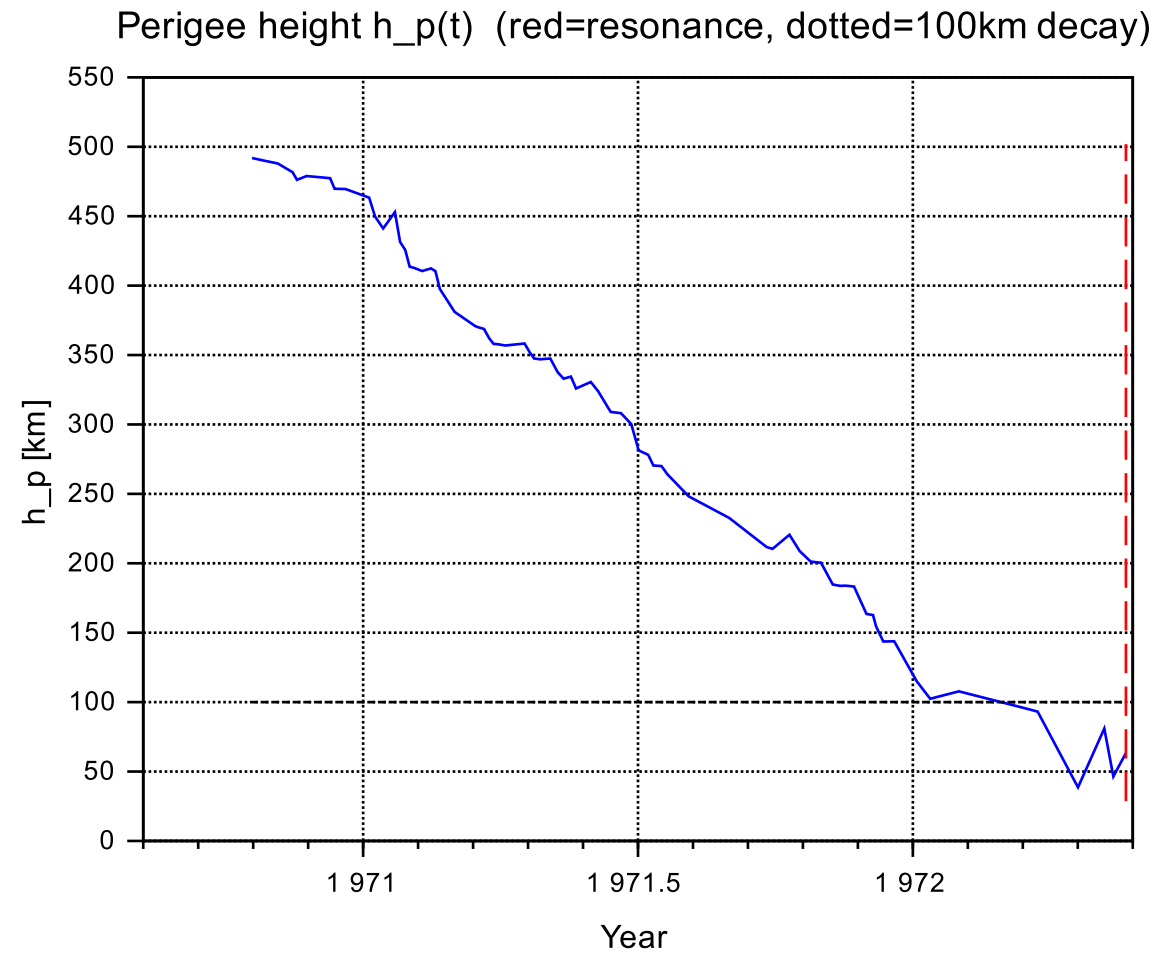
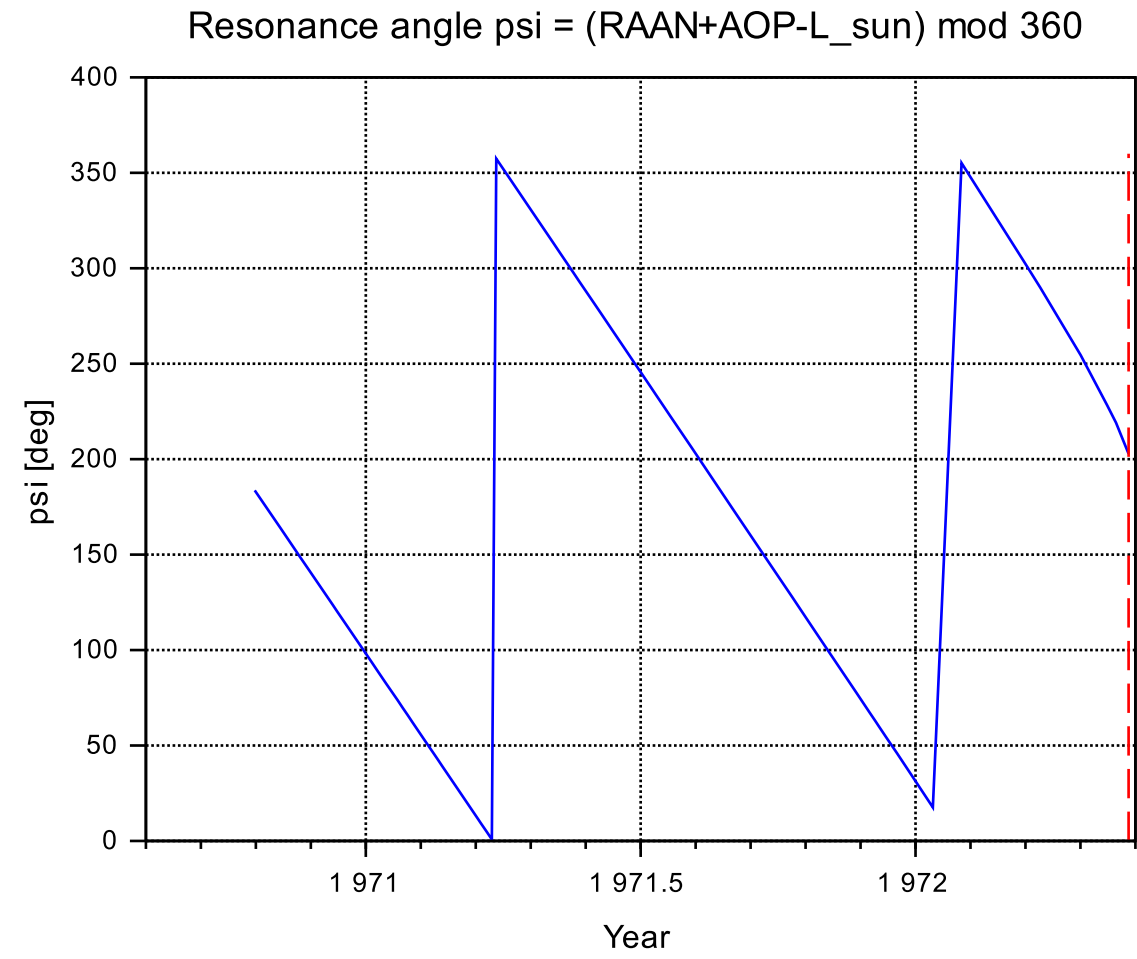
NORAD 28141 CZ-2C R/B
Mean inclination: 25.03 deg
 λ_{crit} (Eq.12): 10904.92
Resonance epoch: year 2007.78
TLE record: 2004.03 to 2007.78
Observed decay: year 2007.7823



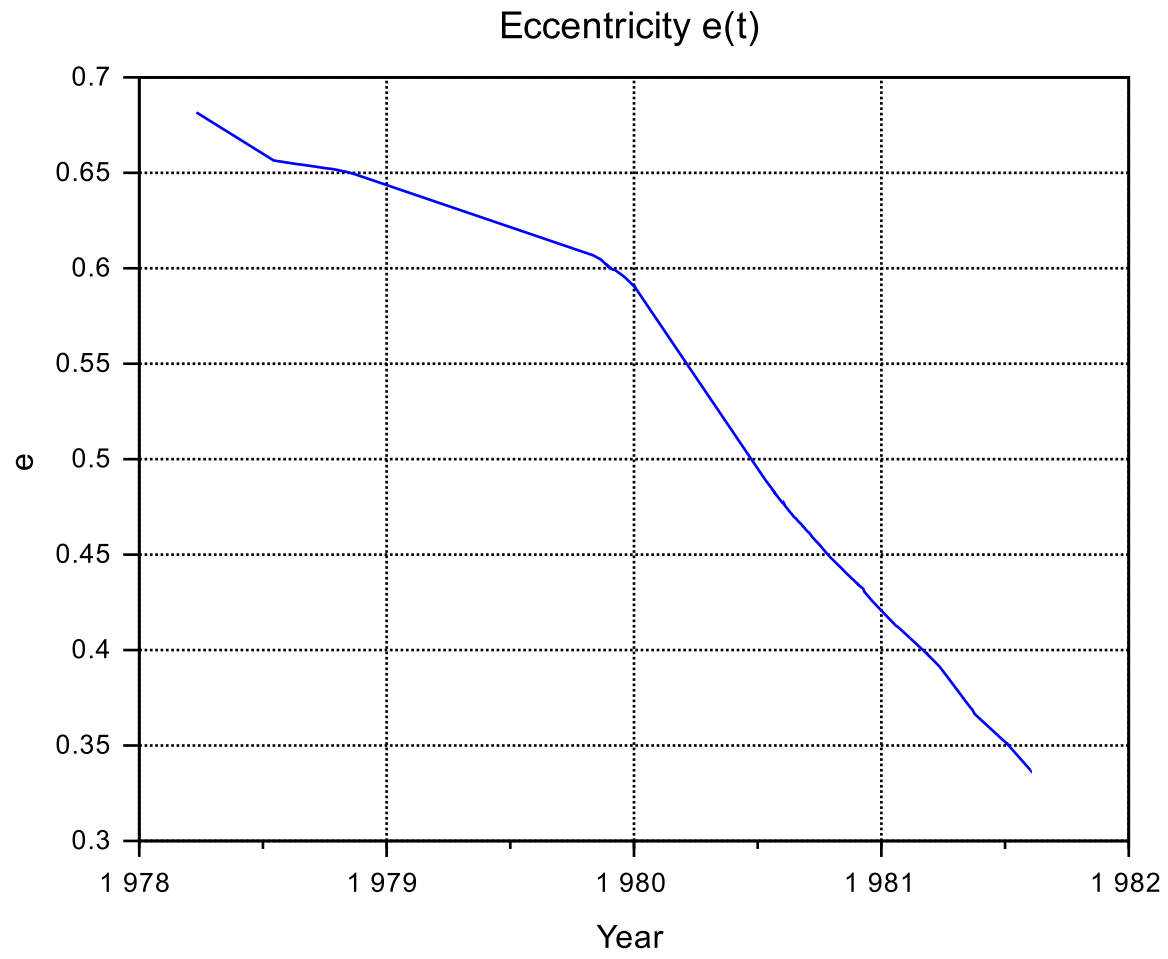
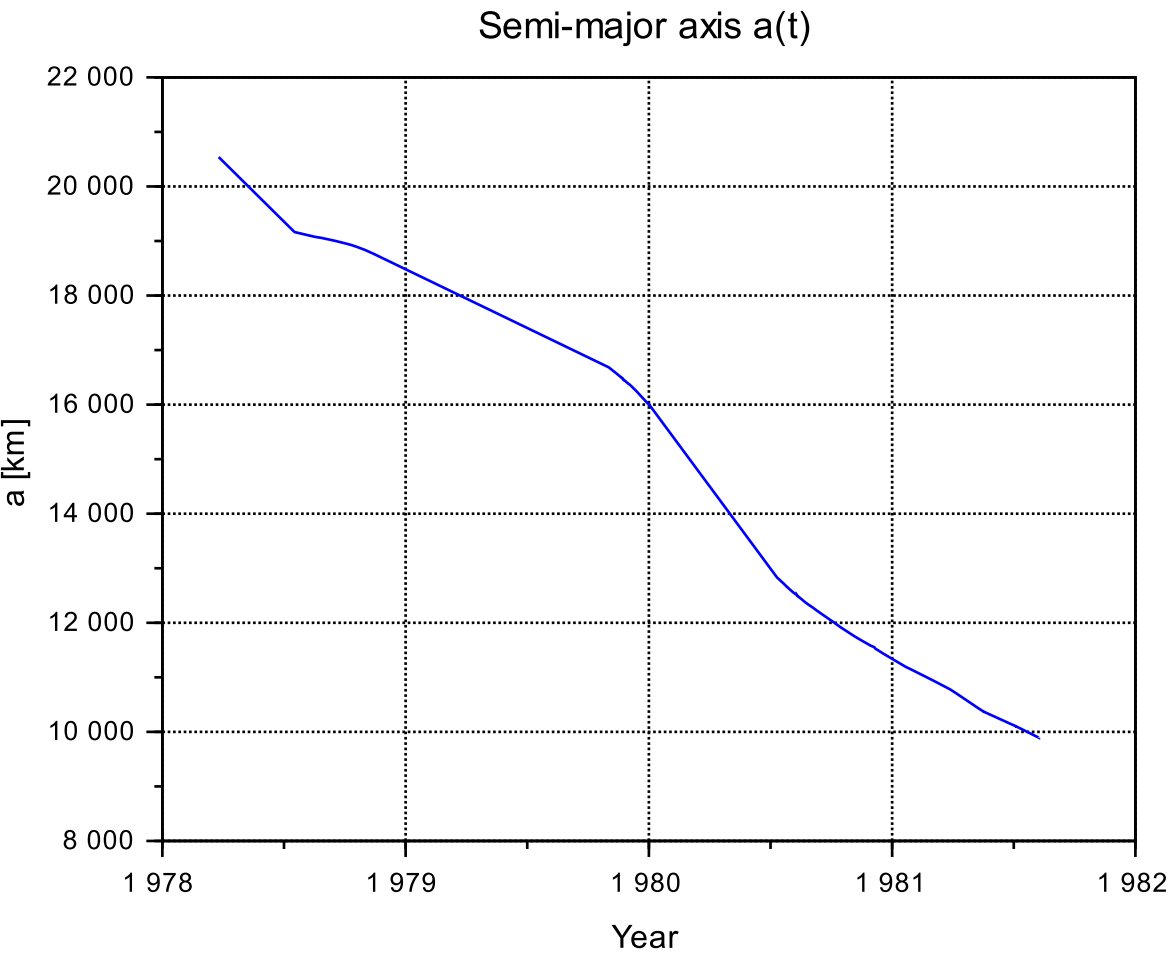
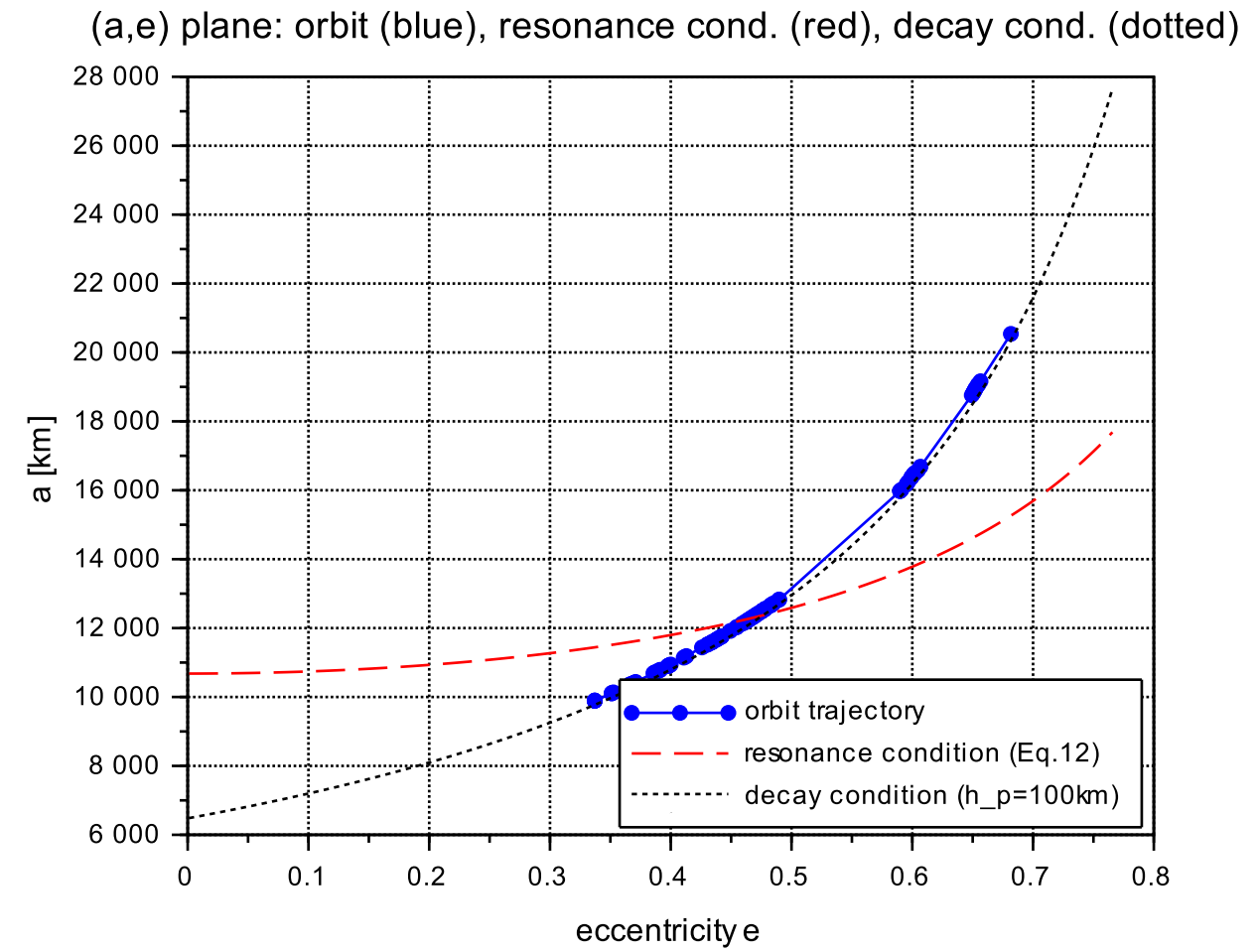
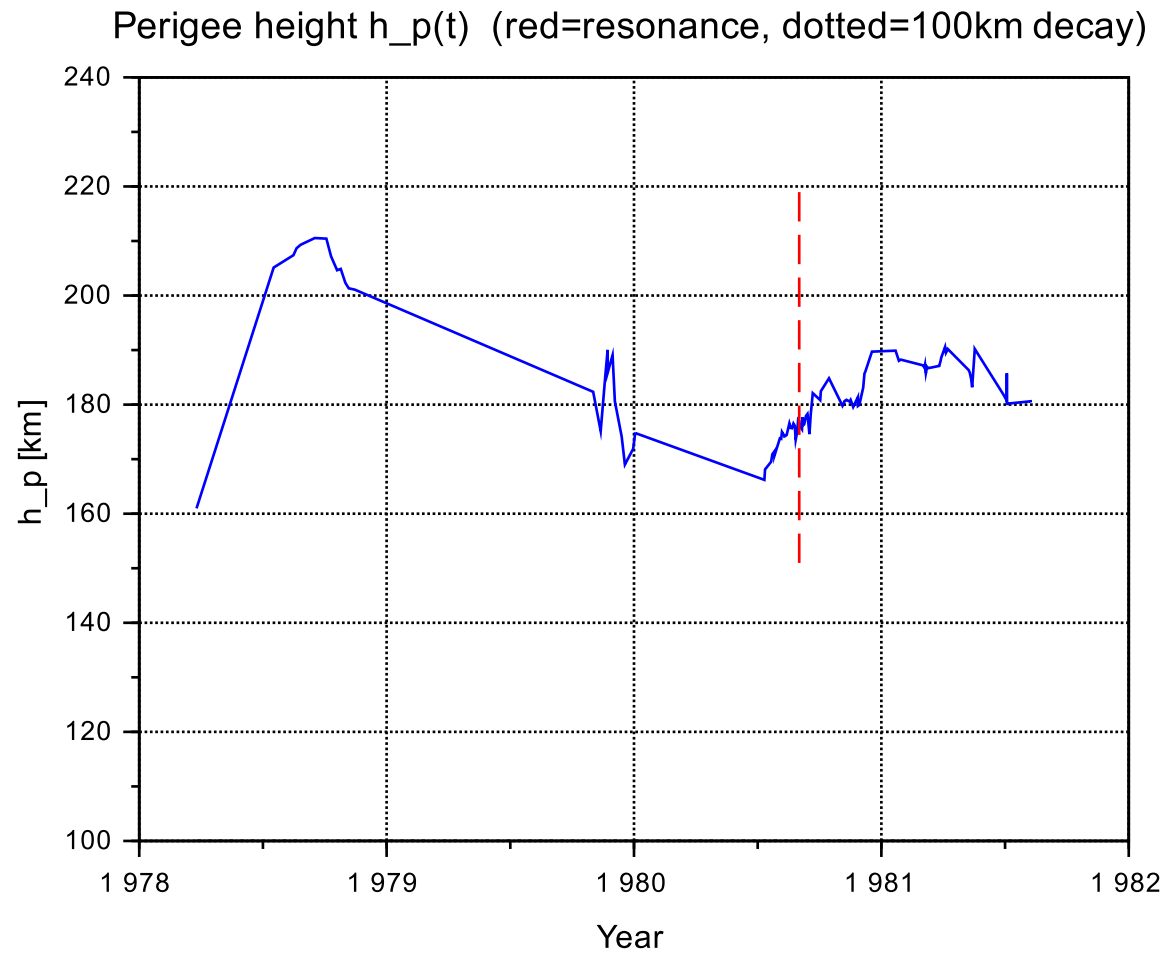
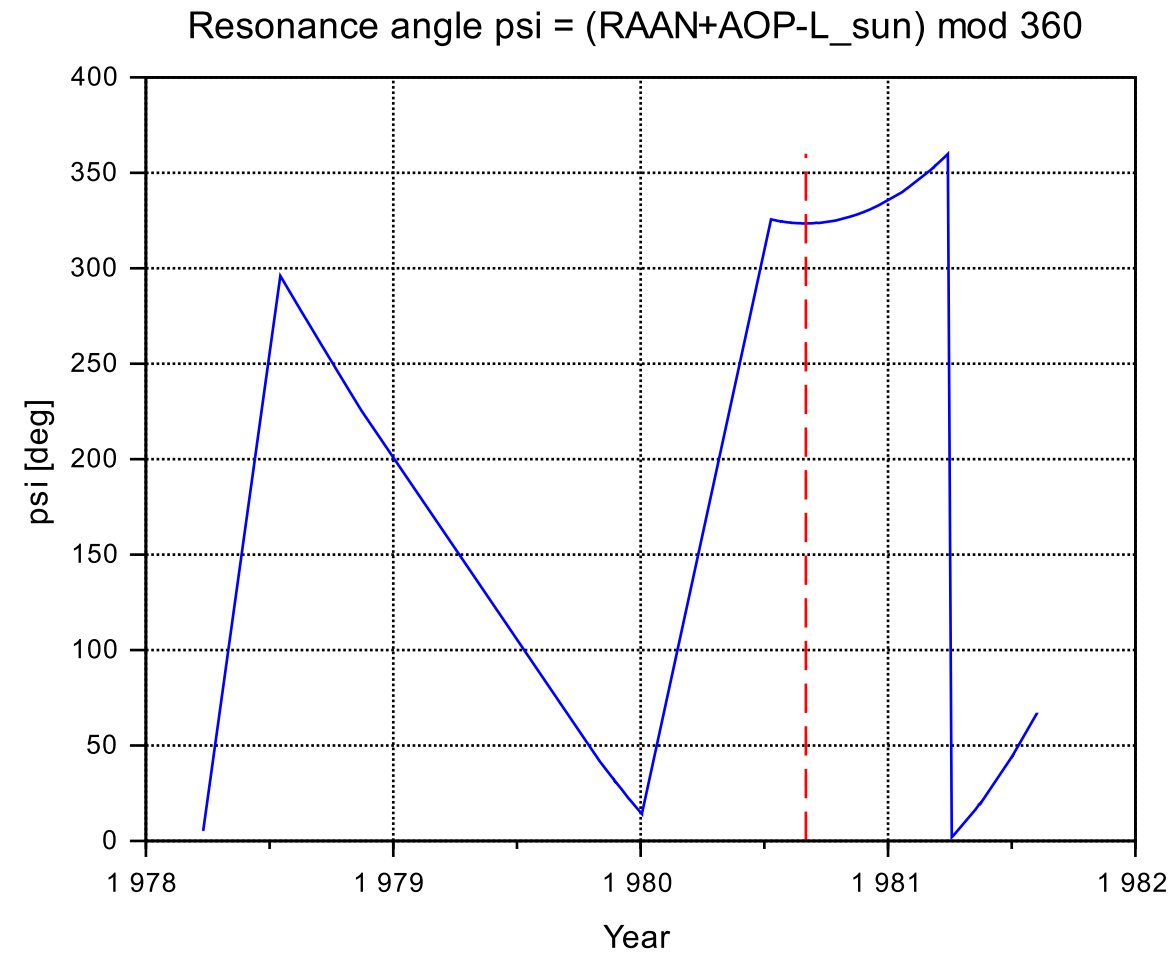
NORAD 19881 Cosmos931 deb
Mean inclination: 59.51 deg
 λ_{crit} (Eq.12): 9252.86
Resonance epoch: year 1993.15
TLE record: 1979.73 to 1993.16
Observed decay: year 1993.1622



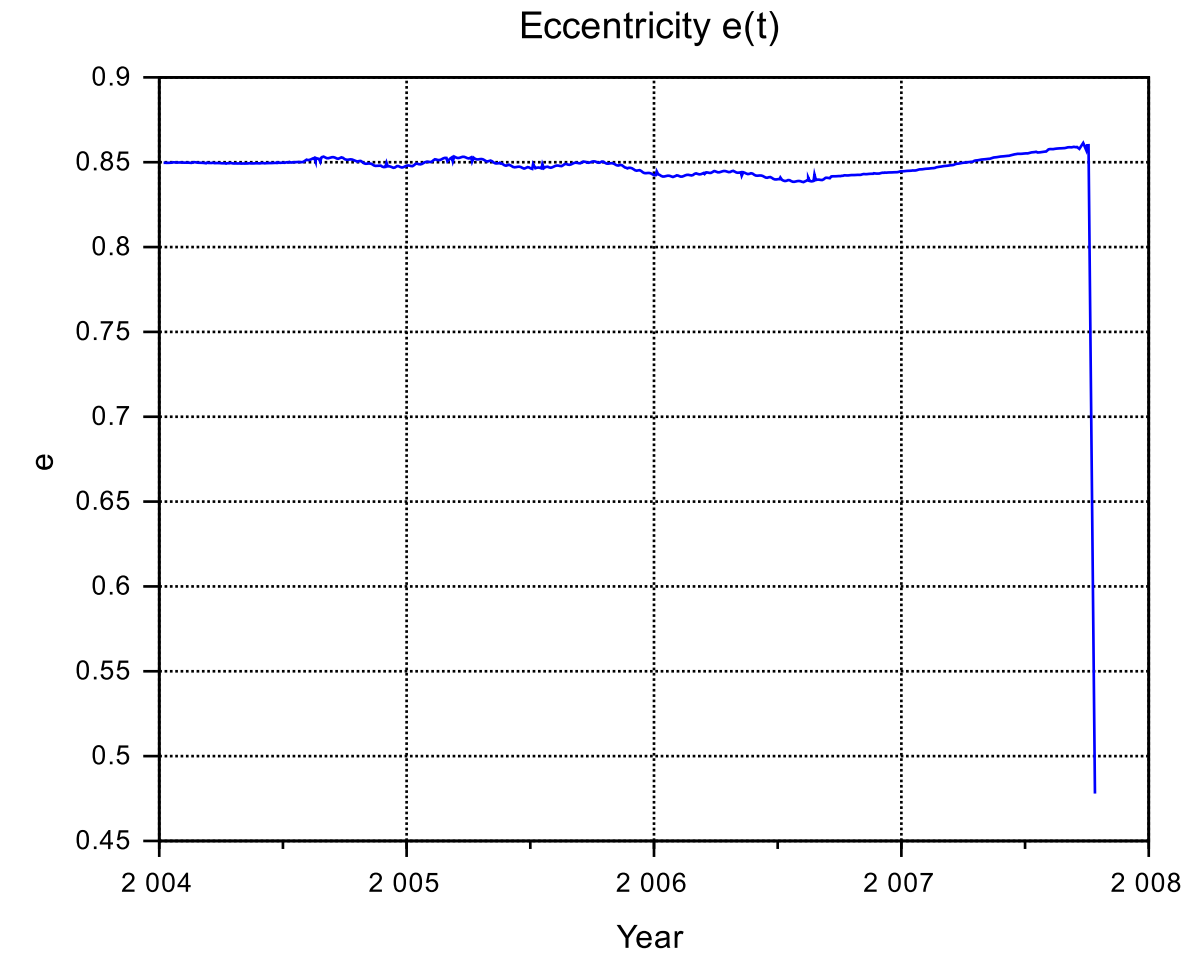
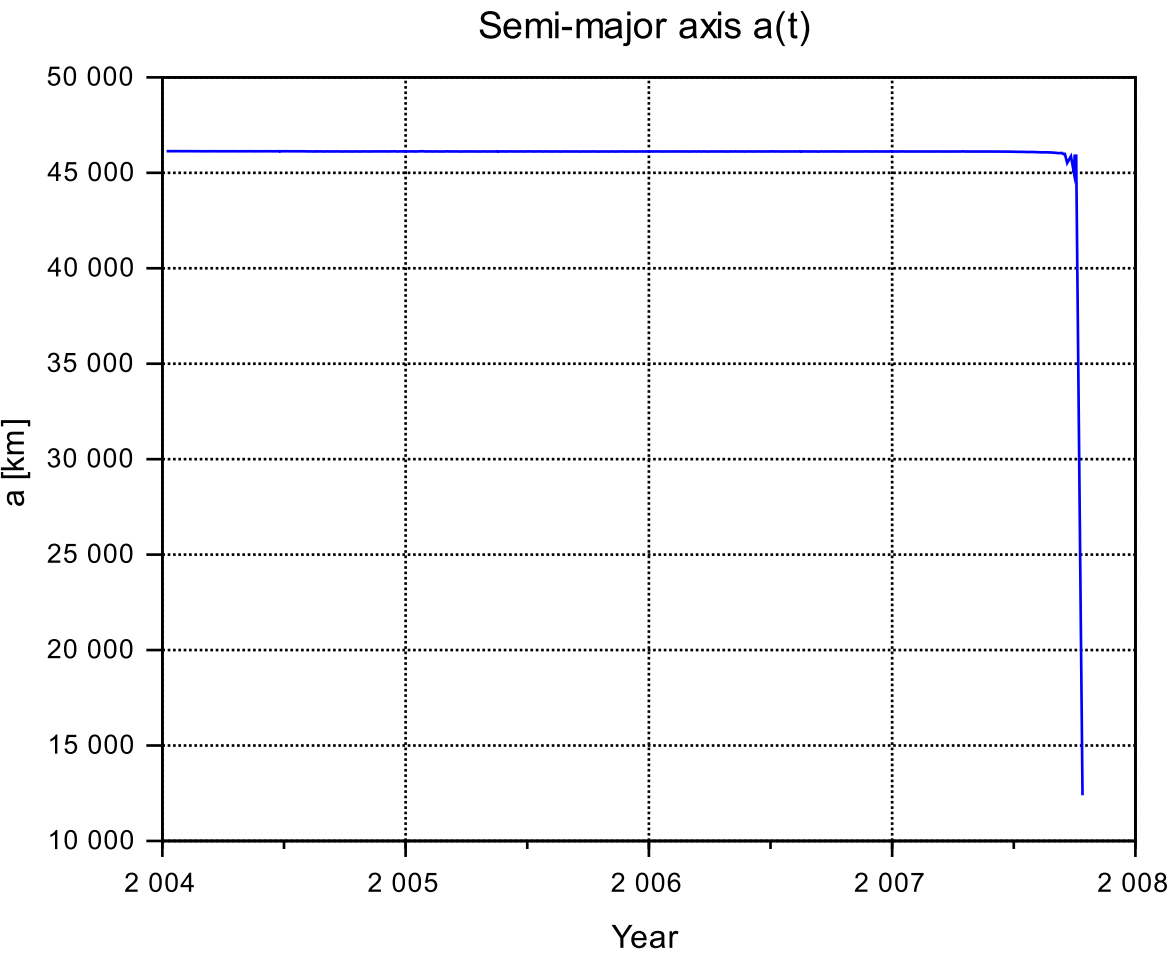
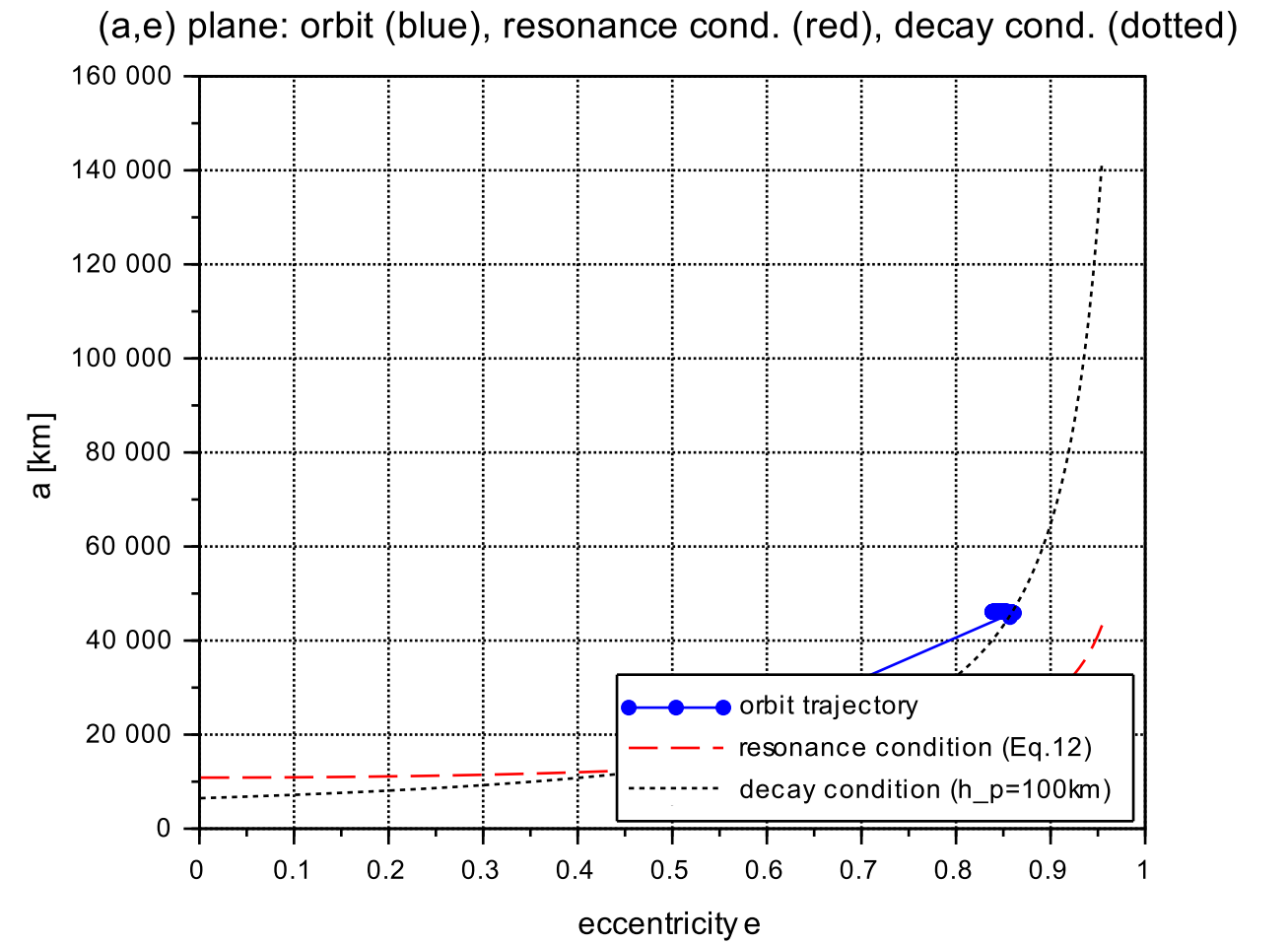
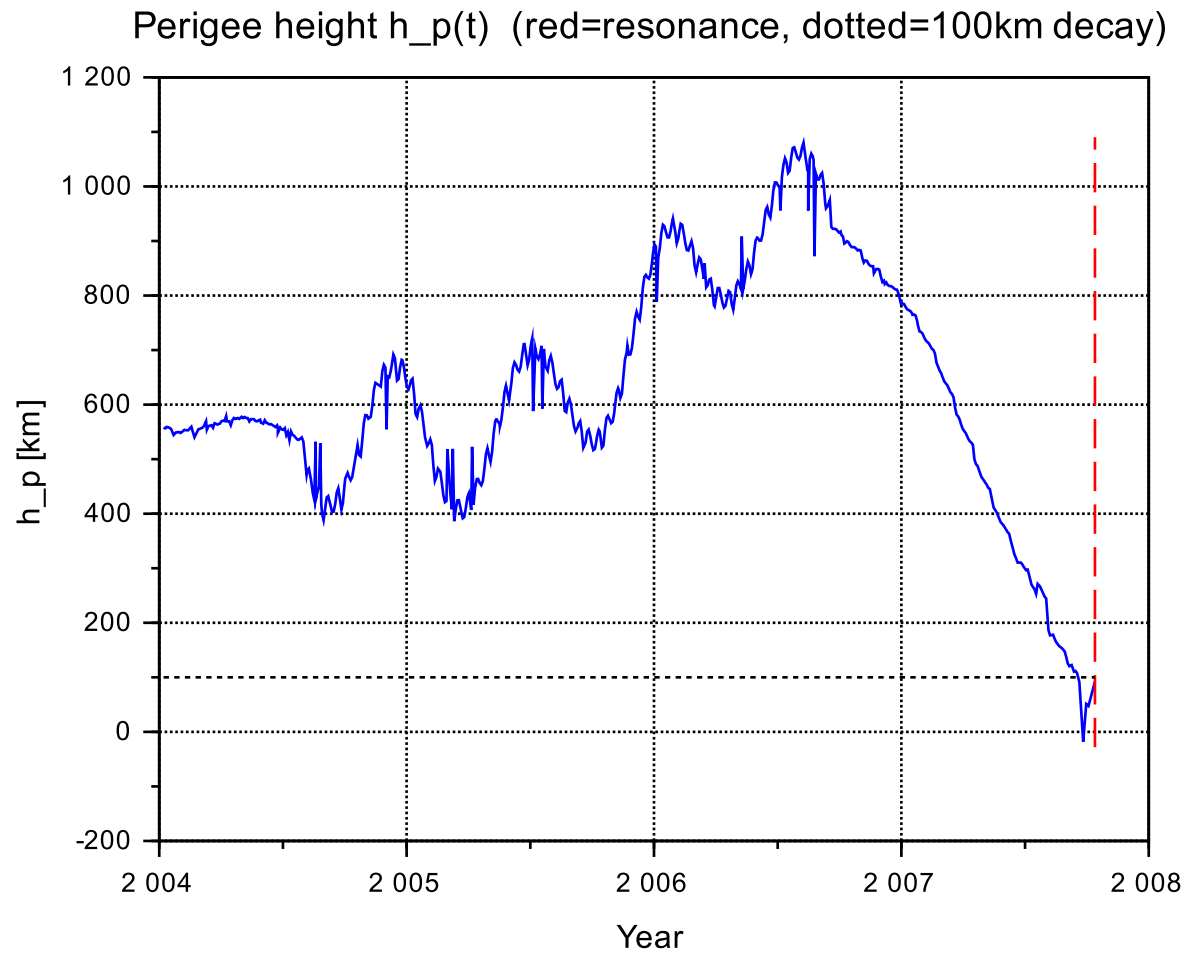
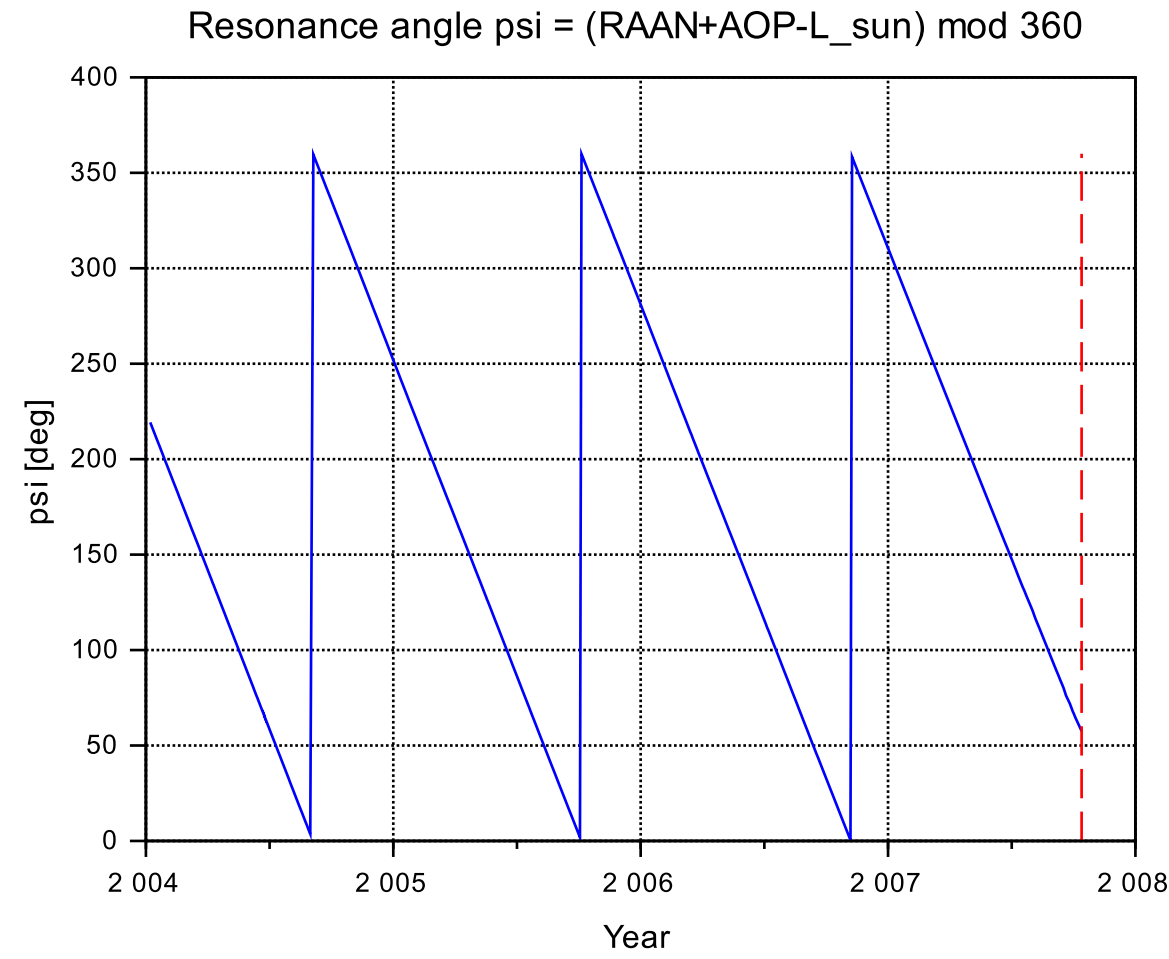
NORAD 5980 Titan3C TS deb
Mean inclination: 26.36 deg
 λ_{crit} (Eq.12): 10731.49
Resonance epoch: year 1975.27
TLE record: 1971.7 to 1975.27
Observed decay: year 1976.2779



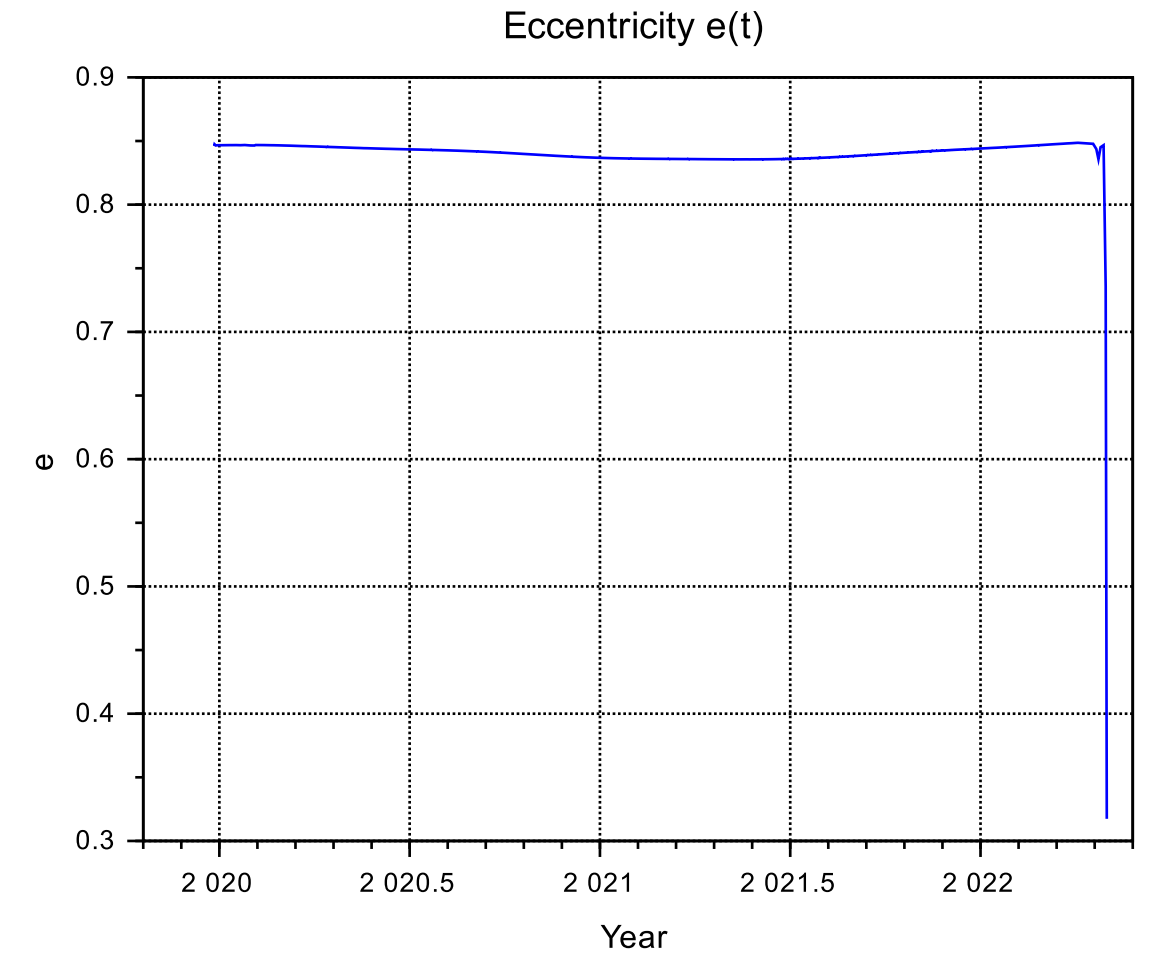
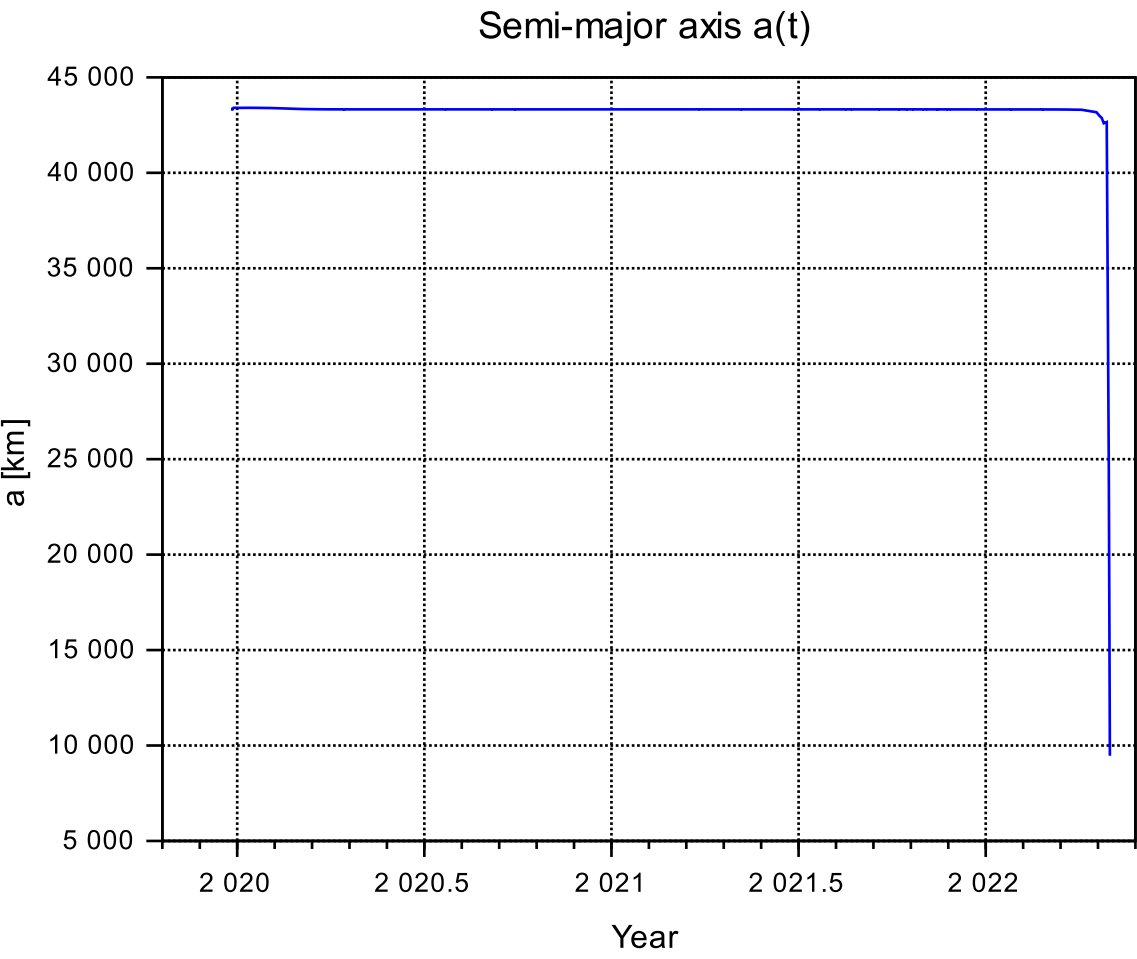
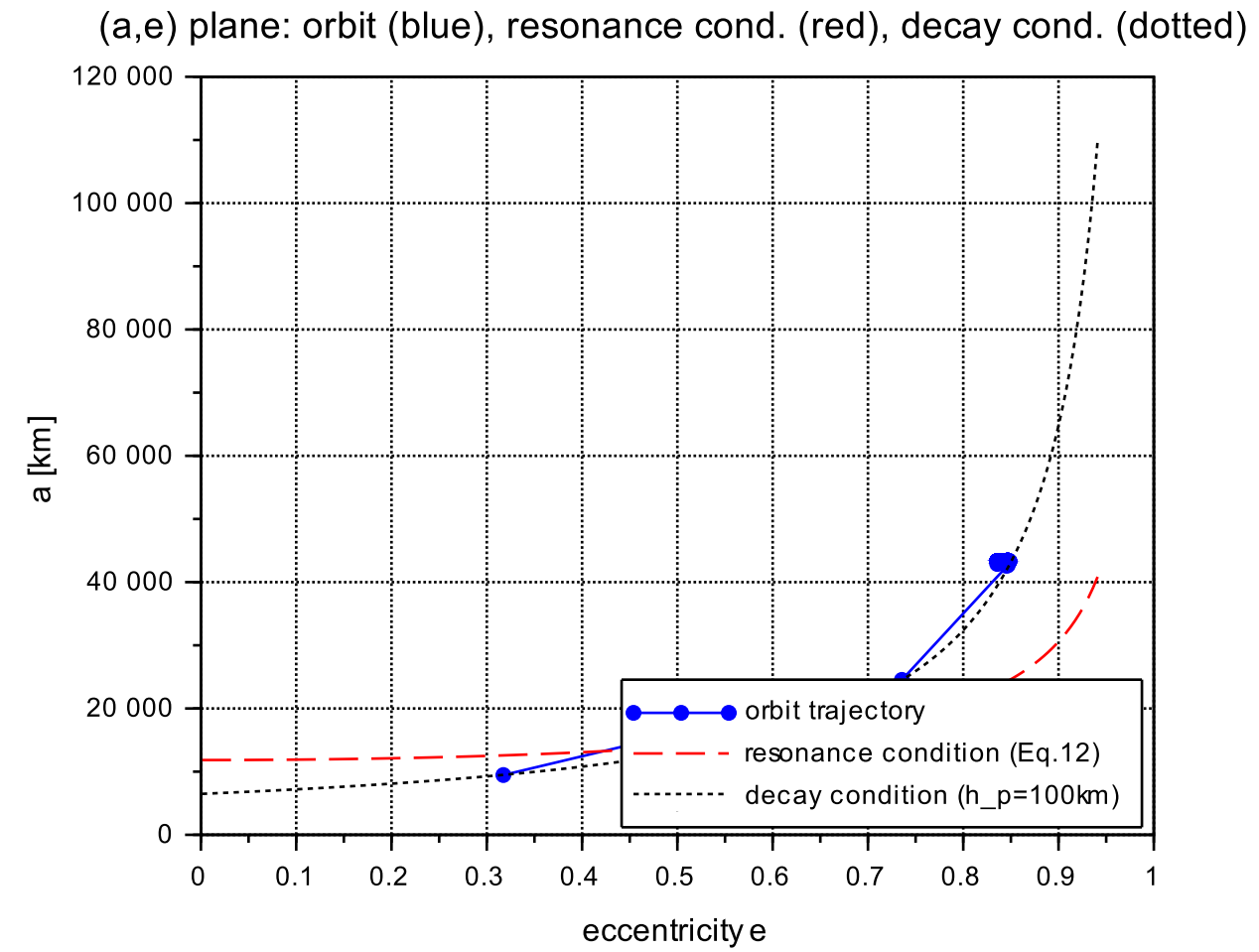
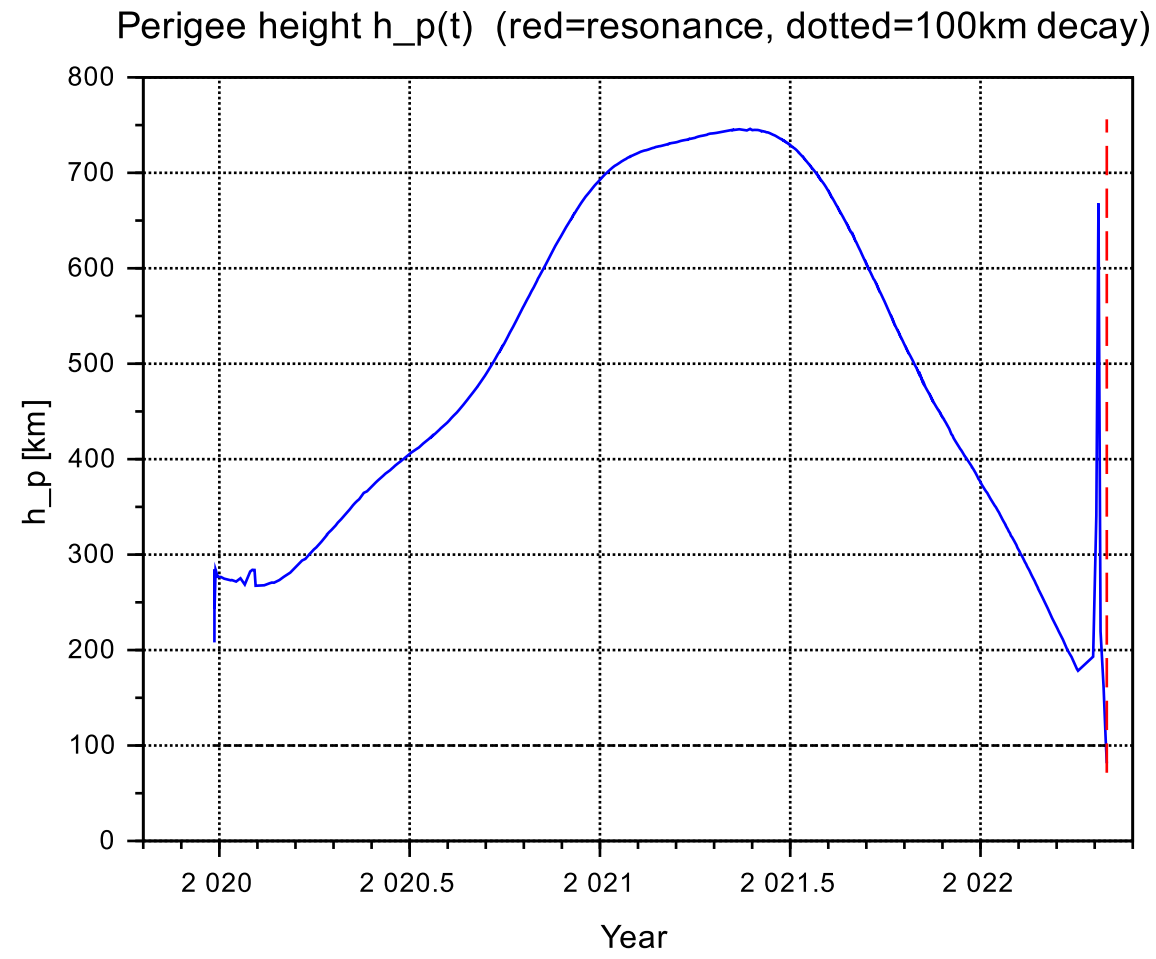
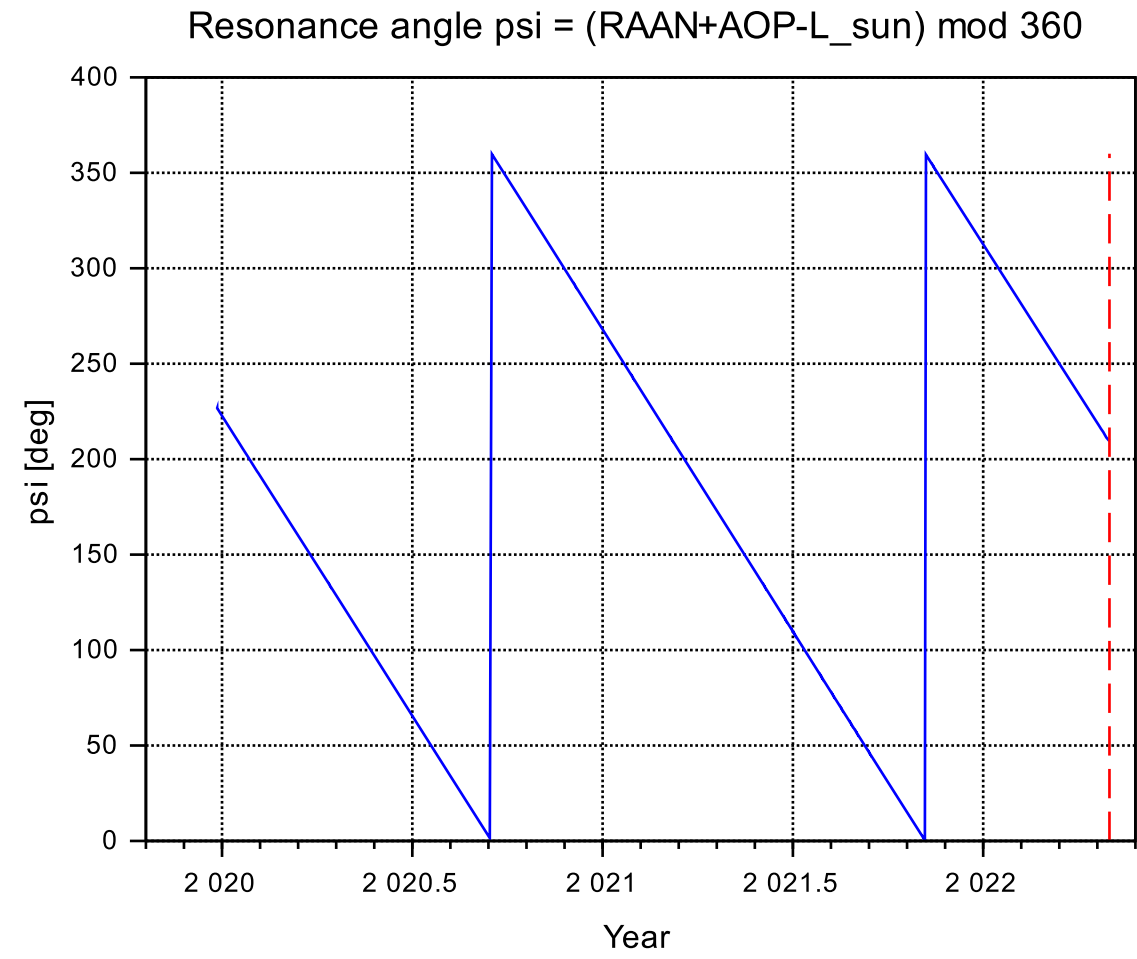
NORAD 4570 SL-6 R/B(2)
Mean inclination: 65.77 deg
 λ_{crit} (Eq.12): 10070.61
Resonance epoch: year 1972.39
TLE record: 1970.8 to 1972.39
Observed decay: year 1972.4120



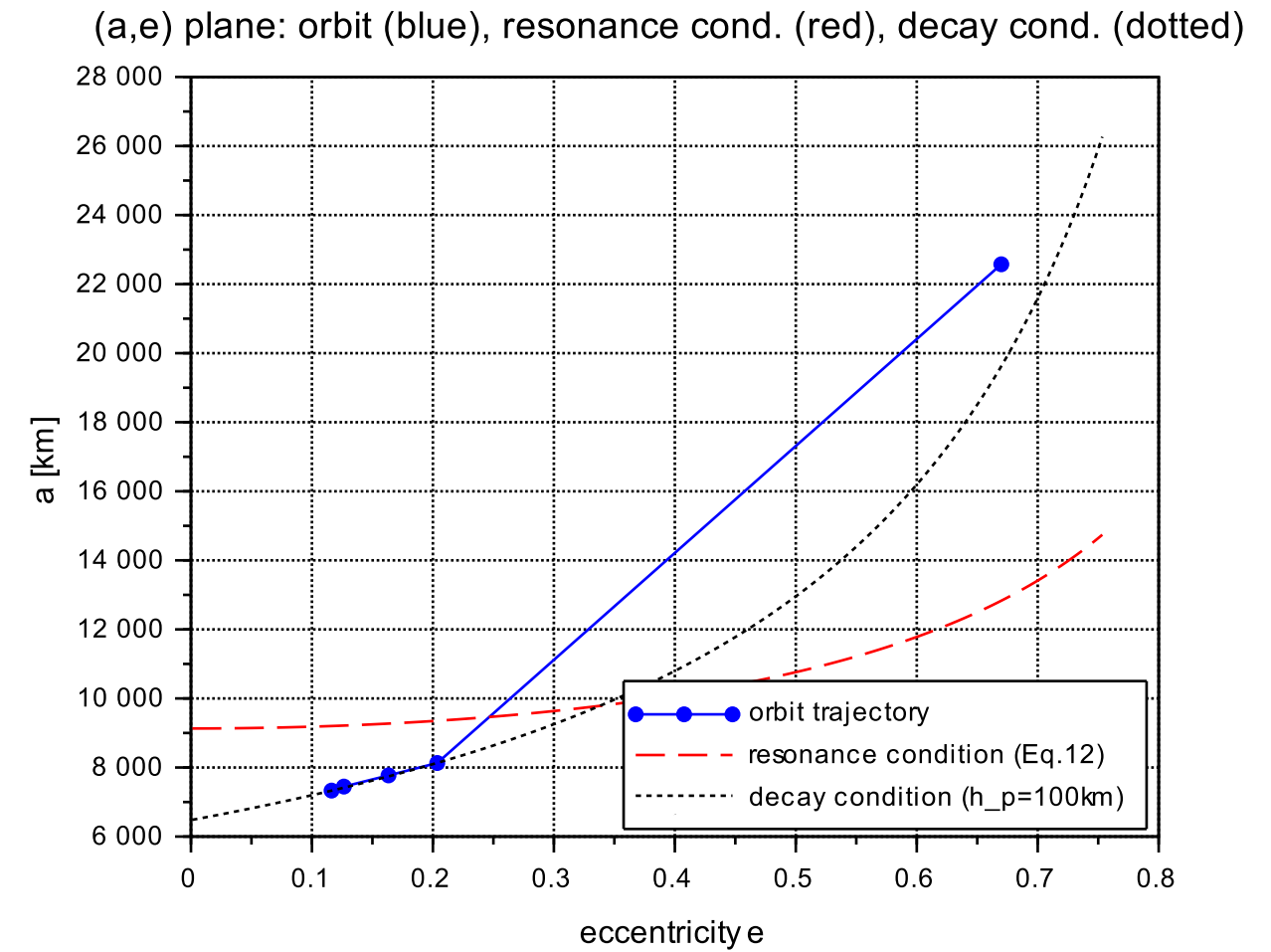
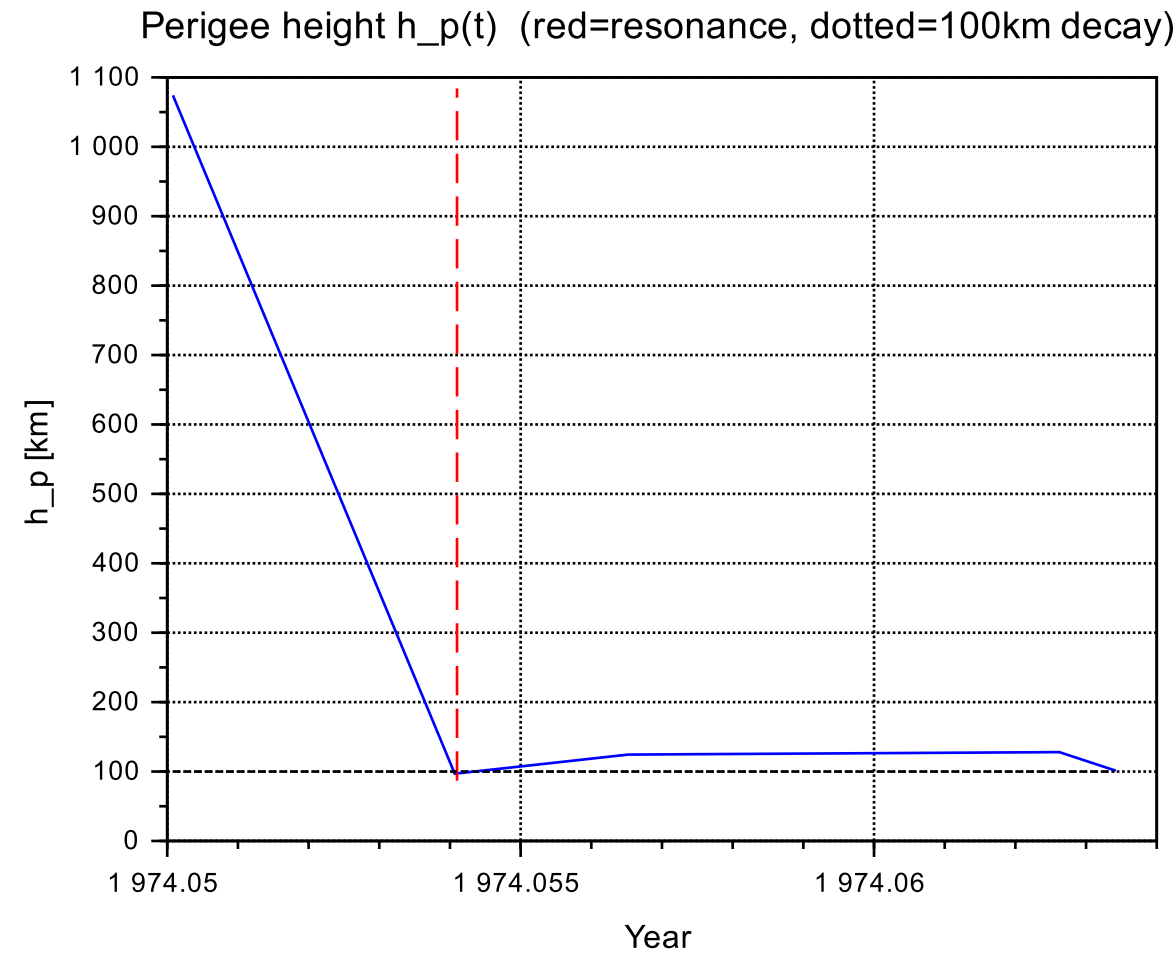
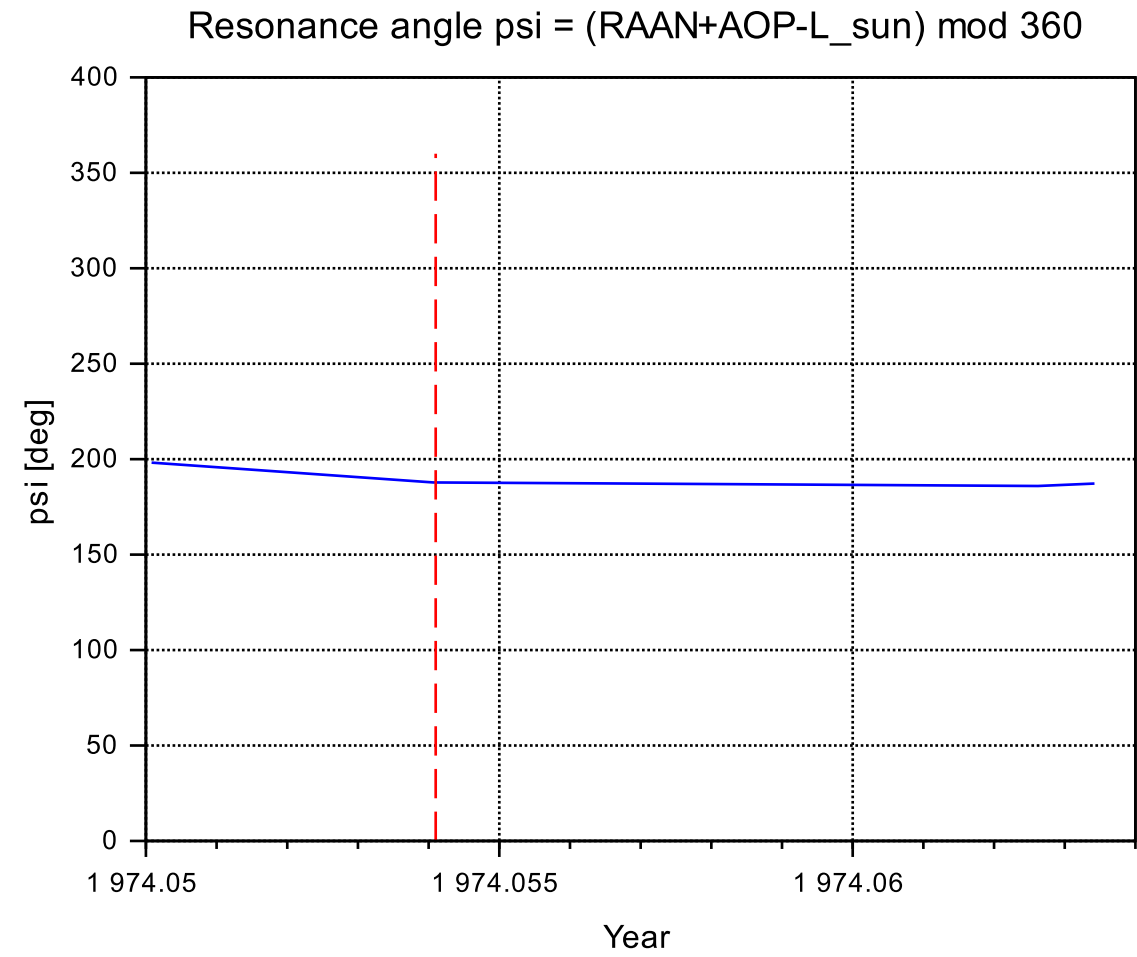
NORAD 10724 AtlasCentaur deb
Mean inclination: 26.74 deg
 λ_{crit} (Eq.12): 10679.56
Resonance epoch: year 1980.67
TLE record: 1978.23 to 1981.6
Observed decay: year 1982.6626



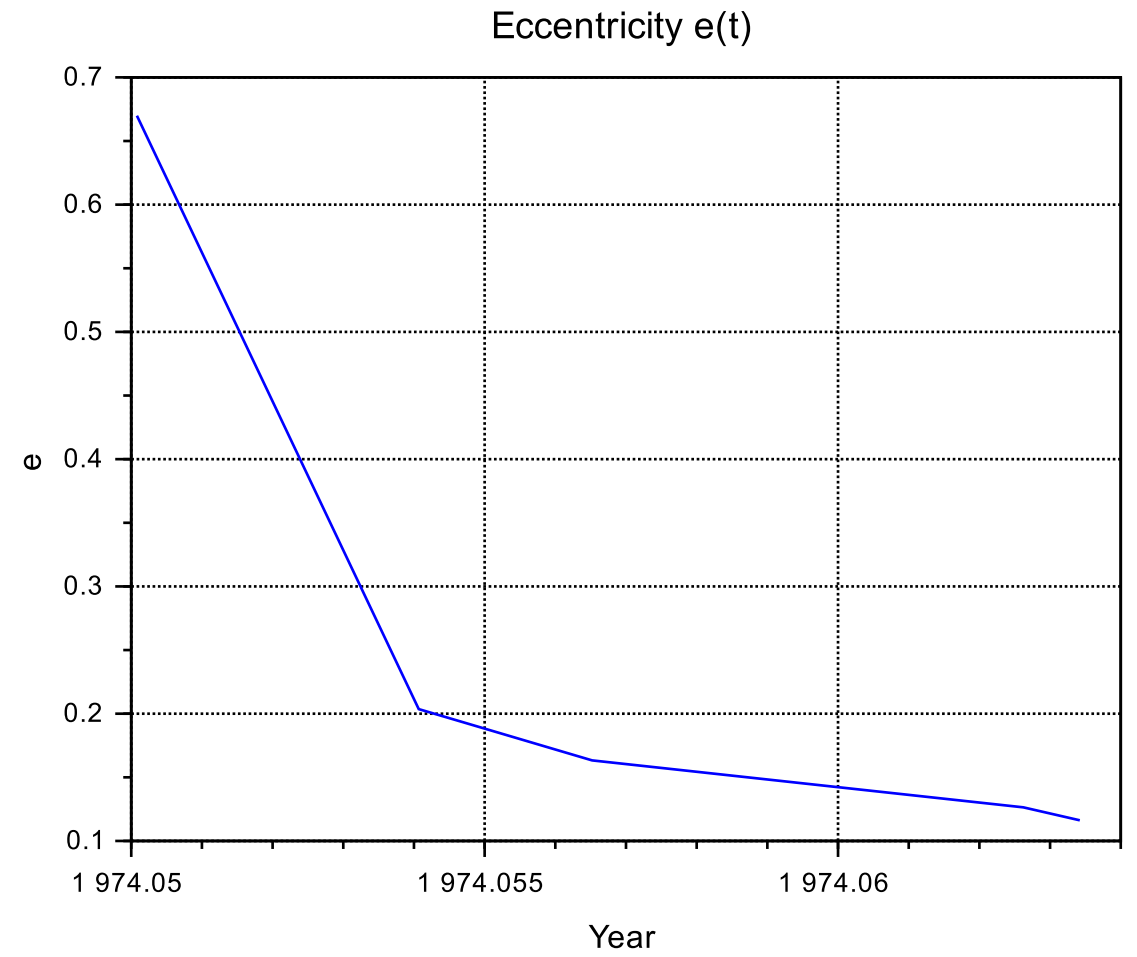
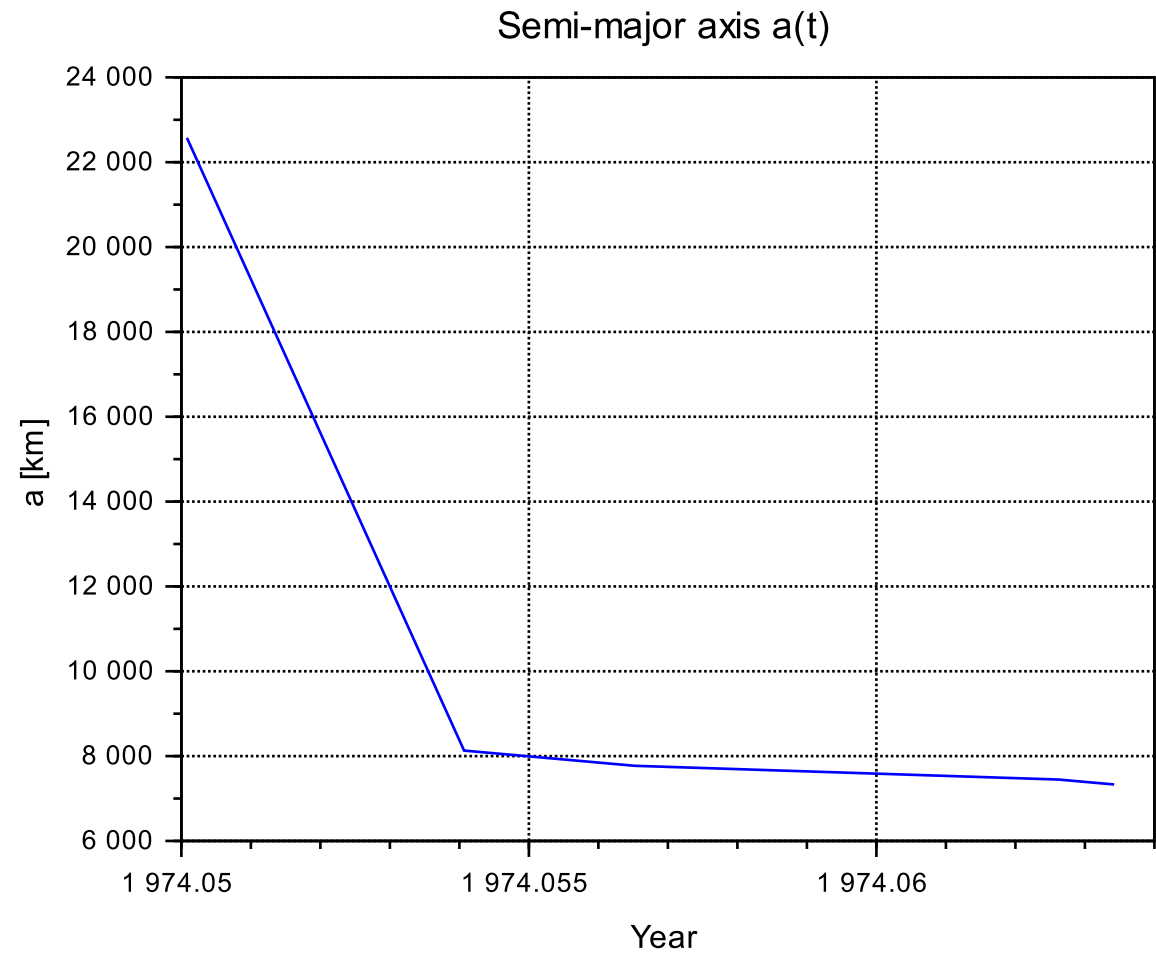
NORAD 28140 Doublestar TC-1
Mean inclination: 25.37 deg
 λ_{crit} (Eq.12): 10861.43
Resonance epoch: year 2007.78
TLE record: 2004.02 to 2007.78
Observed decay: year 2007.7823

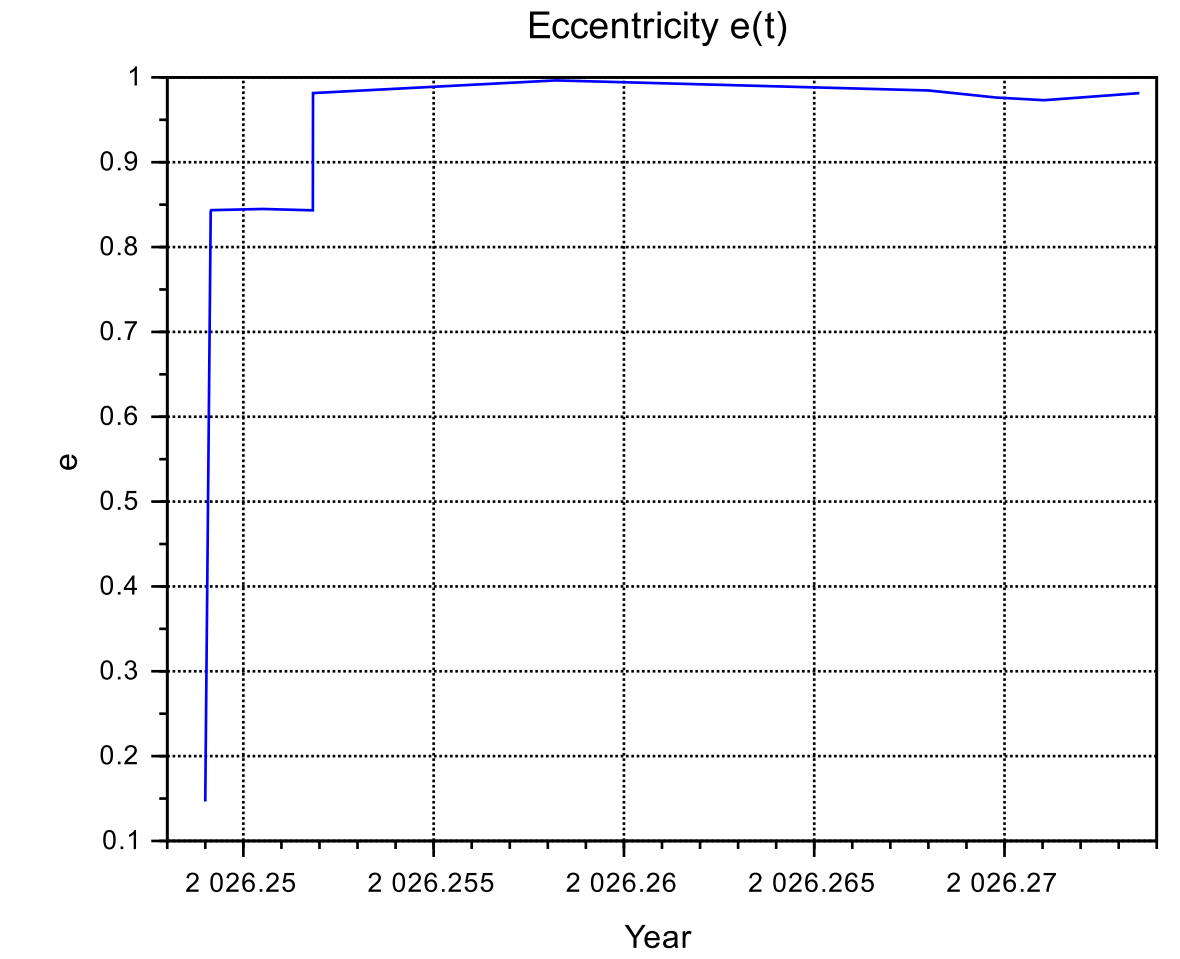
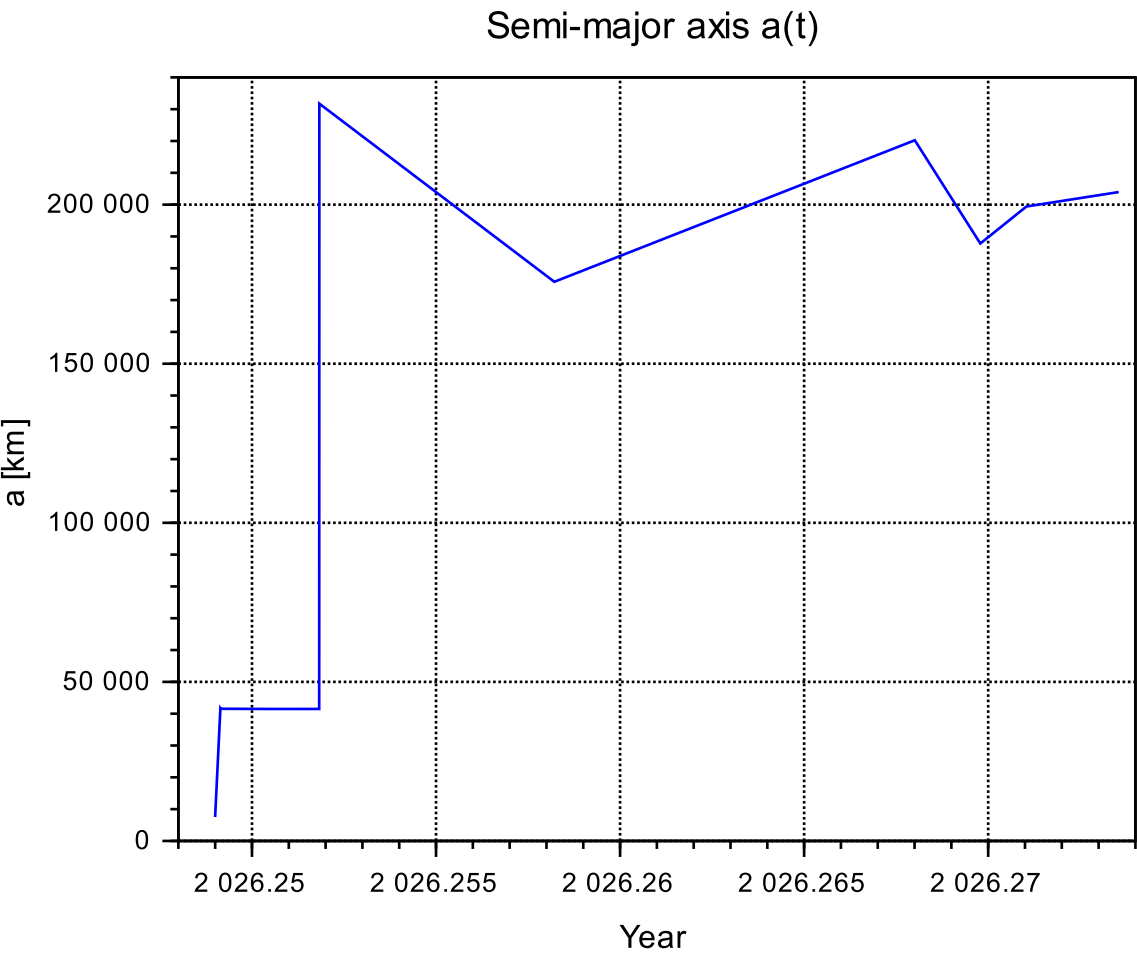
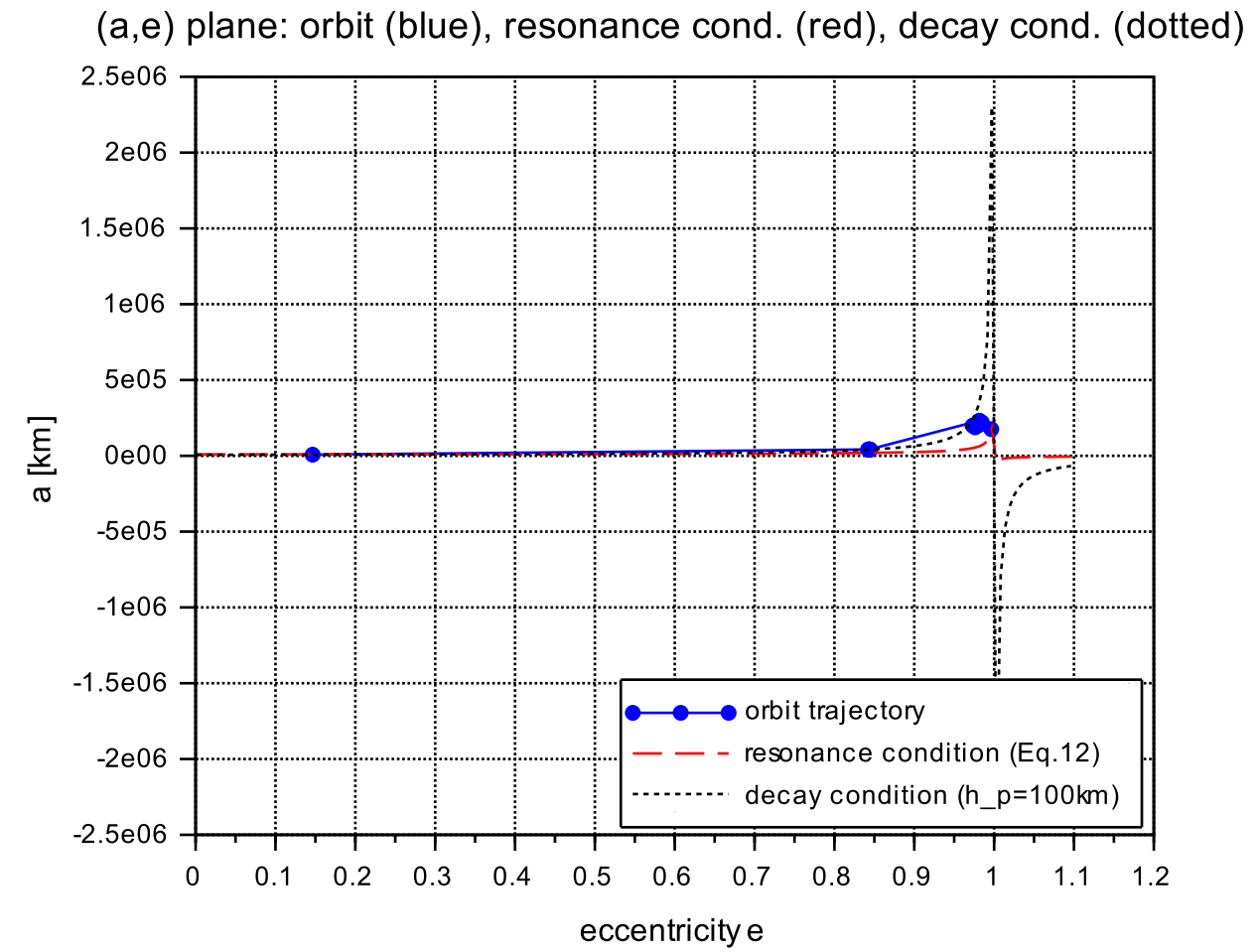
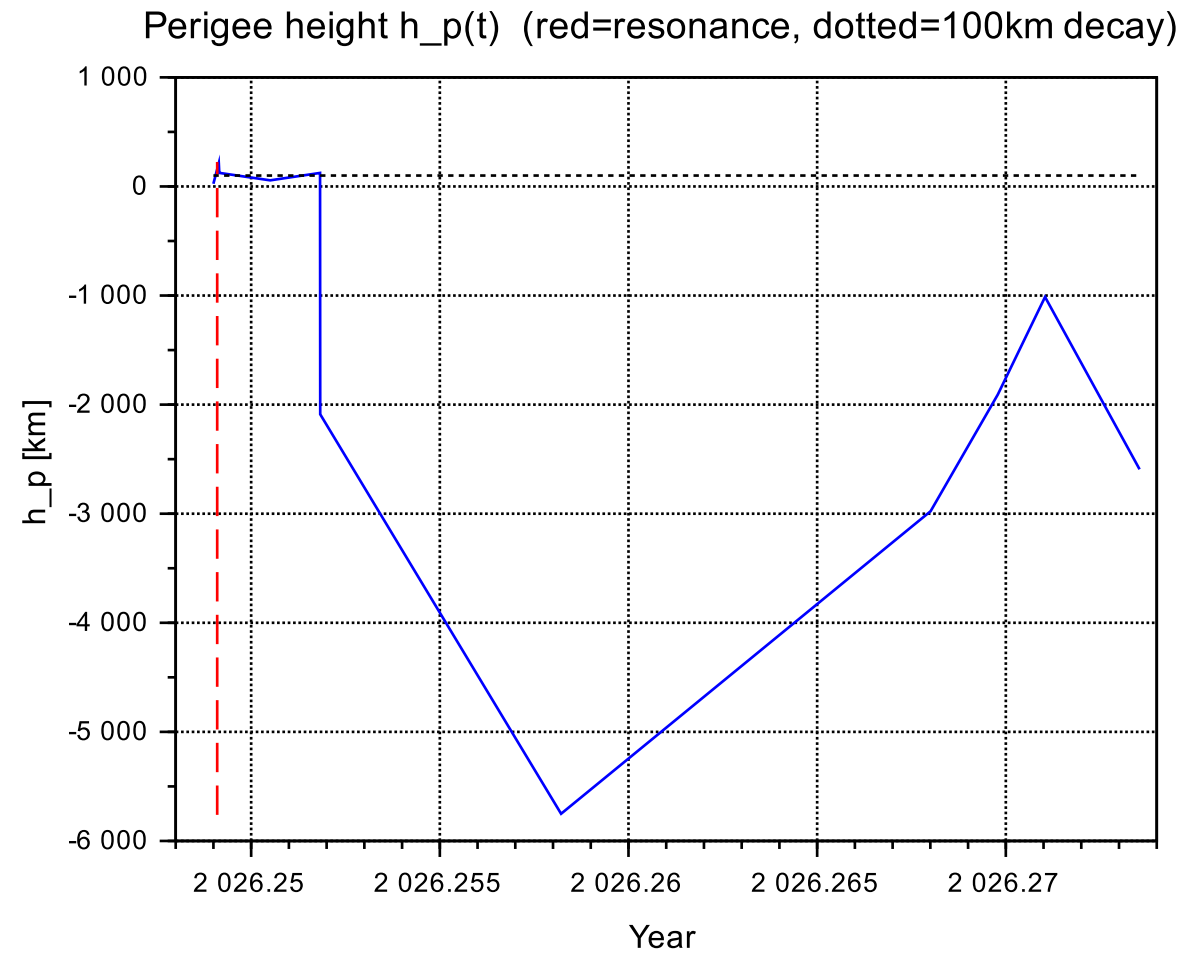
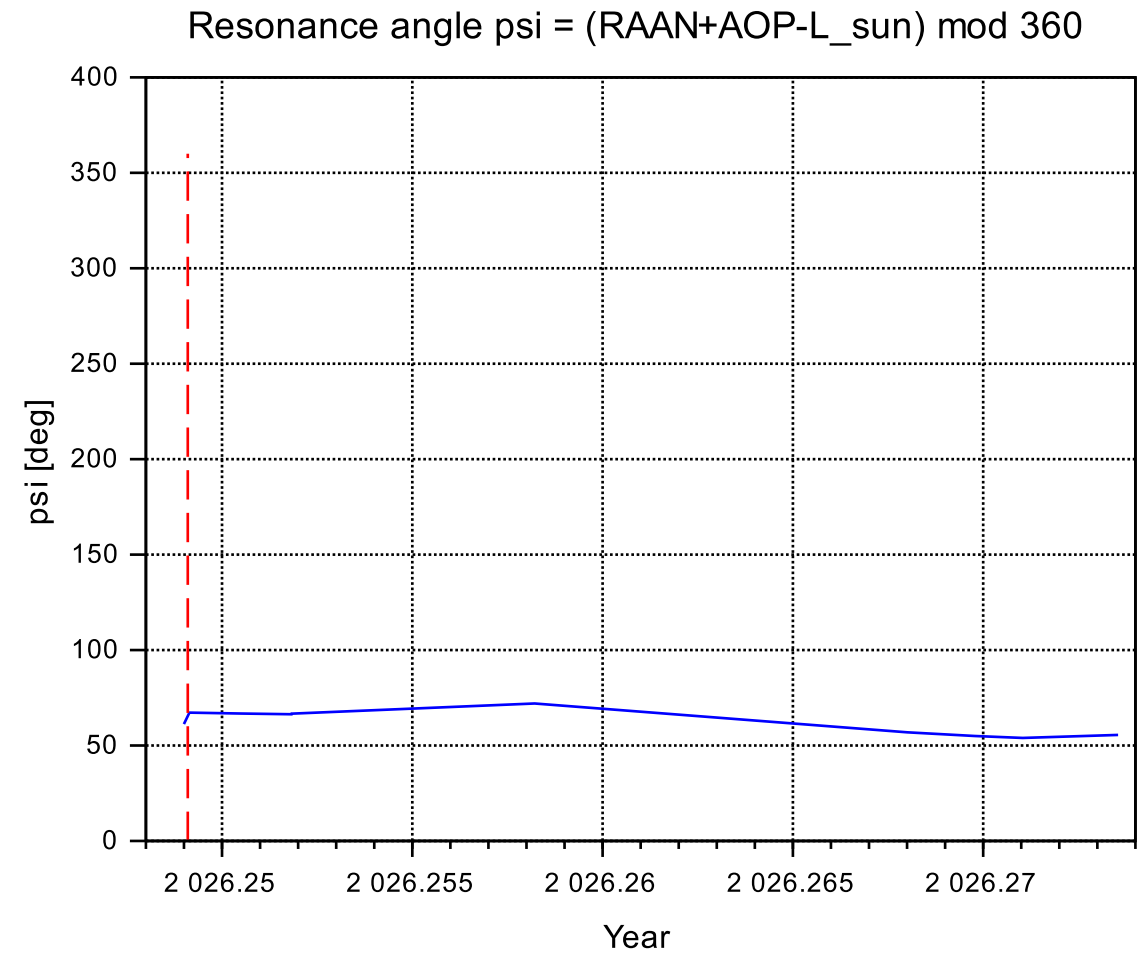


NORAD 44911 CZ-5 R/B
Mean inclination: 15.31 deg
 λ_{crit} (Eq.12): 11835.94
Resonance epoch: year 2022.33
TLE record: 2019.99 to 2022.33
Observed decay: year 2022.3368

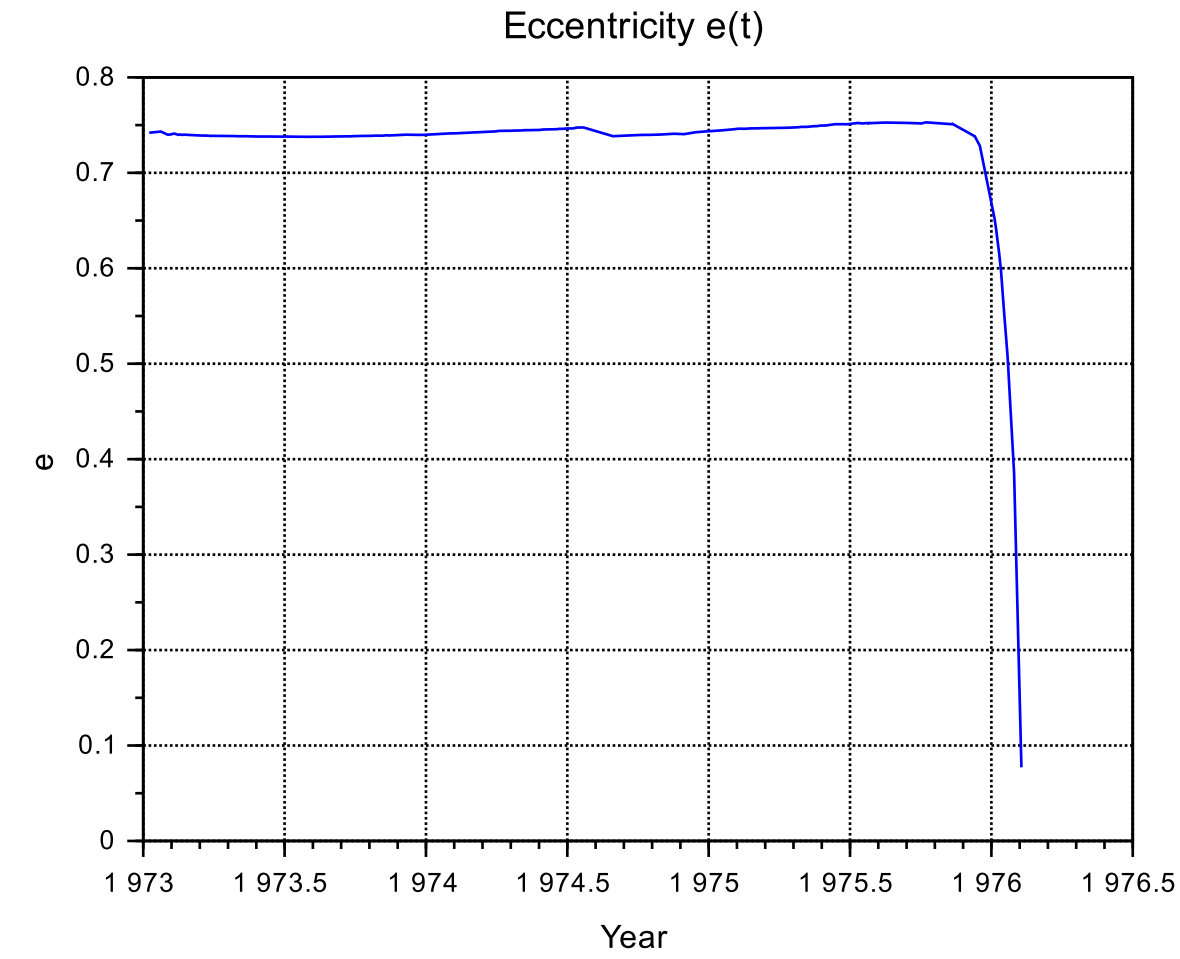
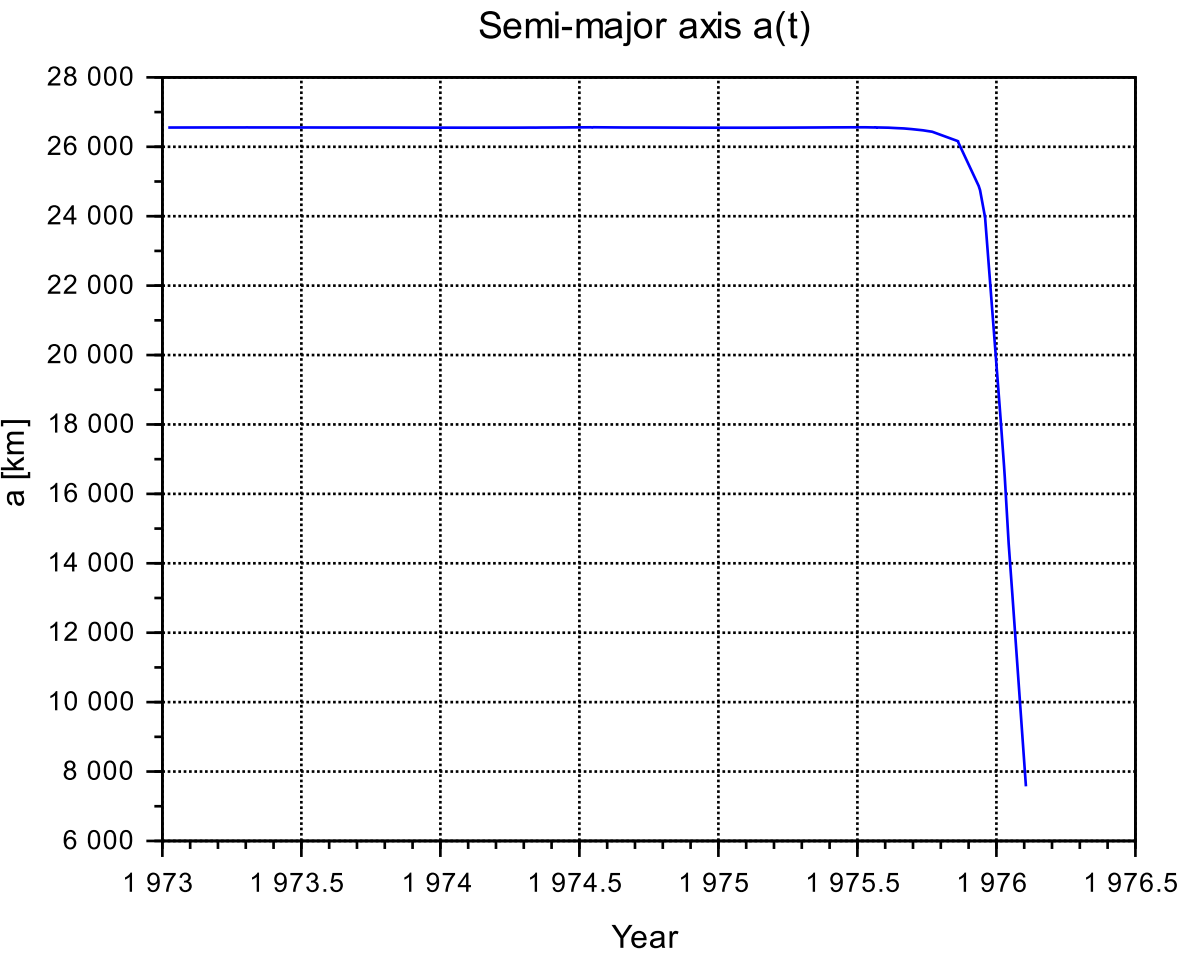
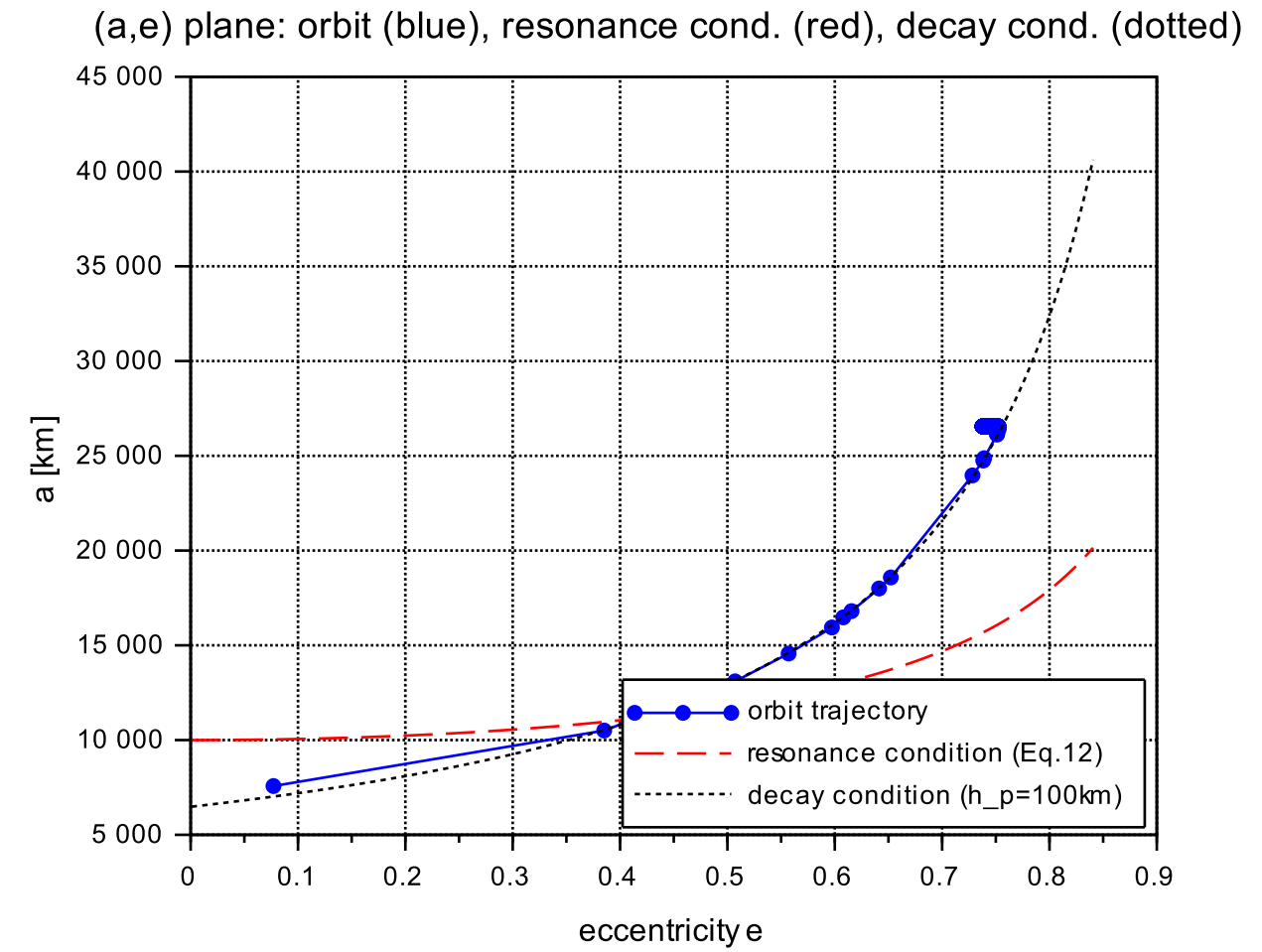
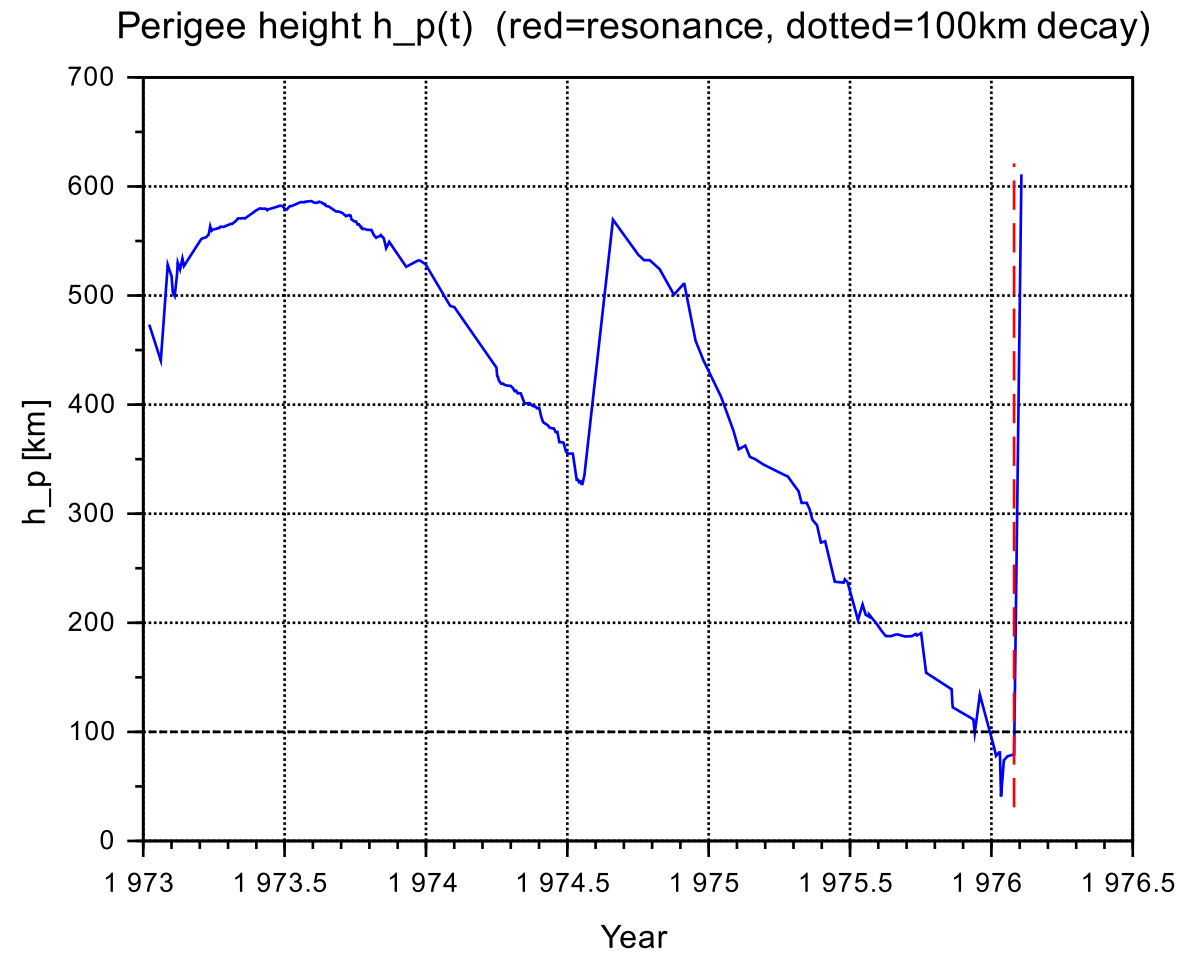
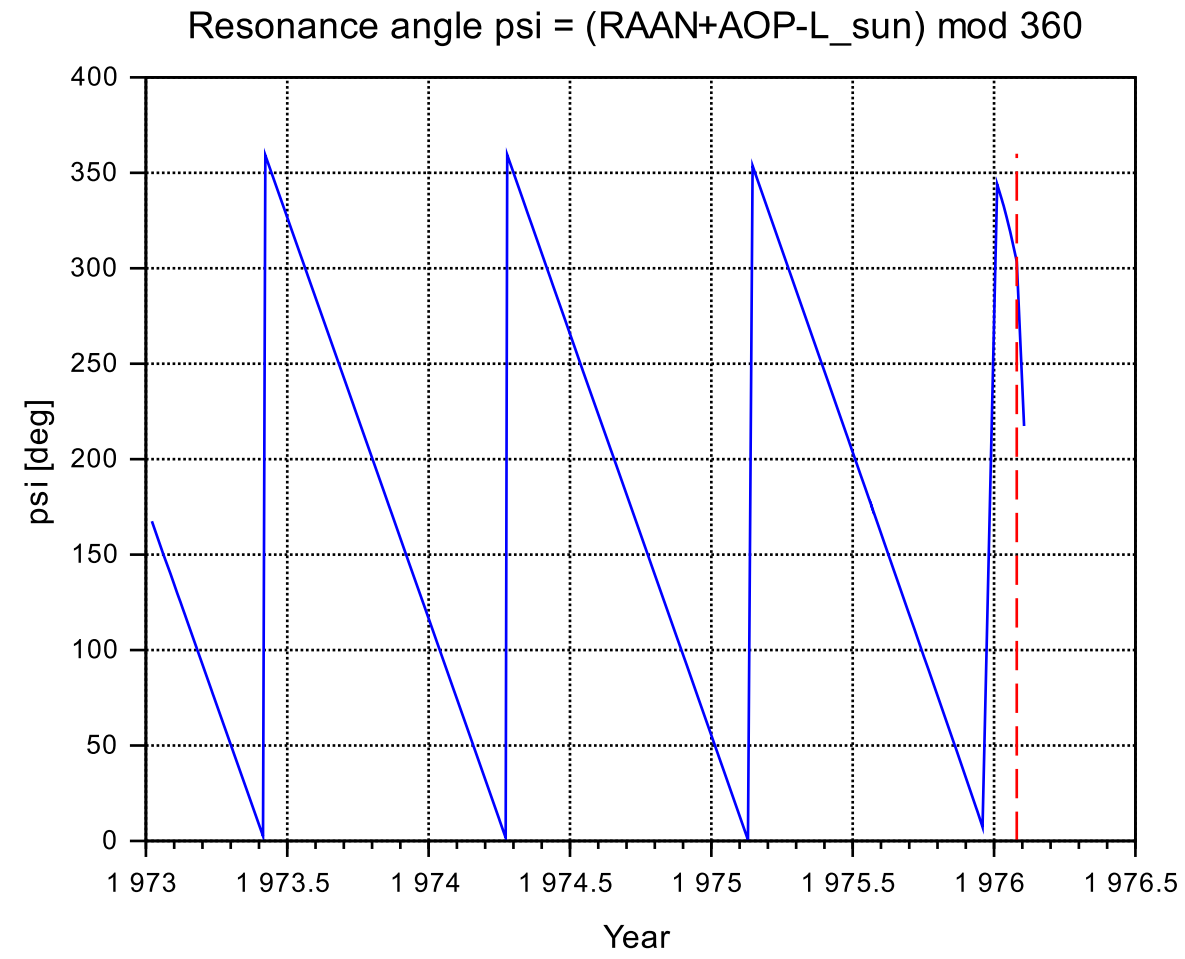


NORAD 7096 SKYNET 2A
Mean inclination: 35.36 deg
 λ_{crit} (Eq.12): 9130.11
Resonance epoch: year 1974.05
TLE record: 1974.05 to 1974.06
Observed decay: year 1974.0657

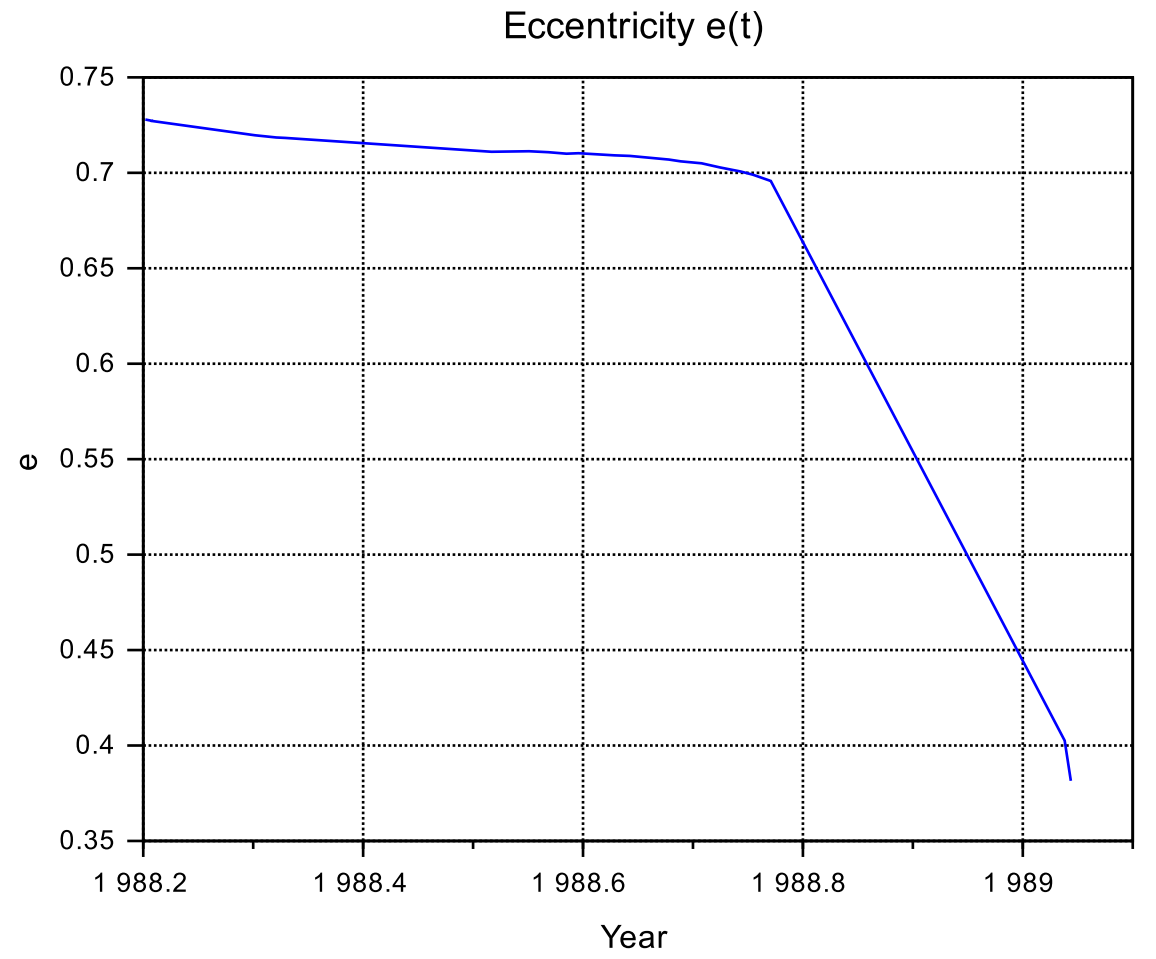
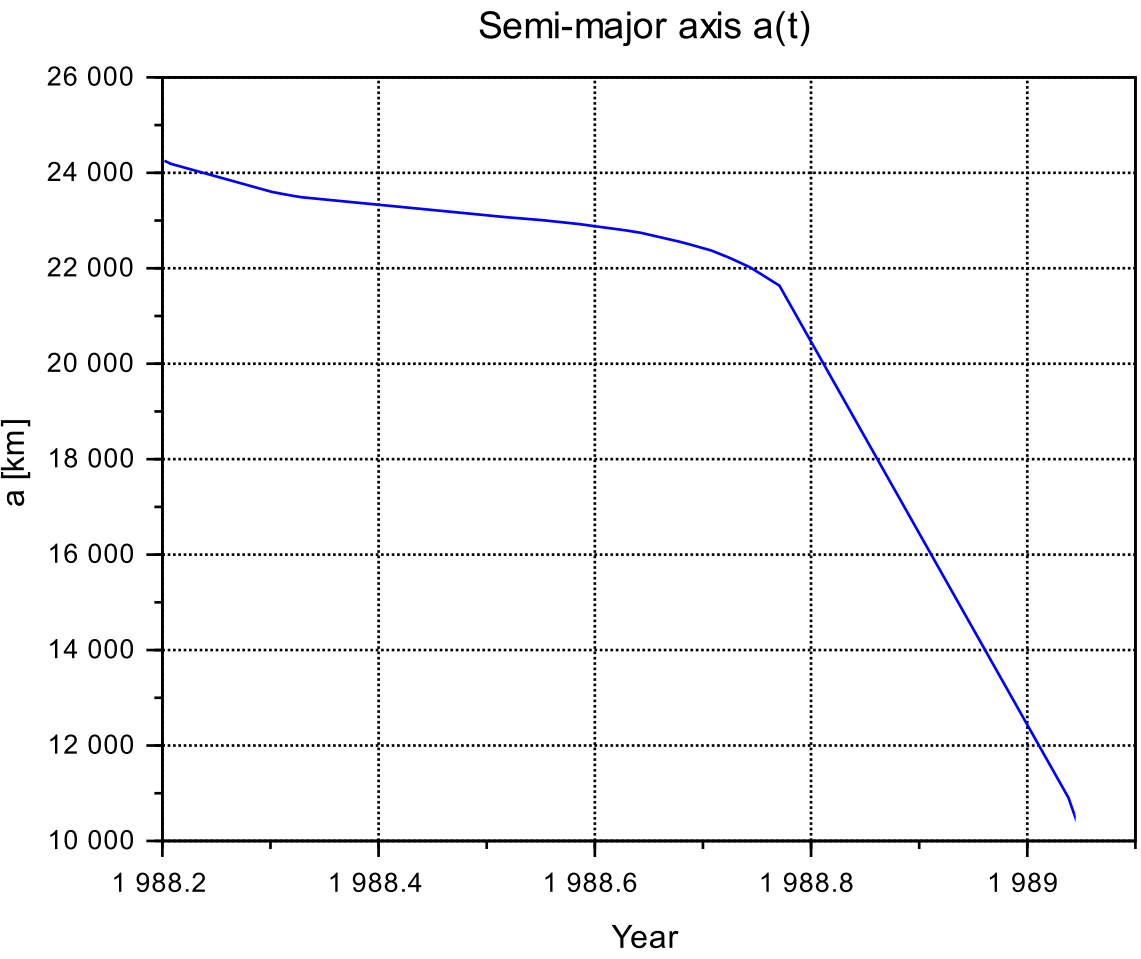
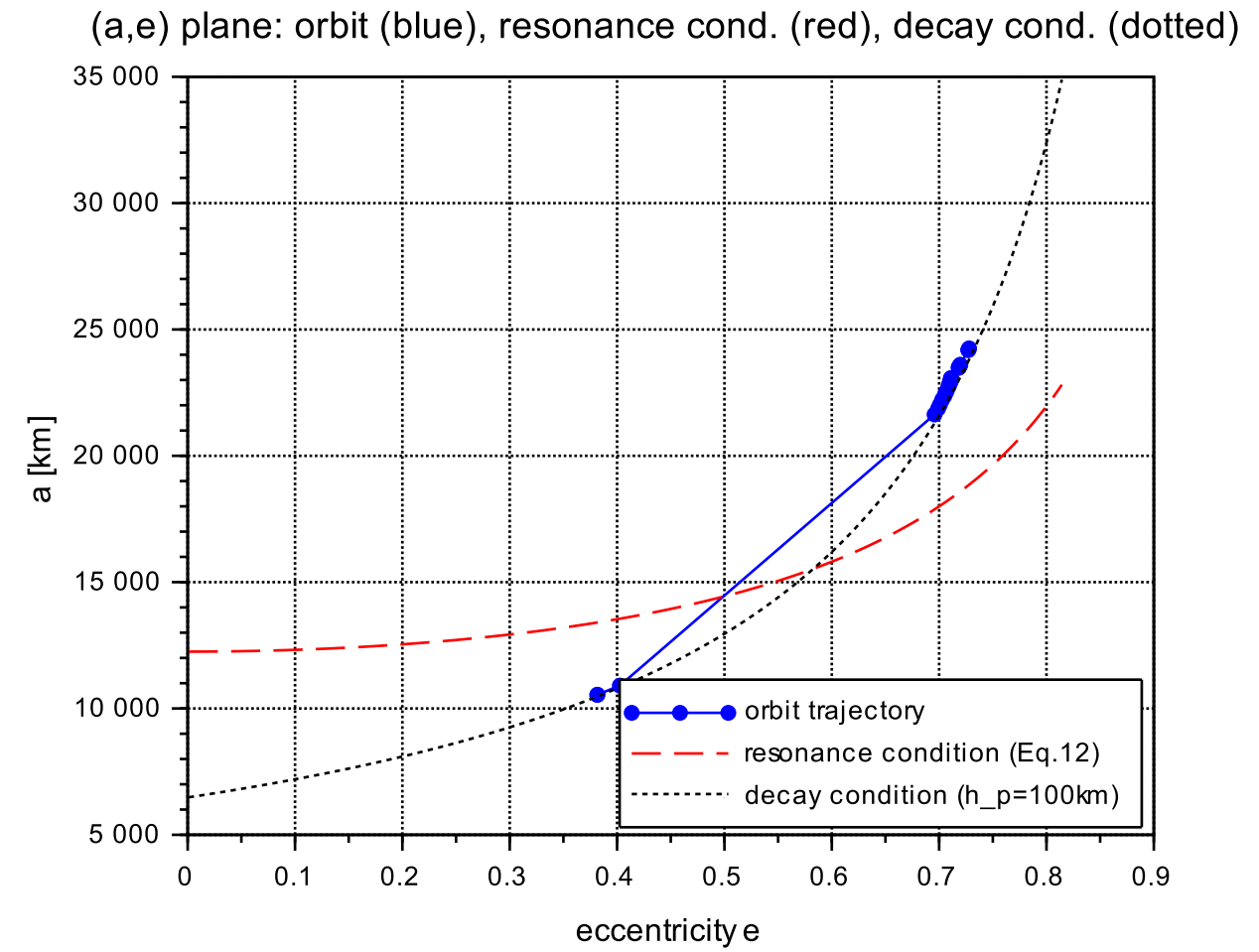
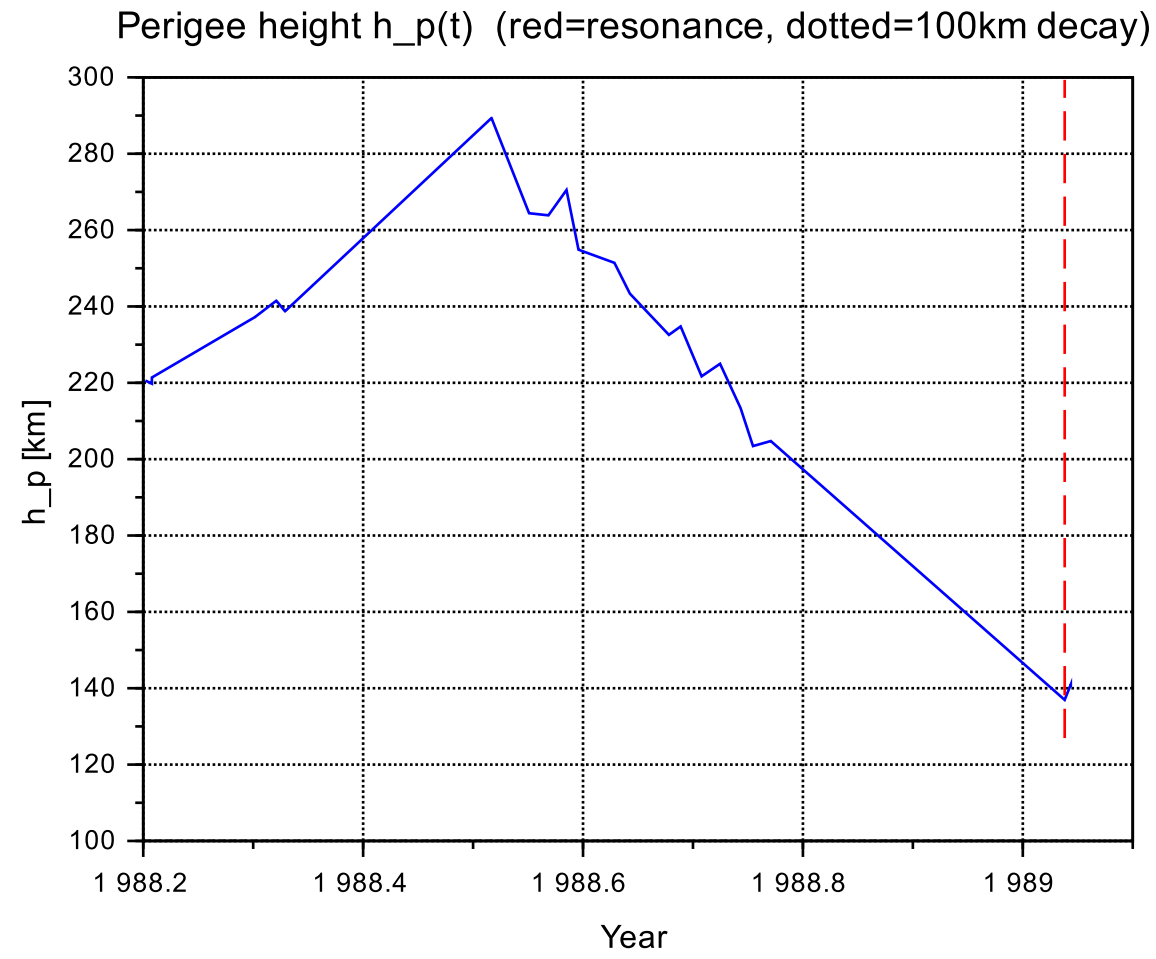
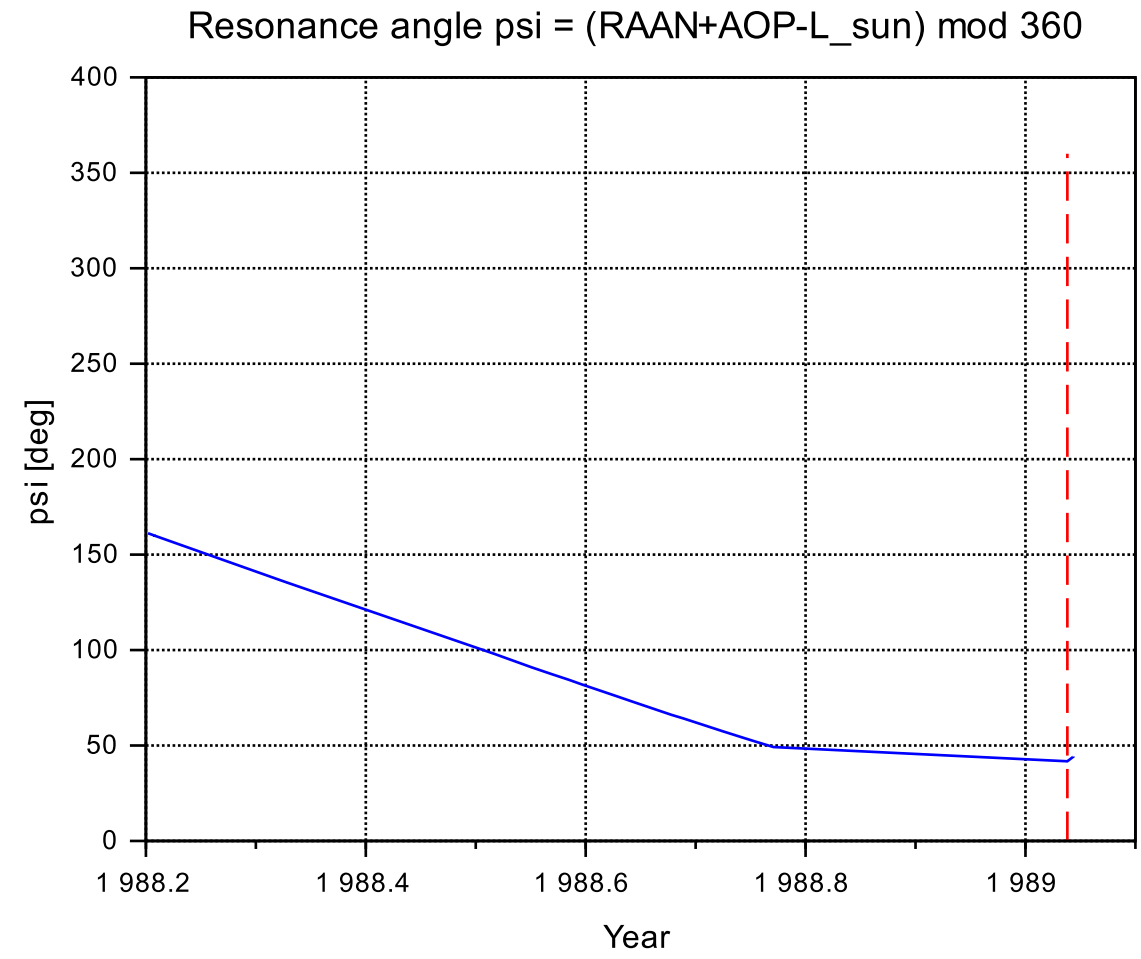




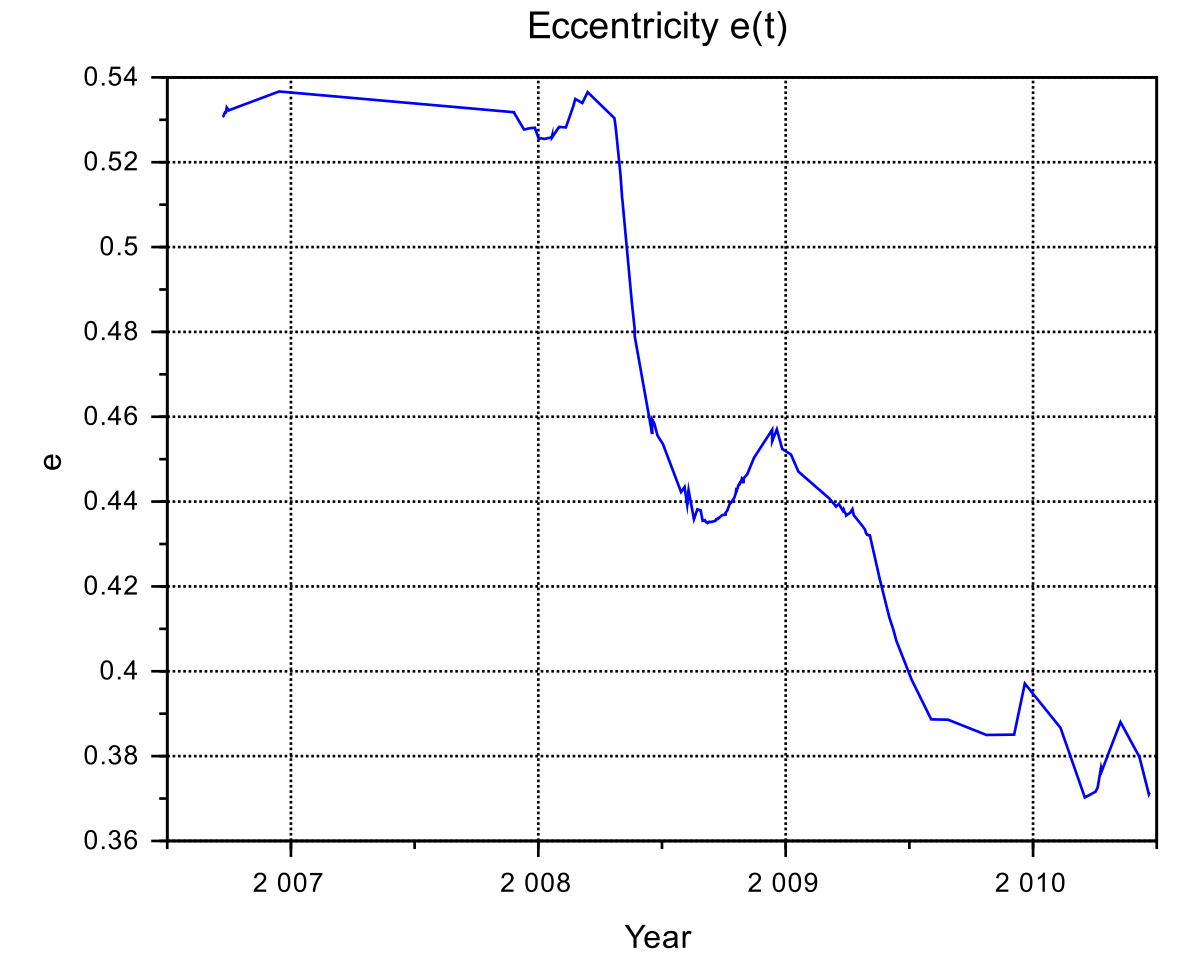
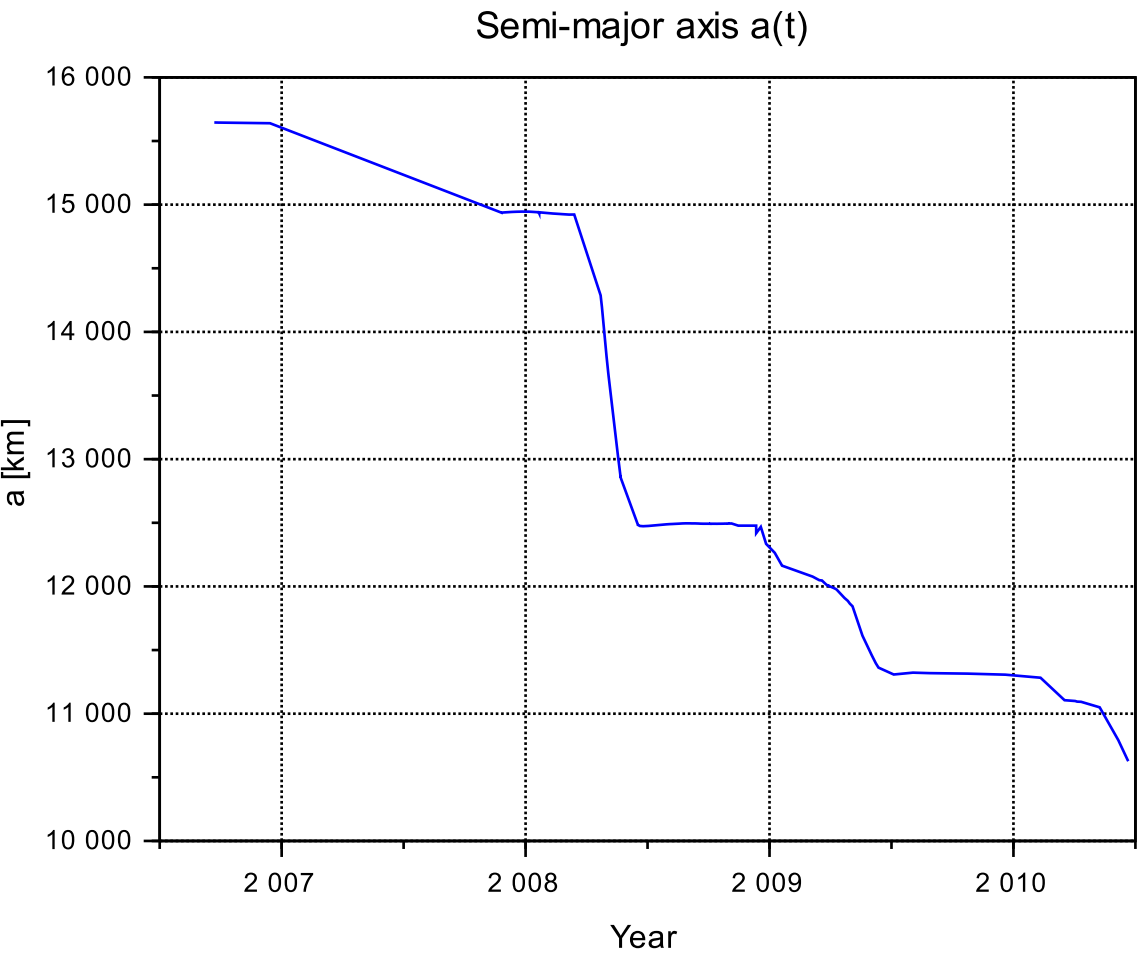
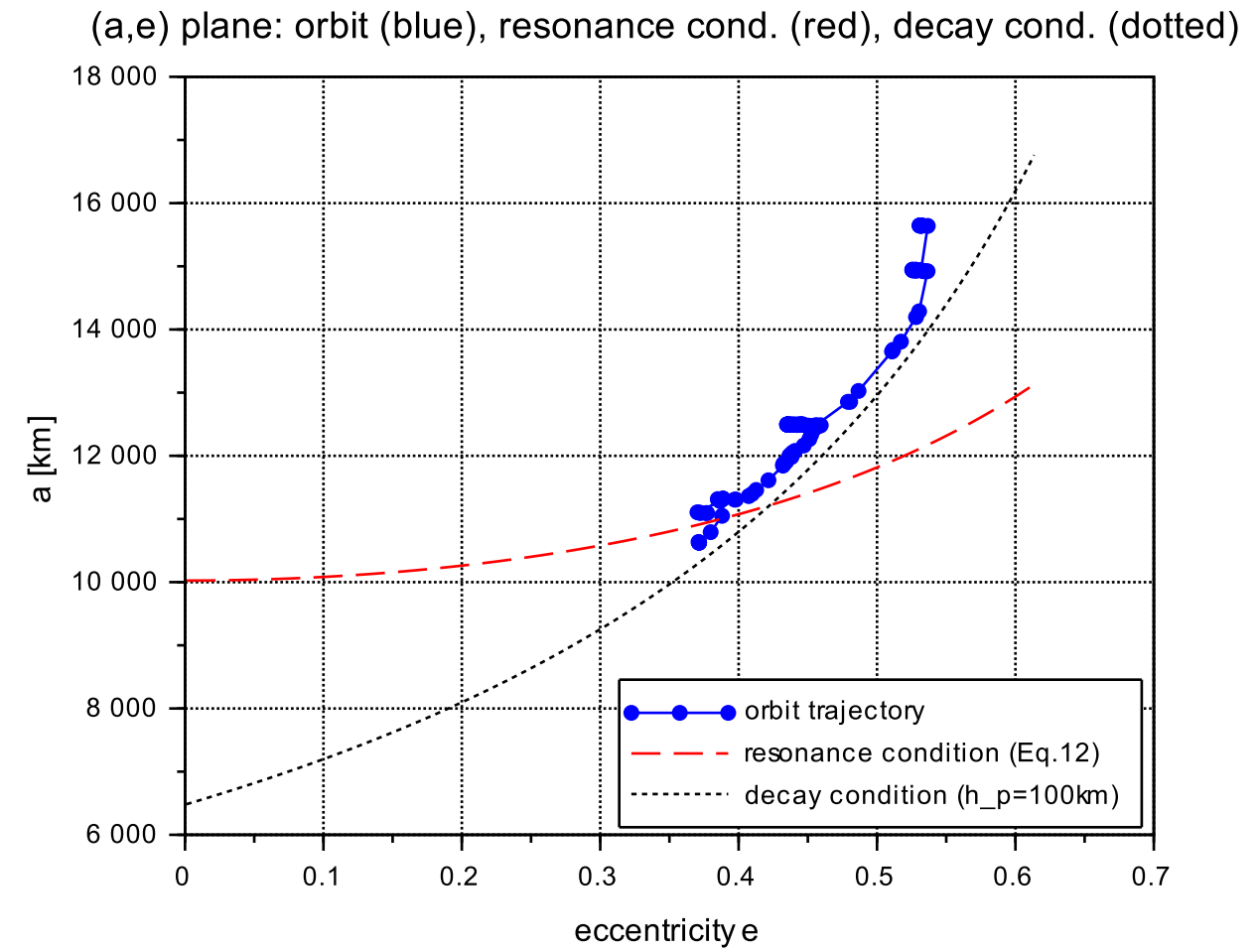
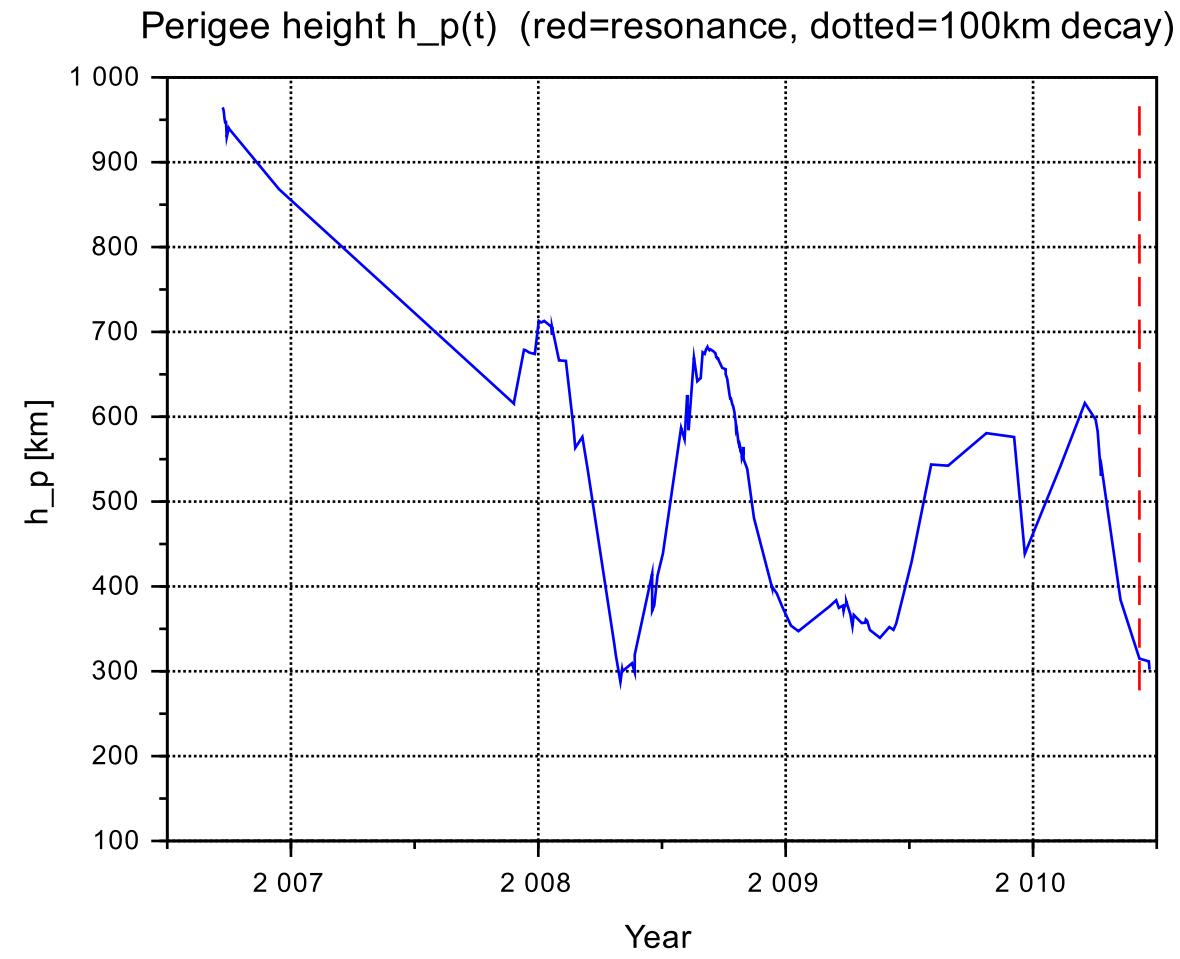
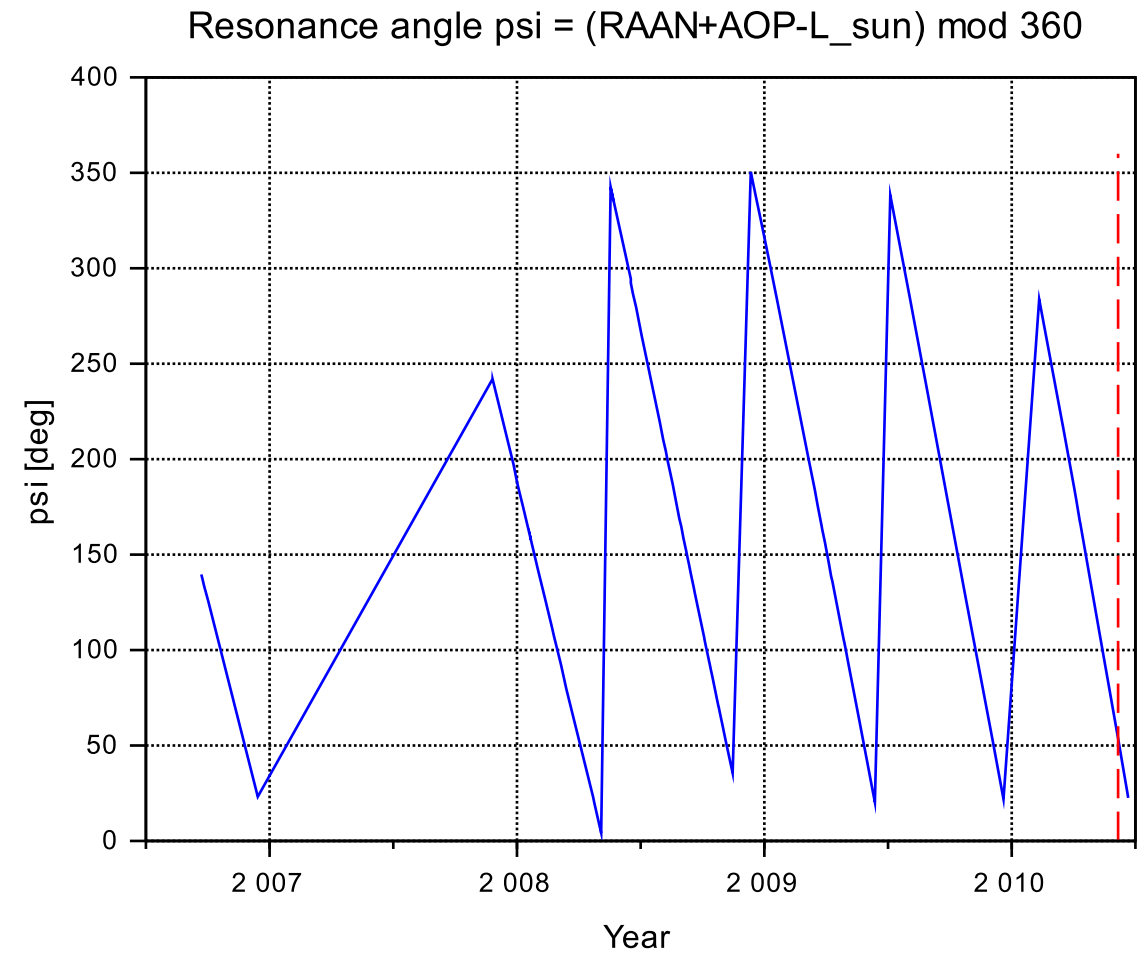
NORAD 68538 ARTEMIS 2 (INTEGRITY)
Mean inclination: 33.62 deg
 λ_{crit} (Eq.12): 9514.11
Resonance epoch: year 2026.25
TLE record: 2026.25 to 2026.27
Observed decay: year 2026.2710



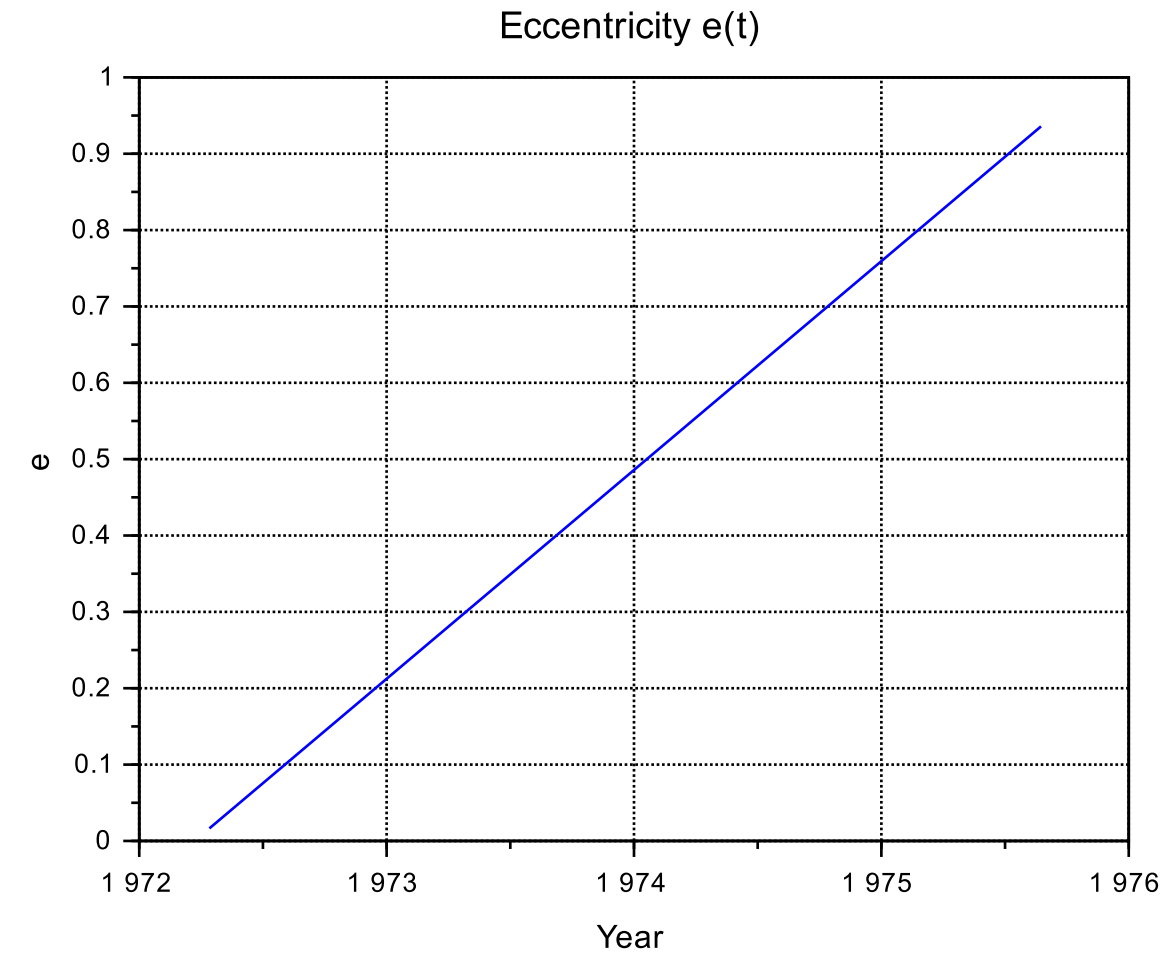
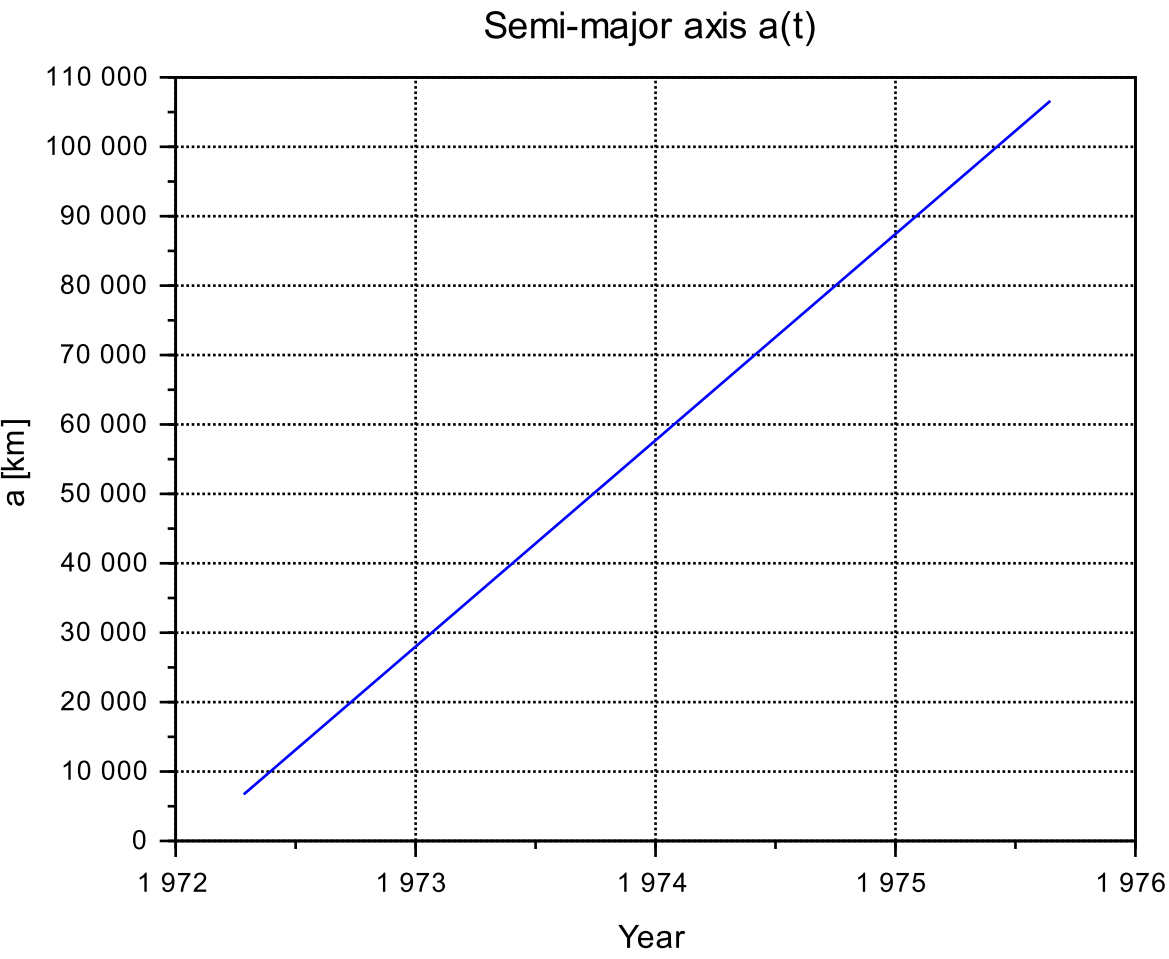
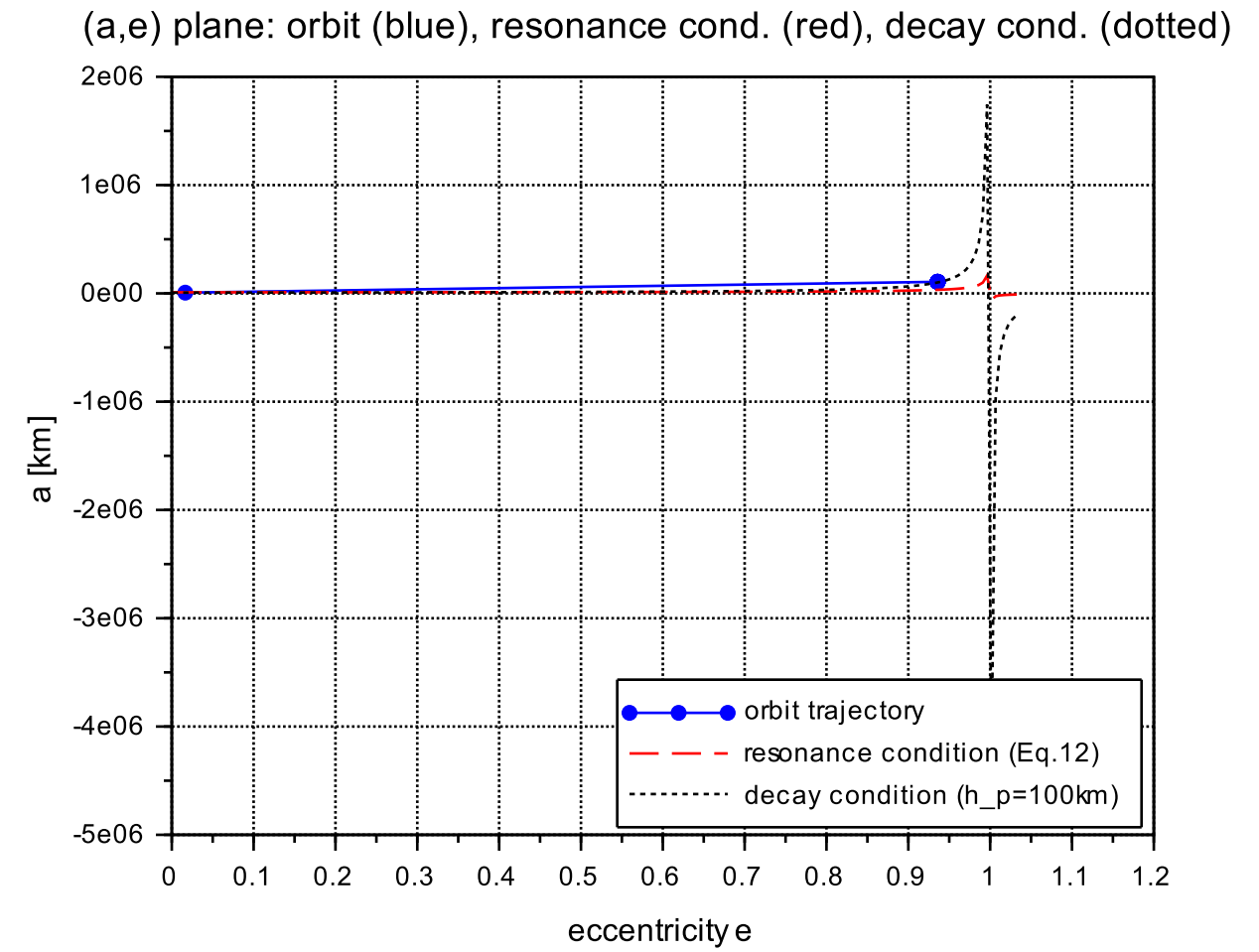
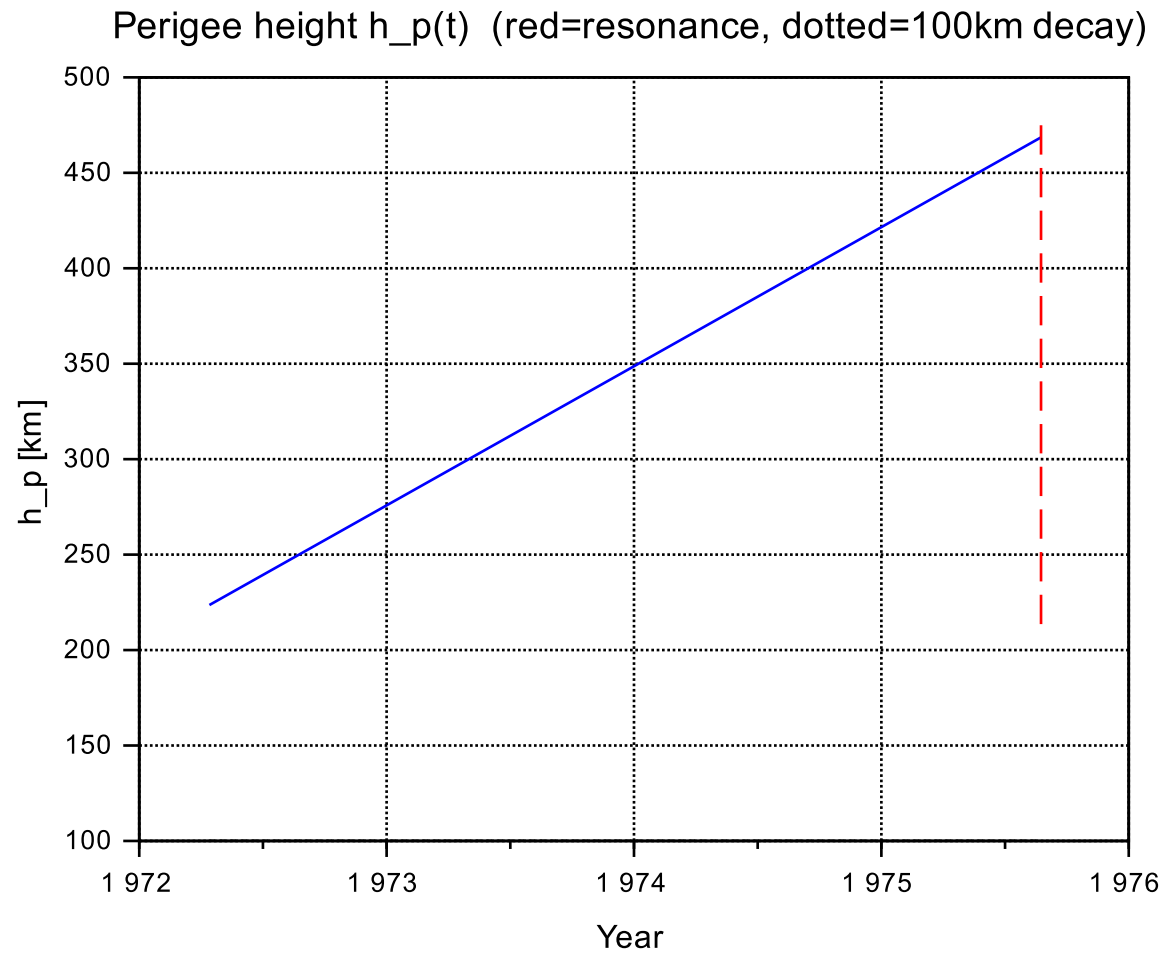
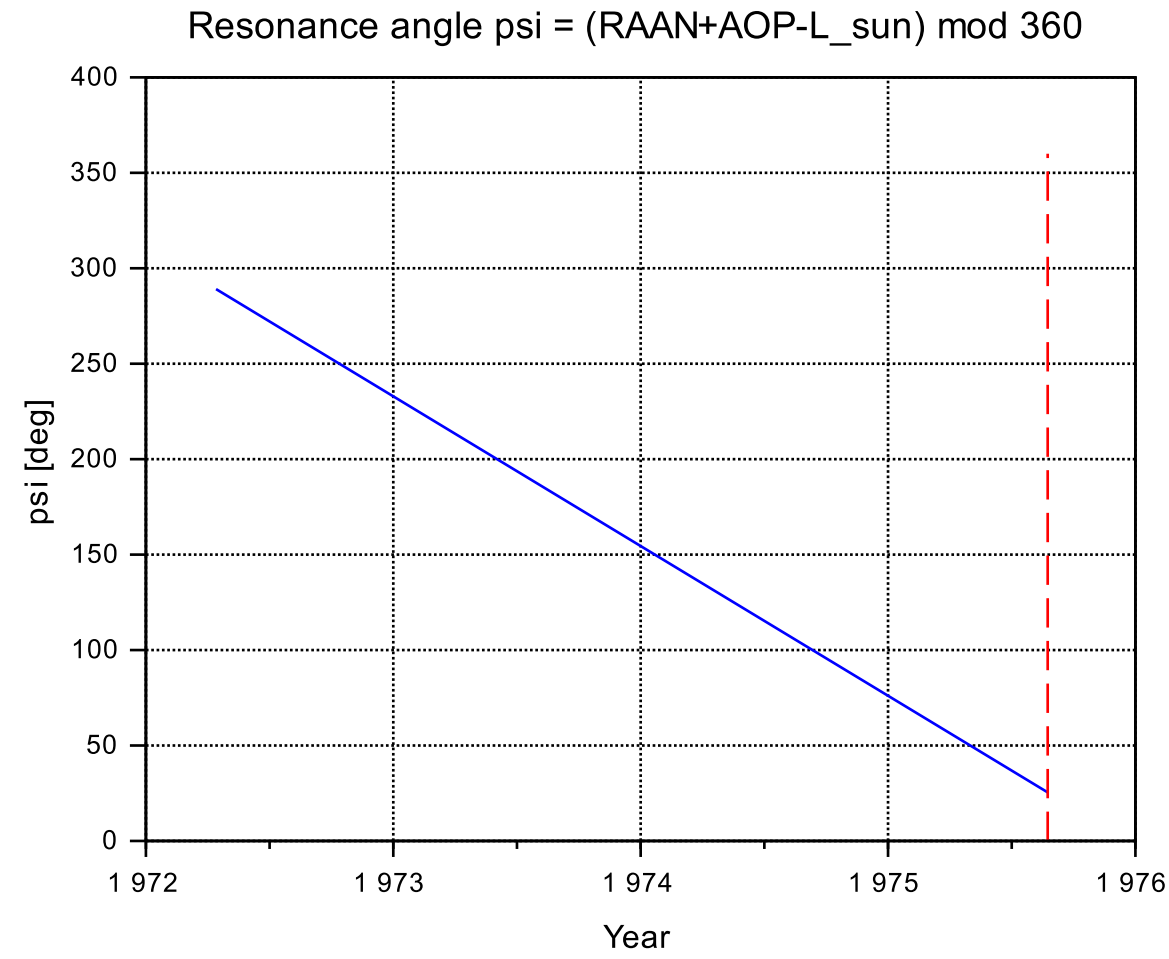
NORAD 6294 MOLNIYA 1-22
Mean inclination: 65.06 deg
 λ_{crit} (Eq.12): 9998.83
Resonance epoch: year 1976.08
TLE record: 1973.02 to 1976.11
Observed decay: year 1976.1109



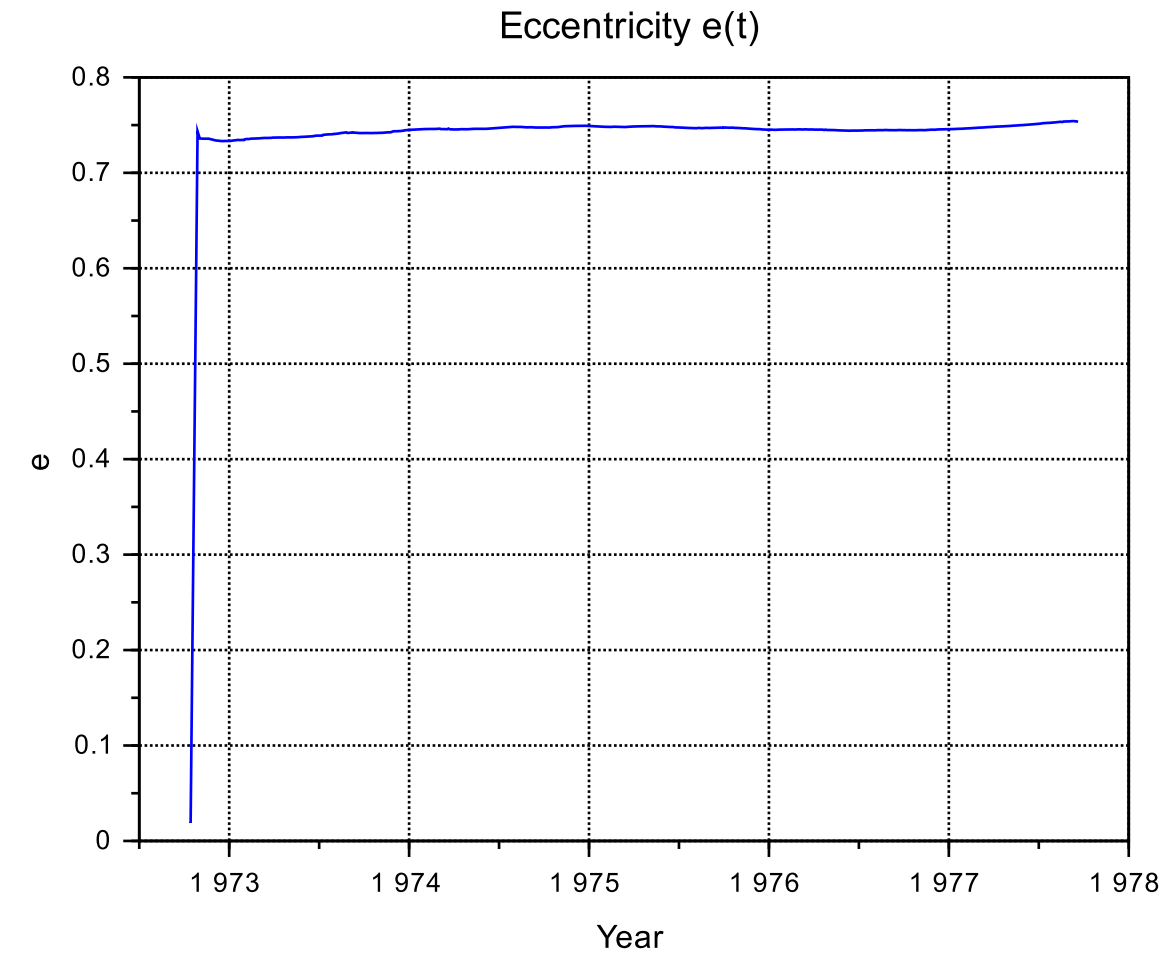
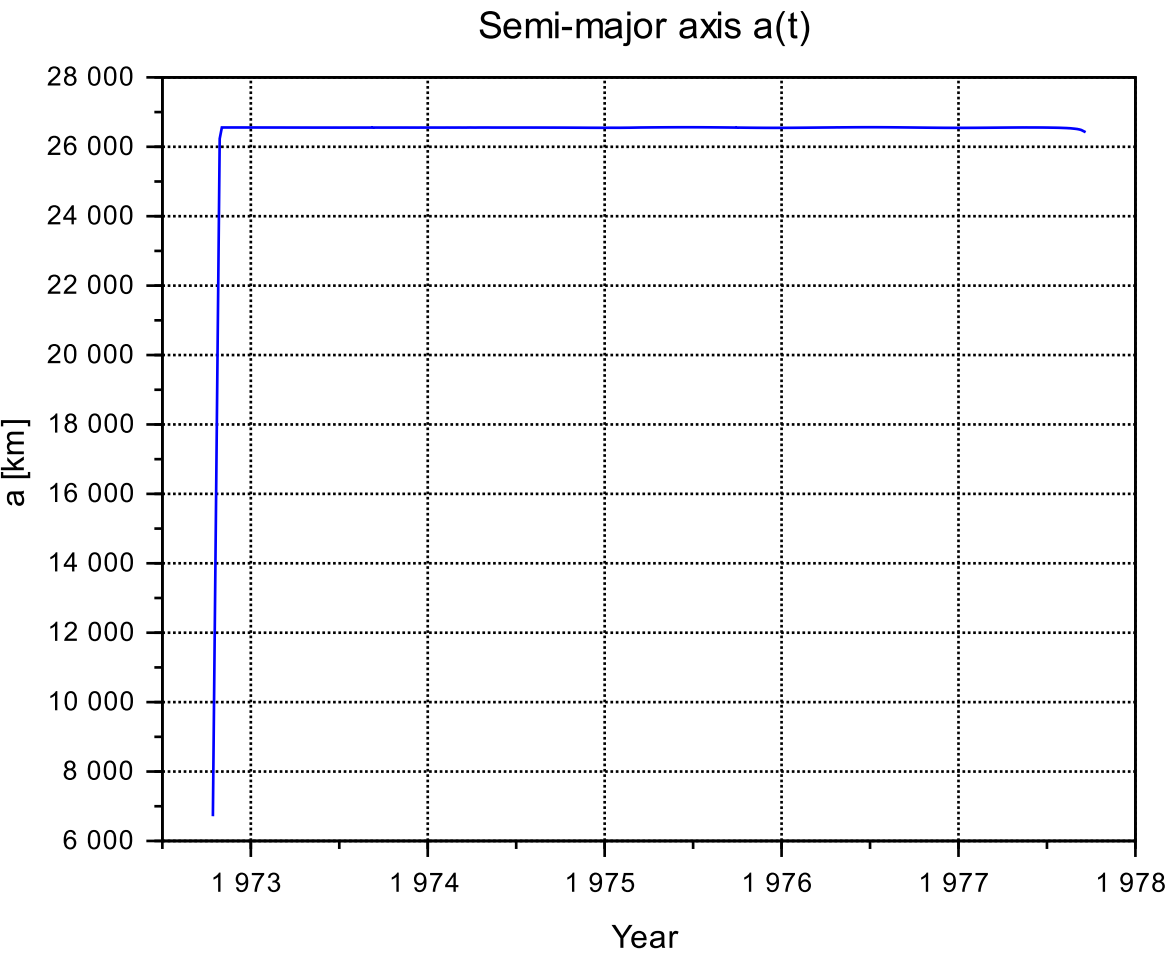
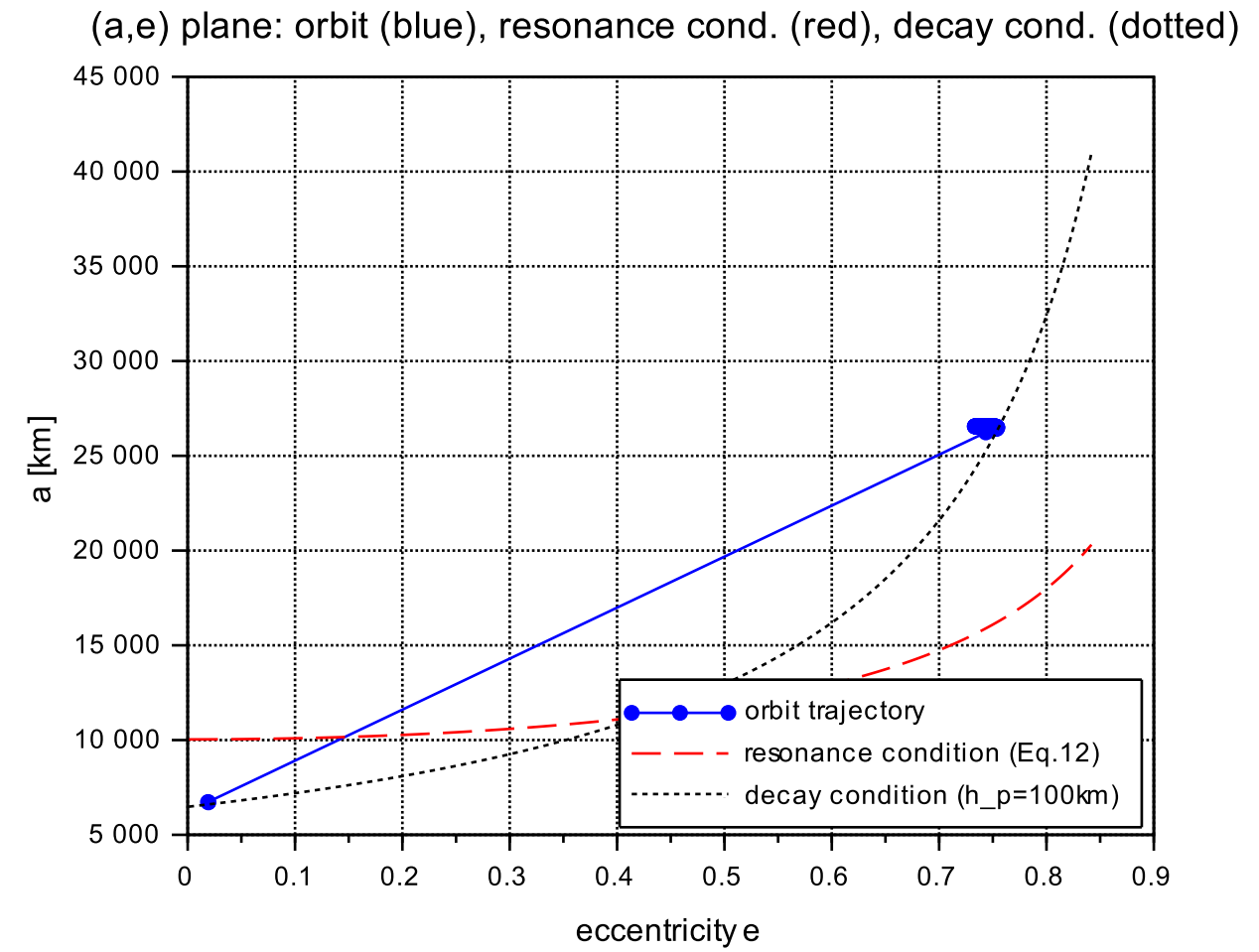
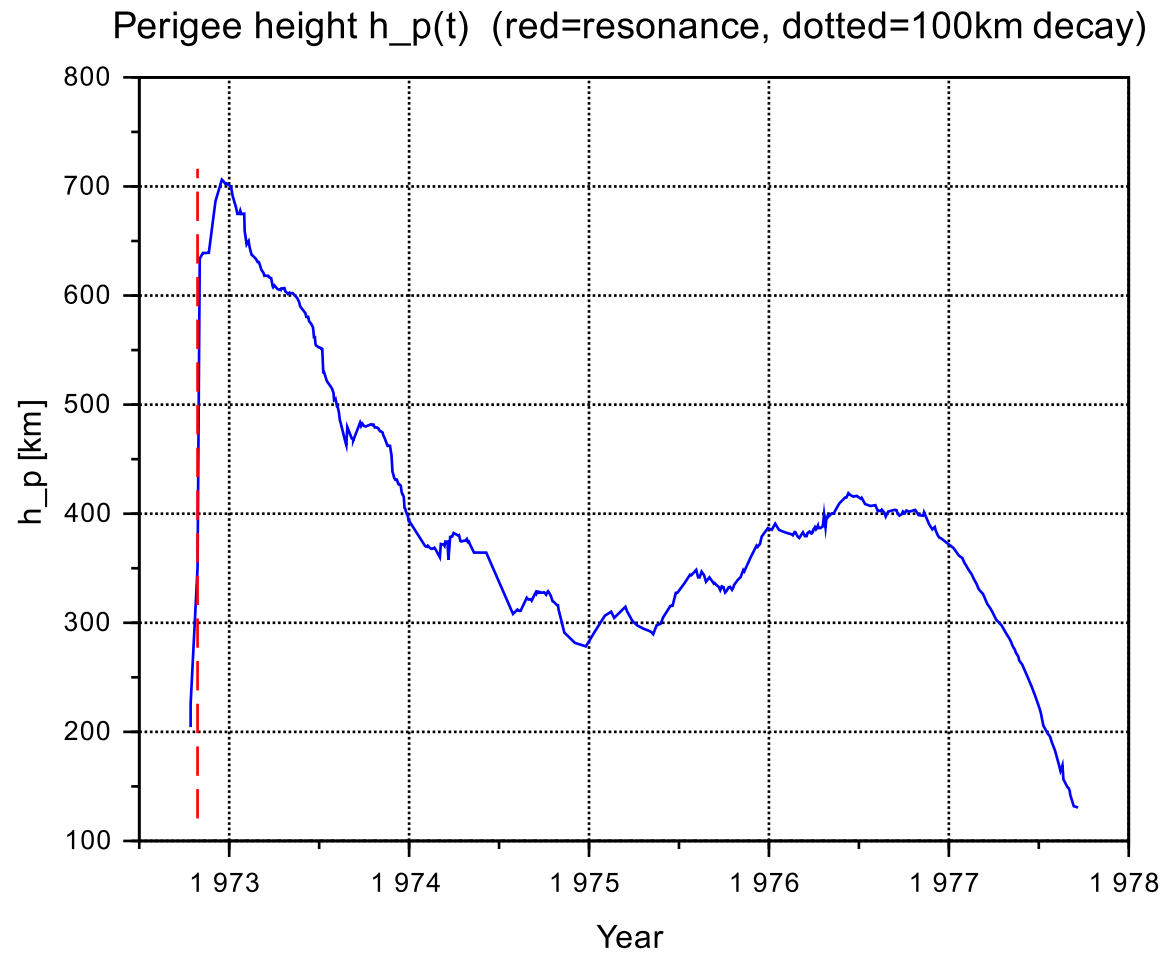
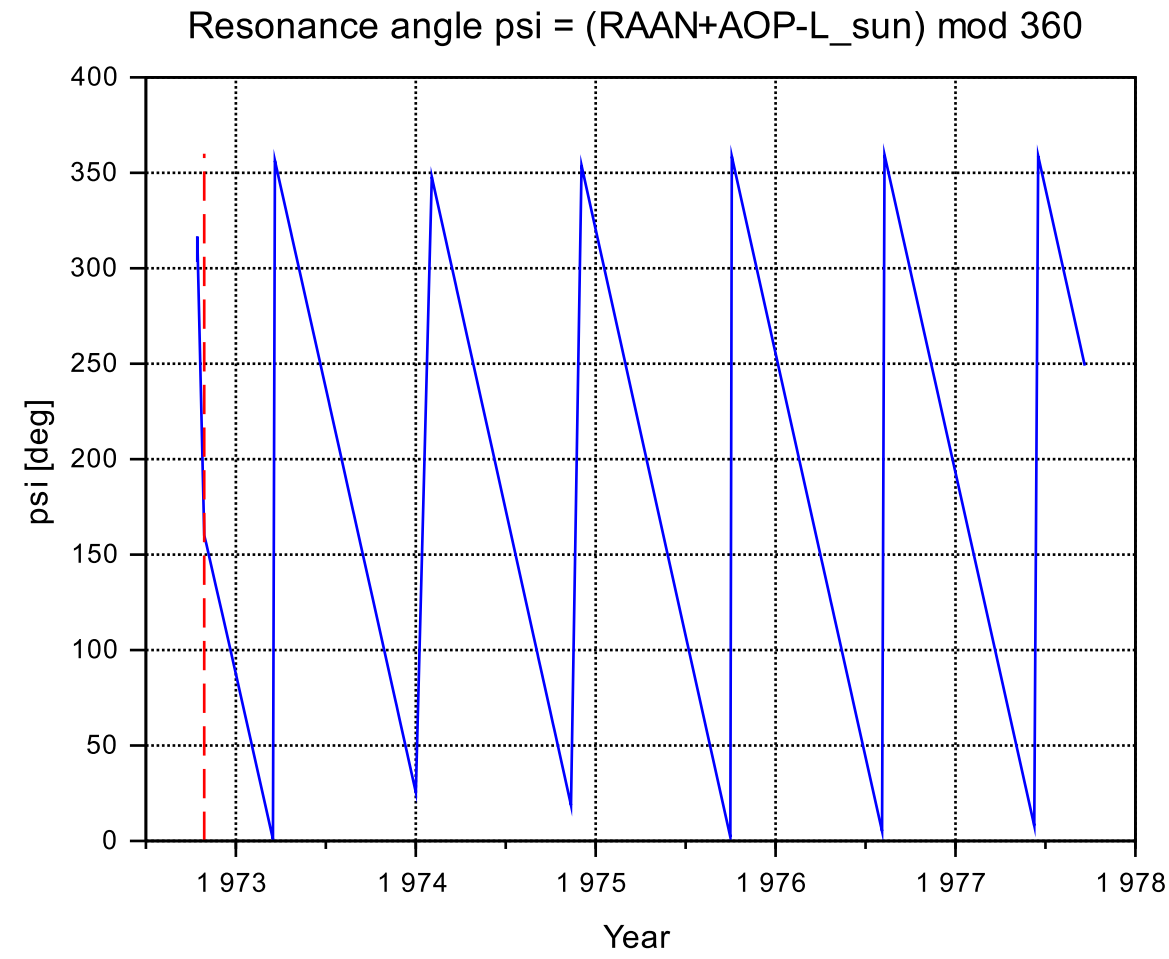
NORAD 18954 ARIANE 3 DEB (SYLDA)
Mean inclination: 6.9 deg
 λ_{crit} (Eq.12): 12249.72
Resonance epoch: year 1989.04
TLE record: 1988.2 to 1989.04
Observed decay: year 1989.0856



NORAD 29462 SL-12 DEB
Mean inclination: 65.31 deg
 λ_{crit} (Eq.12): 10024.89
Resonance epoch: year 2010.43
TLE record: 2006.72 to 2010.47
Observed decay: year 2012.7543



NORAD 5941 PROGNOZ 1
Mean inclination: 64.93 deg
 λ_{crit} (Eq.12): 9984.8
Resonance epoch: year 1975.65
TLE record: 1972.28 to 1975.65
Observed decay: year 1980.0479



NORAD 6231 MOLNIYA 1-21
Mean inclination: 65.41 deg
 λ_{crit} (Eq.12): 10035.02
Resonance epoch: year 1972.82
TLE record: 1972.79 to 1977.72
Observed decay: year 1977.8330